

Institute for Statistics and
Foundations of Machine Learning

Master's Thesis

End-to-End Calibration:

Theory and Empirical Evaluation in Biomedical Context

by

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Overall Research Question:

Can calibration, when embedded into the training objective as a structural constraint, be improved by solving the optimization problem

$$\min_{\theta\in\Theta} \widehat{\mathcal{C}}(f_\theta) + \lambda\widehat{\mathcal{R}}(f_\theta) = \min_{\theta\in\Theta}(1-\lambda)\widehat{\mathcal{C}}(f_\theta) + \lambda\widehat{\mathcal{E}}(f_\theta),$$

while maintaining statistical accuracy (by means of choosing $\lambda > 0$ appropriately)?

Tasks:

- Does augmenting ERM with a calibration objective act as an implicit regularizer, and how does it reshape the bias–variance trade-off relative to predictive-only training followed by post-hoc calibration?
- Can the effect of end-to-end calibration on generalization be characterized within statistical learning theory?
- In simulation-supervised settings with a known true posterior, how do end-to-end calibrated models compare to baselines on exact measures of calibration, refinement, and generalization across model capacity and regularization strength?
- In GRN inference, does end-to-end calibration yield more trustworthy probabilistic predictions than predictive-only training, without sacrificing the scalability that motivates the use of deep models?

Optional task: What does "calibrated" mean under distribution shift? How does it impact marginal calibration and conditional calibration? Does enforcing calibration during training improve robustness under covariate shift?

Statement of Originality

I hereby confirm that I wrote this final thesis independently and without use of prohibited foreign help. I have not used any sources and aids other than those specified. The literal or paraphrased reproduction of paragraphs of third parties' texts as well as from my own previous publications has been properly indicated/referenced.

Furthermore, I assure that this is an original work that has not yet been submitted to another examination office.



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Declaration of Generative AI usage

I hereby declare that during the preparation of this thesis, ChatGPT (GPT-6) was used as an auxiliary tool to improve spelling, grammar, sentence structure, and clarity, as well as to assist with LaTeX formatting, code debugging, and the clarification of technical concepts.

I reviewed and revised the generated outputs as appropriate and take full responsibility for the content, methodology, analysis, and conclusions presented in this thesis.



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Abstract

Reliable probability estimates are important in machine learning, yet most predictive models can be accurate but remain poorly calibrated. This thesis investigates **end-to-end calibration**, in which a differentiable calibration penalty is incorporated into the learning objective during training rather than applying it after training. It examines the approach theoretically, evaluates it in controlled simulation-supervised experiments where the true posterior is known, and studies its application to gene regulatory network (GRN) edge prediction.

In the controlled one-dimensional experiments, combining cross-entropy with a soft calibration penalty improved both calibration and posterior estimation for suitable objective weights. Across 30 random seeds, setting $\lambda = 0.1$ reduced posterior mean-squared error at model capacities $m = 10, 100$, and 1000 from 0.0189, 0.0482, and 0.0378 to 0.0101, 0.0127, and 0.0129, respectively. Expected calibration error also decreased at all three capacities. A controlled covariate-shift experiment showed similarly encouraging results when the input distribution changed but the conditional label mechanism remained fixed. However, ablation studies indicated that some of the improvement was linked to changes in effective step size, and calibration-only training performed poorly. These findings therefore support end-to-end calibration in the studied setting, rather than a universal benefit from adding a calibration penalty.

The GRN experiments produced more mixed results. On DREAM5 Network 1, end-to-end calibration improved the medium-capacity model's AUROC from 0.857 to 0.882, AUPR from 0.787 to 0.833, and Brier score from 0.136 to 0.120. The gains were not consistent across datasets and model capacities, and cross-network evaluation showed that calibration did not automatically transfer to a different network. Post-hoc methods also reduced selected calibration errors, indicating that they remain useful alternatives or complements to end-to-end training.

Overall, this thesis finds that calibration-aware training can improve probability estimates and predictive performance under appropriate conditions, but its effectiveness depends on factors like objective weight, optimization dynamics, dataset, and evaluation criteria. In GRN inference, calibration must be assessed alongside ranking metrics and proper scoring rules. Since the true edge posterior is unknown in biological datasets, the experiments do not establish exact conditional calibration; broader validation across networks and stronger baselines are needed to determine how reliably these methods generalize. The implementation and experimental results are available on GitHub[1].

Keywords: • Probability Calibration • End-to-End Calibration • Deep Learning • Gene Regulatory Networks • Trustworthy AI.

[1]The code implementation for all experiments conducted in this thesis along with all resulting figures and tables are available at GitHub.

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Acronyms

NN	Neural Network
ANN	Artificial Neural Network
ML	Machine Learning
DCA	Difference of Confidence and Accuracy
MSE	Mean Squared Error
MAE	Mean Absolute Error
ERM	Empirical Risk Minimization
BN	Bayesian Network
KDE	Kernel Density Estimation
CE	Cross-entropy
ECE	Expected Calibration Error
MCE	Maximum Calibration Error
ECE_{em}	Equal-mass ECE
ECE_{KDE}	KDE-based Estimator of Canonical Calibration Error
SCE	Static Calibration Error
ACE	Adaptive Calibration Error
TACE	Threshold Adaptive Calibration Error
cwCE	Class-wise ECE
cwCE_{em}	Class-wise Equal-mass ECE
RMSCE	Root Mean Squared Calibration Error
KSCE	Kolmogorov–Smirnov Calibration Error
NLL	Negative Log-Likelihood
DECE	Differentiable Expected Calibration Error
MMCE	Maximum Mean Calibration Error
RKHS	Reproducing Kernel Hilbert Space
RLCR	Reinforcement Learning with Calibration Rewards
GRN	Gene Regulatory Network
DGP	Data Generating Process
GD	Gradient Descent
TS	Temperature Scaling
AUPRC	Area Under the Precision-Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
TF	Transcription Factor

List of Notations

Data, Random Variables, and Sample

X, Y	Input and label random variables; $(X, Y) \sim \mathcal{D}$.
$\mathcal{X}$	Input space, $X \in \mathcal{X}$.
$\mathcal{Y}$	Label space. Convention: $Y \in \mathcal{Y} = \{e_1, \dots, e_K\} \subset \Delta_K$
$\mathcal{D}$	Joint data-generating distribution, with density $\pi(x, y) = \pi(y \mid x)\,\pi(x)$.
$\pi(\cdot)$	Density function. Used for the joint density $\pi(x, y)$, the conditional $\pi(y \mid x)$, the marginal $\pi(x)$, and the posterior-as-function $\pi(\cdot \mid x)$.
Δ_K	Probability simplex, $\Delta_K := \{p \in [0,1]^K : \mathbf{1}^\top p = 1\}$.
K	Number of classes.
e_k	Standard basis (one-hot) vector; the k-th vertex of Δ_K.
n	Sample size.
$S = \{(x_i, y_i)\}_{i=1}^n$	Training sample, drawn i.i.d. from D^n.

Predictors and Hypothesis Class

f	Probabilistic classifier, $f : \mathcal{X} \to \Delta_K$.
$f(x) = (f_1(x), \dots, f_K(x))$	Predicted probability vector at input x.
$\hat{y}(x)$	Predicted class, $\arg\max_k f_k(x)$.
$\hat{p}(x)$	Predicted confidence, $\max_k f_k(x)$.
$\eta(x)$	True posterior / conditional class-probability vector, $\eta(x) := \mathbb{E}[Y \mid X = x]$.
$f^\star$	Bayes-optimal predictor.
$\mathcal{F}$	Hypothesis class of probabilistic classifiers.
θ, Θ	Model parameters and parameter space; f_θ denotes the parameterized predictor.
$P := f(X)$	Prediction random variable.
$\gamma(p)$	Calibration function, $\gamma(p) := \mathbb{E}_D[Y \mid P = p]$.

Loss Functions, Scoring Rules, and Calibration–Refinement Decomposition

$\ell(p, y)$	Per-example loss function.
$\ell(p, q) := \mathbb{E}_{Y\sim q}[\ell(p, Y)]$	Conditional expectation extension of the loss to a target distribution q.
$\mathcal{E}(f)$	Population risk.
$\hat{\mathcal{E}}(f)$	Empirical risk.
$\mathcal{S}(p, y)$	Scoring rule (proper if Eq. 2.13 holds).
$d_\ell(p, q)$	Bregman divergence associated with ℓ ("ℓ-divergence"), $d_\ell(p, q) := \ell(p, q) - \ell(q, q)$.
$e_\ell(q)$	Entropy associated with ℓ ("ℓ-entropy"), $e_\ell(q) := \ell(q, q)$.
$\mathcal{C}_\ell(f)$	Population calibration error (calibration term of the decomposition, Eq. 2.14).
$\mathcal{R}_\ell(f)$	Population refinement error.
$U_\ell(Y)$	Irreducible uncertainty term (Eq. 2.15), independent of f.
$S_\ell(f)$	Sharpness / resolution term (Eq. 2.15).

Calibration Metrics (Evaluation-Time, Non-Differentiable)

ECE	Expected Calibration Error (Eq. 2.6-2.7).
ECE_{prec}	Precision-aware variant of binned calibration error (Eq. 2.8).
MCE	Maximum Calibration Error, worst per-bin gap (Eq. 2.9).
ECE_{em}	Equal-mass (adaptive-binning) ECE.
SCE	Static Calibration Error, class-wise fixed-width binning (Eq. 2.10).
ACE	Adaptive Calibration Error, class-wise equal-mass binning (Eq. 2.11).
TACE(t)	Thresholded Adaptive Calibration Error, threshold t (Eq. 2.12).
RMSCE	Root-Mean-Squared Calibration Error (L_2 per-bin gap).
KSCE	Kolmogorov–Smirnov Calibration Error (binning-free).
B_m	m-th bin of a partition $\mathcal{B} = \{B_1, \dots, B_M\}$ of $[0, 1]$.
$\text{conf}(B_m)$, $\text{acc}(B_m)$, $\text{prec}(B_m)$	Mean confidence, accuracy, precision of predictions in bin B_m.
M	Number of (fixed-width) bins.
R	Number of adaptive (equal-mass) ranges.
t	Threshold parameter of TACE.

Differentiable Calibration Estimators (Training-Time)

$w_{j,i}$	Soft bin-membership weight of sample i to bin j (Eq. 2.23).
c_j	Fixed center of soft bin j.
σ	Kernel bandwidth (soft-binning / MMCE kernels).

$\mathcal{L}_{\text{DECE}}$	Differentiable (kernel-smoothed / soft-binned) ECE loss (Eq. 2.24).
$\kappa(c_i, c_j)$	Differentiable kernel function (typically Laplacian).
$\mathcal{L}_{\text{MMCE}}$	Maximum Mean Calibration Error loss (Eq. 2.26).
r_i	Correctness indicator, $r_i = \mathbf{1}[\hat{y}_i = y_i]$.
$CE_p(f)$	L_p canonical calibration error (Eq. 2.27).
κ_{Dir}	Dirichlet kernel used for KDE over the simplex (Eq. 2.29).
α_i	Dirichlet concentration parameters, $\alpha_i = f(x_i)/h + 1$.
h	KDE bandwidth parameter ($h \to 0$ as $n \to \infty$).
$\widehat{CE_p}(f)$	KDE-based empirical estimator of $CE_p(f)$ (denoted ECE_{KDE} in the text).

Statistical Learning Theory

$\hat{\mathfrak{R}}_m(\mathcal{G})$	Empirical Rademacher complexity of function class $\mathcal{G}$ on a sample of size m (Eq. 2.32).
σ_i	i.i.d. Rademacher (± 1) random variables.
δ	Confidence parameter in generalization bounds (bound holds w.p. $\geq 1 - \delta$).
P, Q	PAC-Bayes prior and posterior distributions over hypotheses (Subsection 2.7.3, Eq. 2.34).
$\text{KL}(Q\|\|P)$	Kullback–Leibler divergence between posterior and prior.

Composite Calibration-Regularized Objective (Chapter 3)

$\mathcal{L}_\lambda(f_\theta)$	Composite training objective, $\mathcal{L}_\lambda = (1-\lambda)\hat{\mathcal{C}} + \lambda\hat{\mathcal{E}}$ (Eq. 3.1).
$\lambda \in (0, 1]$	Calibration/refinement trade-off weight. $\lambda = 1$ recovers plain proper-loss ERM.
$\hat{\mathcal{C}}(f)$	Empirical, differentiable calibration surrogate (instantiated as Soft-ECE, MMCE, or ECE_{KDE}).
$\hat{\mathcal{R}}(f) := \hat{\mathcal{E}}(f) - \hat{\mathcal{C}}(f)$	Empirical refinement, defined as the residual.
$\hat{f}_\lambda$	Calibration-regularized ERM estimator, $\hat{f}_\lambda \in \arg\min_\theta \mathcal{L}_\lambda(f_\theta)$.
$\mu = (1-\lambda)/\lambda$	Equivalent penalty weight in the penalized (Lagrangian) formulation.
$\mathcal{F}_c$	Calibration-restricted hypothesis class, $\mathcal{F}_c := \{f \in \mathcal{F} : \hat{\mathcal{C}}_\ell(f) \leq c\}$ (Eq. 3.3).
$c = c(\mu)$	Constraint level corresponding to penalty weight μ; non-increasing in μ.
$\ell \circ \mathcal{F}$	Induced loss class, $\{(x, y) \mapsto \ell(f(x), y) : f \in \mathcal{F}\}$.

$D_{\text{train}}, D_{\text{test}}$	Training and test distributions (Section 2.3, 3.7); equal under the i.i.d. assumption, may differ under covariate shift.

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1. Introduction

1.1. From Accurate Predictions to Trustworthy Probabilities

Over the past decade, deep neural networks have achieved remarkable predictive accuracy across domains ranging from image recognition to natural language understanding. But, accuracy alone cannot indicate how useful a model will be in real-life situations. In cases where the model prediction is used for a downstream decision under uncertainty, we should not only ask *whether the model is right on average*, we should also see *how much its stated confidence can be trusted*. A model that assigns a probability of 0.9 to an event, ideally should be correct about 90% percent of the time it makes such prediction. The model is said to be *calibrated*, when this holds true.

Calibration is an important requirement for informed decision-making, especially in biomedical context. For example, a clinician deferring borderline cases to further testing, an autonomous vehicle choosing when to cede control to a human, and a scientist deciding which predictions are worth further experimental validation, all these depend on the numerical confidence attached to a prediction, not just the predicted label [1, 2]. If these confidences are too high, the decision system inherits the model's overconfidence and acts on evidence that is weaker than it appears.

Unfortunately, despite higher accuracy, modern neural networks are often miscalibrated. In a study, Guo *et al.* [1] showed that modern deep network architectures, despite showing superior accuracy compared to their predecessors, are significantly overconfident: their average predicted confidence considerably exceeding their empirical accuracy. He attributed this phenomenon to several factors such as increased depth and width, weight decay, and the use of normalization layers, all of which improve discrimination while distorting the probability scale. Subsequent work has attempted to refine this picture. Minderer *et al.* [3] revisited the question for newer architectures and found that some recent, non-convolutional models are truly among the best calibrated, suggesting that architectural design is itself a determinant of calibration. Many benchmark studies have since investigated how calibration interacts with accuracy, robustness, and architecture across several models [4]. There are two common messages across all studies: miscalibration is neither universal nor inevitable, but it is common and consequential enough to be treated as a first-class objective rather than an afterthought. Figure 1.1 shows confidence histograms and reliability diagrams of two different models on CIFAR-100, reproduced from [1]. The dotted diagonal line in the reliability diagram denote the perfect calibration.

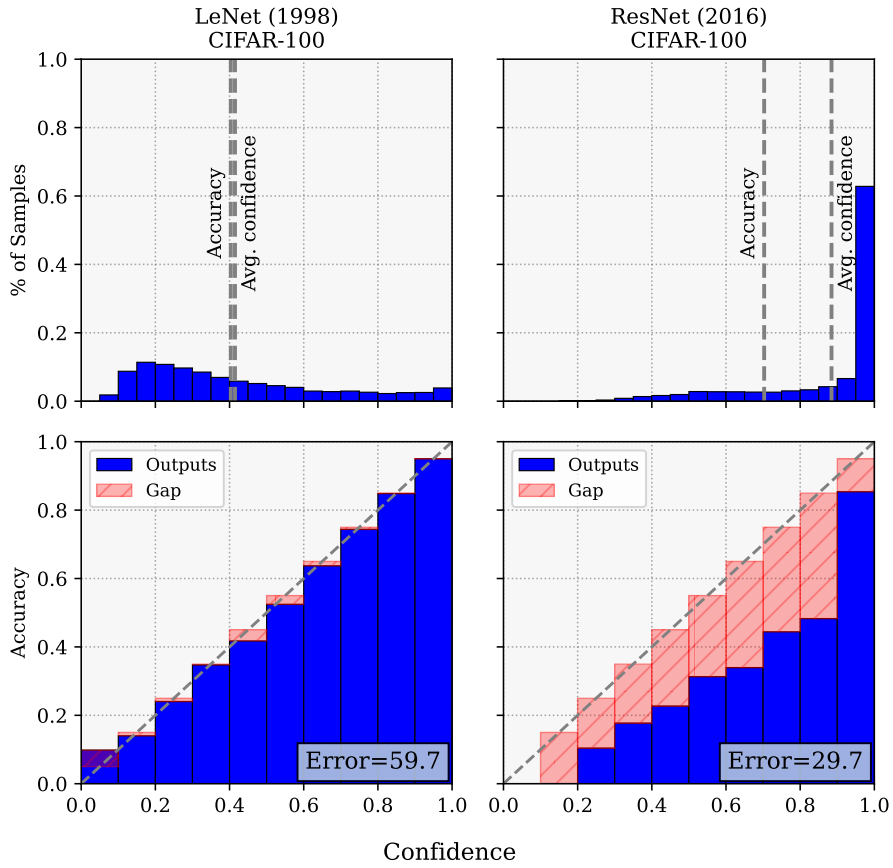


Figure 1.1.: Confidence histograms (top) and reliability diagrams (bottom) for a 5-layer LeNet (left) and a 110-layer ResNet (right) on CIFAR-100, reproduced from [1].

1.2. The Limits of Post-hoc Calibration

The most frequently used strategy to calibrate a model has been *post-hoc calibration*: train a model in the usual way to maximize predictive performance, then adjust its outputs afterwards using a small held-out dataset. Classical post-hoc calibration methods include *Platt scaling*, which fits a sigmoid to a classifier's scores [5]; *histogram binning* and *isotonic regression*, which map scores to empirical frequencies [6]; *Bayesian Binning into Quantiles* (BBQ), a non-parametric method that averages over binning schemes [7]; and *temperature scaling*, a single-parameter simplification of Platt scaling that Guo *et al.* [1] found to be surprisingly effective for deep networks. These methods are model-agnostic and inexpensive: they treat the trained predictor as a black box and reshape only its output probabilities.

This modularity is also their central limitation. Since these post-hoc calibration methods operate on a frozen model, they leave the *hypothesis class* and the *learning dynamics* unchanged. It rescales the output probabilities, but it cannot alter the learned representation of the model, the decision boundary, or the inductive biases that shaped them. This has three consequences:

(1) since the learned representation is unchanged, the expressive power of the correction is limited. For example, temperature scaling can only apply a monotone, global transformation and does not work when calibration varies across input space, (2) the corrected calibration is weak under distribution shift, because a transformation fit on one distribution may transfer to another [3], and most fundamentally, (3) calibration is decoupled from representation learning. Since it is treated as an afterthought of training, the model does not organize its internal computation during training to yield meaningful probabilities.

A recent research direction makes this explicit. Proper losses such as cross-entropy admit a decomposition into a *calibration* term and a *refinement* term. Berta *et al.* [8] demonstrate, both theoretically and empirically, that these two terms are not minimized simultaneously during training. Selecting a model by validation loss thus lands on a compromise that is optimal for neither. Their proposed remedy is to optimize refinement during training and to handle calibration afterwards. While this approach proves to be effective, this also confirms that, under standard training, calibration and the learned representation pull in different directions. It is natural to ask the converse question: *what happens if calibration is enforced during the training via a differentiable penalty folded into the training objective*, so that the learner is shaped by it from the beginning of the training?

1.3. End-to-End Calibrated Learning

This thesis investigates **end-to-end calibrated learning**: it integrates calibration directly into the training objective, where it act as a *structural constraint* on the hypothesis space instead of correcting after training. So, we minimize a composite objective during training, in which a calibration penalty also contributes in empirical risk minimization and influences which hypothesis is selected.

The most important challenge is that the standard calibration metric, expected calibration error (ECE), is piecewise constant and hence has zero gradient almost everywhere, so it cannot be optimized directly. Different papers have proposed differentiable alternatives that overcome this challenge. The Difference of Confidence and Accuracy (DCA) term penalizes the gap between mean confidence and accuracy as an auxiliary loss [2]; the Maximum Mean Calibration Error (MMCE) restructure calibration in a reproducing kernel Hilbert space, yielding a trainable measure [9]; soft (continuous) binning replaces hard bin assignments with smooth weights, making the calibration error differentiable [10]; and consistent kernel-density estimators of the L_p canonical calibration error provide low-bias, differentiable penalties suitable for direct risk regularization [11]. Beyond supervised classification, reinforcement-learning approaches this problem by adding a correctness reward with a bounded proper scoring rule (e.g. Brier score) for training that are simultaneously accurate and calibrated [12]. We need to note that the boundedness of the scoring rule is a conscious design choice because unbounded rules (e.g. log-loss) can be dominated by a few over-confident wrong predictions.

The conceptual claim of this thesis is that introducing such a calibration component into empirical risk minimization (ERM) objective act as *implicit regularization*. By forcing the model's

confidences to agree with empirical frequencies, the calibration term restricts the set of admissible solutions: it rejects hypotheses that fit the training labels well but make over- or under-confident probabilistic claims. In theoretical sense, calibration acts as an *inductive bias*, reshaping the bias–variance trade-off, same as weight decay or early stopping, and so theoretically it should be analyzable through the lens of generalization theory. Our hypothesis is that, under suitable conditions, end-to-end calibration reduces the effective capacity exploited by the learner and can be related to tighter generalization guarantees. The theoretical motivation of this thesis is to establish the precise form of this relationship, and test it where ground truth is known.

For the empirical study, we intentionally use the *simulation supervised* settings. We generate the data with *known* true conditional distribution (and hence the perfectly calibrated predictor). Such settings are crucial here because calibration quality and generalization gaps can only be measured exactly when the ground truth is known, which is rare for real data, specially in biomedical context.

1.4. Gene Regulatory Network Inference as a Testbed

To stress-test our idea in a realistic and meaningful setting, we use the inference of *gene regulatory networks* (GRNs) as an example. A GRN is a directed graph where genes are the nodes and edges denote regulatory influence: usually the action of a transcription factor on the expression of a target gene. Reconstructing such a network from an expression measurements is a central problem in computational biology, because the relationship between regulatory circuits is crucial for several biomedical inference [13, 14]. This task can also be seen as probabilistic reasoning with uncertain edges and indirect evidences. And the practical question is *which predicted interactions are confident enough to pursue experimentally*: the answer to which is meaningful only if the predicted probabilities are calibrated.

GRN inference is important for three reasons. *Firstly*), the data are high-dimensional and corrupted. Modern single-cell RNA-sequencing measure thousands of genes but suffer from increased noise, sparsity, and "dropout" events, where expressed transcripts are recorded as zero. Several simulation studies proved that such noises can affect the network inference accuracy [15]. *Secondly*, the field provide a mathematically rigorous, but expensive way to model gene expression. They represent relationships between genes using a directed graph and help identify which genes are conditionally dependent or independent of each other. With certain assumptions, they also help infer causal relationships between genes [13]. However, learning the Bayesian network structure from data is usally NP-hard, and the structure learning does not scale to the thousands of genes [16]. And *thirdly*, the methods that can actually scale, often optimize the wrong thing. Tree-ensemble regressors such as GENIE3 [17] and its gradient-boosted successor GRNBoost2 [18], information-theoretic scores [19], and a growing family of deep models (convolutional classifiers of gene-pair co-expression [20] and graph attention networks [21]) are evaluated almost entirely on *ranking-based predictive metrics* such as the areas under the ROC and precision–recall curves [14, 22]. Such metrics reward correct order

of candidate edges but hardly say anything about the reliability of the associated probabilities or whether the recovered edges indicate genuine regulatory (causal) structure.

This gap motivates the main idea of this thesis. Deep learning provides the scalability missing in classical Bayesian structure learning, but standard deep models are trained for predictive accuracy and evaluated on such metrics; they provide neither calibrated uncertainty nor guarantee causal validity. GRN inference is therefore a crucial area to ask whether end-to-end calibration can close the gap by producing trustworthy probabilistic inferences by training the model with calibration as part of the learning objective.

It is important for the thesis's methodological aims that GRN inference comes with controlled, simulation-supervised benchmarks. The DREAM5 challenge provides standardized expression datasets with known gold-standard networks [22]; the BEELINE framework systematizes the evaluation of single-cell GRN methods against curated and synthetic ground truth [14]; and the SERGIO simulator generates single-cell expression data from a user-specified GRN [15]. These resources allow us to build settings with known true network and the calibrated predictor, so that calibration and generalization can be measured. Figure 1.2 provides a conceptual diagram of how typically we can predict the network from the expression dataset.

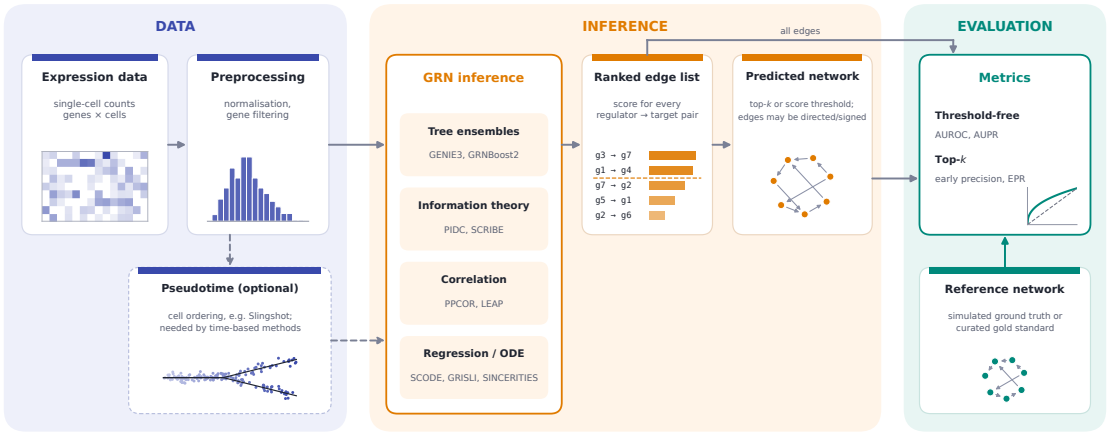


Figure 1.2.: A conceptual schematic of the GRN inference pipeline: expression matrix (genes × cells) → learned model → predicted edge probabilities → comparison against a gold-standard reference network. This shows calibration governs which predicted edges are "trustworthy".

1.5. Problem Statement, Research Questions, and Hypotheses

The thesis is organized around a single overarching question:

> *Can calibration, when embedded into the training objective as a structural constraint, improve the generalization and trustworthiness of scalable probabilistic predictors?*

And can this effect be both characterized theoretically and validated empirically in simulation-supervised settings, including gene regulatory network inference?

We decompose this overarching objective into the following research questions.

RQ1 (Mechanism). Does augmenting empirical risk minimization (ERM) with a calibration objective act as an implicit regularizer, and how does it reshape the bias–variance trade-off relative to predictive-only training followed by post-hoc calibration?

RQ2 (Theory). Can the effect of end-to-end calibration on generalization be characterized within the scope of statistical learning theory, e.g. by relating the calibration-constrained hypothesis space to complexity-based or PAC-Bayesian generalization bounds?

RQ3 (Controlled validation). In simulation-supervised settings with a known true posterior, how do end-to-end calibrated models compare to baselines on exact measures of calibration, refinement, and generalization across model capacity and regularization strength?

RQ4 (Application). If applied to GRN inference, does end-to-end calibration yield more trustworthy probabilistic predictions than predictive-only training, without losing the scalability that motivates the use of deep models?

These questions are guided by our central **hypothesis** that embedding calibration into empirical risk minimization constrains the hypothesis space in a way that behaves similar to regularization, and this constraint can improve generalization while producing reliable uncertainty estimates, under appropriate conditions.

1.6. Contributions

This thesis makes the following contributions.

1. **A unifying framing of end-to-end calibration as a structural constraint.** We consolidate differentiable calibration objectives [2, 9, 10, 11] and reward-based proper-scoring-rule formulations [12] into a single perspective in which calibration acts as an inductive bias on the hypothesis space, and we articulate why bounded proper scoring rules are preferable at training time.

2. **A theoretical analysis linking calibrated training to generalization.** We study how a calibration constraint interacts with classical generalization machinery (complexity-based and PAC-Bayesian bounds) and with the calibration–refinement decomposition of proper losses [8], clarifying the sense in which calibration can tighten the bias–variance trade-off.

3. **A controlled, simulation-supervised study.** Using data-generating processes with

a known true posterior, we empirically characterize how end-to-end calibration affects calibration error, refinement, and the generalization gap as a function of model capacity and penalty strength, isolating effects that cannot be measured on real data.

4. **An application case study in GRN inference.** We apply end-to-end calibrated learning to scalable GRN inference and evaluate it on simulation-supervised and challenge benchmarks [22, 14, 15], assessing not only predictive ranking but the trustworthiness and structural validity of the inferred probabilities.

1.7. Thesis Outline

The remainder of this thesis is organized as follows. **Chapter 2** reviews the necessary background and related work along with the formal definitions and notions of calibration, estimators and their biases, proper scoring rules and the calibration–refinement decomposition, post-hoc and in-training calibration methods, the statistical learning theory used to reason about generalization, and the gene regulatory network inference literature. **Chapter 3** develops the theoretical analysis of end-to-end calibration as a structural constraint and connect it to generalization. **Chapter 4** presents the simulation-supervised experiments with known ground truth. **Chapter ??** applies the concept in gene regulatory network inference as a case study. Finally, **chapter 6** concludes with a discussion of limitations and directions for future work.

2. Background and Literature Review

This chapter lays the conceptual and technical foundation for the rest of the thesis. Sections 2.1–2.4 introduce the theory of calibration and cover its formal definition and evaluation metrics, different methods to measure it, and its relationship to proper scoring rules as well as the calibration–refinement decomposition. Section 2.5 reviews most common post-hoc recalibration methods along with a comparative summary, while section 2.6 reviews in-training (end-to-end) calibration. These approaches frame the central research question addressed in this thesis. Section 2.7 presents the statistical learning theory needed to analyze generalization and to evaluate the hypothesis that calibration can serve as a structural constraint. Finally, section 2.8 introduces gene regulatory network (GRN) inference as an application domain, and section 2.9 connect these threads together to identify the research gaps which motivates the thesis.

2.1. Notation and Problem Setup

We consider a supervised classification setting.

Convention: Throughout the rest of the thesis, random variables are denoted by uppercase letters while their realizations (or generic arguments of a function) by the corresponding lowercase letters. So $f(x)$ is the prediction at a fixed input x while $f(X)$ is the (random) prediction at the random input X.

Let (X, Y) be random variables with joint distribution $\mathcal{D}$ with density $\pi(x, y) = \pi(y \mid x)\,\pi(x)$, where $X \in \mathcal{X}, Y \in \mathcal{Y} \subset \Delta_K$, and

$$\Delta_K := \{p \in [0,1]^K \;:\; \mathbf{1}^\top p = 1\}$$

denote the probability simplex.

A *probabilistic classifier* is a measurable function $f : \mathcal{X} \to \Delta_K$ mapping inputs to the probability simplex, so that for an input x, it returns $f(x) = (f_1(x), \dots, f_K(x))$ with $f_k(x) \geq 0$ and $\sum_k f_k(x) = 1$ for $k = 1, \dots, K$.

We write $\hat{y}(x) = \arg\max_k f_k(x)$ for the predicted class at x and $\hat{p}(x) = \max_k f_k(x)$ for the associated *confidence*. We also let $\ell : \Delta_K \times \mathcal{Y}_K \to \mathbb{R}_+$ be a loss function. We define its conditional expectation as

$$\ell(p, q) := \mathbb{E}_{Y \sim q}[\ell(p, Y)] \quad \text{for } p \in \Delta_K,\ q \in \Delta_K.$$

The (population) risk of a classifier f is defined as

$$\mathcal{E}(f) := \mathbb{E}_{(X,Y)\sim D}\big[\ell(f(X), Y)\big] = \mathbb{E}_X\big[\ell(f(X), \eta(X))\big],$$

where $\eta(x) := \mathbb{E}[Y \mid X = x] \in \Delta_K$ denotes the *conditional class-probability vector* or the *true posterior*.

Given a training sample $S = \{(x_i, y_i)\}_{i=1}^n \sim \mathcal{D}^n$ (drawn i.i.d.), we select f from a hypothesis class $\mathcal{F}$ by minimizing an empirical risk

$$\widehat{\mathcal{E}}(f) \;=\; \frac{1}{n}\sum_{i=1}^{n} \ell\big(f(x_i), y_i\big), \qquad \mathcal{E}(f) \;=\; \mathbb{E}_{(X,Y)\sim\mathcal{D}}\big[\ell(f(X), Y)\big], \tag{2.1}$$

where ℓ is the defined loss function and $\mathcal{E}(f)$ is the population (true) risk.

The Bayes-optimal probabilistic predictor, the *true posterior*, is $f^\star(x) = \eta(x) = \pi(\cdot \mid x)$; it is the unique minimizer of any *strictly proper* loss (explained in detail in Section 2.4.4). The central problem explored in this thesis is that accuracy, governed by $\arg\max_k f_k(X)$, and the quality of the full probability vector $f(X)$ are distinct objectives, and that standard training optimizes the former far more reliably than the latter.

2.2. What It Means for a Model to be Calibrated

A model is said to be calibrated if its stated probabilities match observed frequencies: a requirement formalized for sequential predictors by the notion of the well-calibrated Bayesian [23]. For a probabilistic classifier f we define the prediction random variable $P := f(X)$. The classifier f is said to be *(perfectly) calibrated* if

$$\mathbb{E}[Y \mid P] = P \quad \text{almost surely.}$$

Equivalently, for all $p \in \Delta_K$ with $\mathbb{P}(P = p) > 0$,

$$\mathbb{P}(Y = e_k \mid P = p) = p_k \quad \text{for all } k = 1, \dots, K.$$

We now provide several formalizations of this notion in increasing strictness [24, 11], also summarized in table 2.1 along with their implication hierarchy and the typical estimator used.

Confidence (top-label) calibration. The weakest and most common notion concerns only the predicted class and its confidence [1]. A model is confidence-calibrated if

$$\mathbb{P}\big(Y = \hat{y}(X) \,\big|\, \hat{p}(X) = p\big) \;=\; p, \qquad \forall\, p \in [0, 1]. \tag{2.2}$$

This means, among all inputs assigned confidence p, a fraction p are classified correctly. The expected calibration error and reliability diagrams used by Guo *et al* [1] used this notion.

Table 2.1.: Notions of calibration in increasing order of strictness. Each notion conditions on a different statistic of the predicted vector $f(X)$ and implies all weaker notions above it. The conditioning event becomes higher-dimensional from top to bottom, which is why consistent estimation of canonical calibration is the hardest. The comparison has been adapted from Popordanoska *et al* [11]. Canonical calibration implies marginal calibration, which implies confidence calibration; the converses do not hold.

Notion	**Conditions on**	**Formal condition**	**Hierarchy**	**Typical estimator**
Confidence (top-label)	Top confidence $\hat{p}(X) = \max_k f_k(X)$	$\mathbb{P}\big(Y = \hat{y}(X) \mid \hat{p}(X) = p\big) = p$	Weakest; implied by the two below	Binned ECE; reliability diagram [1, 7]
Marginal (class-wise)	Each class score $f_k(X)$	$\mathbb{P}\big(Y = e_k \mid f_k(X) = p\big) = p,\ \forall k$	Implies confidence; implied by canonical	Class-wise ECE; Dirichlet diagnostics [24, 26]
Canonical (distribution)	Full vector $f(X) = \mathbf{p}$	$\mathbb{P}\big(Y = e_k \mid f(X) = \mathbf{p}\big) = p_k,\ \forall k$	Strongest; implies both above	Kernel/KDE canonical error $\mathrm{ECE}_{\mathrm{KDE}}$ [11, 25]

Marginal (class-wise) calibration. A stronger requirement is that each class probability be calibrated one-vs-rest [24]: for every class k,

$$\mathbb{P}\big(Y = e_k \,\big|\, f_k(X) = p\big) \;=\; p, \qquad \forall\, p \in [0,1]. \tag{2.3}$$

Canonical (distribution) calibration. The strongest notion requires the *entire* probability vector to be calibrated [11, 25]:

$$\mathbb{P}\big(Y = e_k \,\big|\, f(X) = \mathbf{p}\big) \;=\; p_k, \qquad \forall\, \mathbf{p} \in \Delta_K,\ \forall k. \tag{2.4}$$

Estimating canonical calibration is statistically challenging because it requires conditioning on points in the $(K-1)$-dimensional probability simplex. As a consequence, most practical estimators focus on weaker notions; while these are consistent, low-bias estimators for canonical calibration have recently become available as a significant advance [11].

It is crucial to understand (and a recurring theme of this thesis) that calibration is *necessary but not sufficient* for a good probabilistic model. A model can be perfectly calibrated and still inaccurate. One example is that if a model always outputs the marginal base rates $\mathbb{P}(Y = k)$, it is perfectly calibrated yet useless, as it is not *sharp*: it fails to discriminate between inputs. The complementary virtue, resolution or sharpness, is formalized in Section 2.4.5.

2.3. Calibration under Distribution Shift

The calibration notions of table 2.1 are defined with respect to a single fixed distribution $\mathcal{D}$. In real life, often the test distribution $\mathcal{D}_{\text{test}}$ may differ from the training distribution $\mathcal{D}_{\text{train}}$, a phenomenon broadly known as *distribution shift*. Ovadia *et al.* [27] and Minderer *et al.* [3] studied this setting empirically at scale and evaluated both predictive accuracy and calibration (via ECE) as models are exposed to out-of-distribution (OOD) data, and found that miscalibration under shift is severe and not well predicted by in-distribution calibration quality.

A more narrowly structured and tractable special case is *covariate shift*, in which (A) the marginal distribution of the input changes between training and test data, while (B) the conditional label distribution is preserved. This means that while the relationship between the input features and the target variable doesn't change, the way the input features are distributed does. Mathematically, we can denote these two criteria as below:

$$\begin{aligned} \mathbb{P}_{\text{train}}(X) &\neq \mathbb{P}_{\text{test}}(X) \\ \mathbb{P}_{\text{train}}(Y \mid X = x) &= \mathbb{P}_{\text{test}}(Y \mid X = x) \end{aligned} \tag{2.5}$$

Here the first equation refers the input distribution change, while the second one indicates the conditional distribution stability, two criteria for covariate shift. In the notation of section 2.1, this states that $\pi_{\text{train}}(\cdot \mid x) = \pi_{\text{test}}(\cdot \mid x)$ for all x, i.e. $\eta(x)$ is invariant across train and test, even though the population of inputs X on which η is evaluated shifts. Following the studies by Shimodaira [28], Sugiyama [29] and Sugiyama & Kawanabe [30], this makes covariate shift a most directly analyzable setting, as opposed to the more general distribution shift considered by Ovadia *et al.* [27] and Minderer *et al.* [3], which permits both $\mathbb{P}(X)$ and $\mathbb{P}(Y \mid X)$ to change.

Covariate shift interacts differently with each level of the calibration hierarchy, mentioned in table 2.1. Confidence calibration (eq. 2.2) can degrade under covariate shift even when $\pi(\cdot \mid x)$ is preserved, because the marginal distribution of the confidence score $\hat{p}(X)$ shifts along with $\mathbb{P}(X)$, changing which inputs populate each confidence bin. Marginal (class-wise) calibration (eq. 2.3) is affected identically per class, whereas canonical calibration (eq. 2.4) is generally the hardest to preserve, since it requires the conditional law of Y given the entire prediction vector to remain stable even as the input population feeding each region of the simplex changes. In summary, calibration under covariate shift is not the invariance of $\pi(\cdot \mid x)$, which the covariate-shift assumption already assumes, it is the invariance of the calibration function $\gamma(p)$ (introduced in subsection 2.4.5), as the distribution feeding it shifts.

2.4. Calibration Metrics and Calibration–Refinement Decomposition

We now turn to how calibration is measured and how it relates to the losses used during model training.

We proceed in five steps. Section 2.4.1 introduces the expected calibration error (ECE) and the reliability diagram, the standards for measuring the weakest (confidence) notion of calibration from table 2.1. Section 2.4.2 discusses the statistical bias of the binned ECE estimator and section 2.4.3 surveys other named calibration metrics beyond ECE, for completeness. This motivates a more principled question, addressed in section 2.4.4: *what properties should a loss function have if it is to be trusted as an evaluation criterion for probabilistic predictions at all?* We answer this by introducing proper scoring rules. Then, section 2.4.5 shows that any proper loss admits an exact calibration–refinement decomposition, which gives build the foundation for chapter 3.

2.4.1. Expected calibration error (ECE) and reliability diagrams

The most common scalar summary is the *expected calibration error* (ECE), which is the expected gap between confidence and accuracy [7, 1]:

$$\mathrm{ECE} \;=\; \mathbb{E}_{\hat{p}}\Big[\,\big|\,\mathbb{P}(Y=\hat{y}\mid\hat{p})-\hat{p}\,\big|\,\Big]. \tag{2.6}$$

Because the conditioning variable $\hat{p}$ is continuous, ECE is estimated via discrete hard binning and then comparing per-bin accuracy and confidence.

Let $\mathcal{B}=\{B_1,\dots,B_M\}$ be a partition of $[0,1]$ into M non-overlapping bins. For a dataset of n predictions $\{(p_i,y_i)\}_{i=1}^{n}$, each sample i is assigned to bin B_m if $p_i\in B_m$. Within each bin, we define:

- $\mathrm{conf}(B_m)=\dfrac{1}{|B_m|}\sum_{i\in B_m}p_i$: the mean predicted confidence,
- $\mathrm{acc}(B_m)=\dfrac{1}{|B_m|}\sum_{i\in B_m}\mathbf{1}[\hat{y}_i=y_i]$: the empirical accuracy (fraction of correct predictions),
- $\mathrm{prec}(B_m)=\dfrac{\sum_{i\in B_m}\mathbf{1}[\hat{y}_i=y_i\wedge y_i=1]}{\sum_{i\in B_m}\mathbf{1}[\hat{y}_i=1]}$: the empirical precision (fraction of positive predictions that are true positives), with the convention that $\mathrm{prec}(B_m)=0$ when the denominator vanishes.

The estimated ECE is then

$$\widehat{\mathrm{ECE}} \;=\; \sum_{m=1}^{M}\frac{|B_m|}{n}\,\big|\,\mathrm{acc}(B_m)-\mathrm{conf}(B_m)\,\big| \tag{2.7}$$

where $\mathrm{acc}(B_m)$ and $\mathrm{conf}(B_m)$ are the average accuracy and confidence of the corresponding bin B_m. A *reliability diagram* (figure 1.1) plots $\mathrm{acc}(B_m)$ against $\mathrm{conf}(B_m)$. In a reliability diagram, a perfectly calibrated model lies on the diagonal, and any deviation below it indicates overconfidence [1, 31].

In the binary classification setting with severe class imbalance, accuracy alone can be misleading. We therefore also track a *precision-aware* variant of the binned calibration error:

$$\mathrm{ECE}_{\mathrm{prec}} = \sum_{m=1}^{M} \frac{|B_m|}{n} \Big| \mathrm{prec}(B_m) - \mathrm{conf}(B_m) \Big| \tag{2.8}$$

which measures how well the model's confidence aligns with the fraction of true positives among its positive predictions, a stricter and more informative criterion for imbalanced binary problems.

Hard, equal-width binning is the most common estimator of ECE but not the only one. *Adaptive (equal-mass) binning* [32] reduces the sensitivity to sparsely populated bins by sizing each bin to contain an equal number of predictions rather than an equal-width interval of $[0, 1]$. This idea along with several other calibration metrics are surveyed separately in section 2.4.3. A further family of differentiable, binning-aware or binning-free estimators, designed specifically to be optimized during training (the methodological approach of our thesis) rather than only evaluated post hoc, is explained in section 2.6.1, where we introduce the soft-binned, MMCE, and $\mathrm{ECE}_{\mathrm{KDE}}$ estimators that this thesis builds on directly.

2.4.2. The bias of binned estimators

Binned ECE is convenient but statistically unstable. Its value depends on the number and placement of bins (equal-width vs equal-mass (adaptive)) [26]. Moreover, with finite samples, it can report a low error for a miscalibrated model, making it a *biased* estimator of the calibration error [25, 33]. Vaicenavicius *et al.* [25] analyzed this dependence and showed that conclusions drawn from binned ECE depend on the binning scheme. These concerns matter for this thesis, because they distinguish the *training-time penalty* from the *evaluation metric*: an estimator suitable as a smooth loss is not necessarily suitable for unbiased evaluation.

2.4.3. Other calibration metrics

There are several other scalar summaries of calibration proposed by researchers. They differ either in the summary statistic applied within each bin, or in whether they evaluate per-class or only top-label calibration, or in whether they require binning at all. We survey these alongside ECE for complete comparison only; none of them is used as a training-time penalty in this thesis (the differentiable estimators actually used for that purpose, Soft-ECE, MMCE, and $\mathrm{ECE}_{\mathrm{KDE}}$, are introduced separately in section 2.6.1).

The *Maximum Calibration Error* (MCE) reports the single worst per-bin gap rather than the weighted average [7]:

$$\mathrm{MCE} = \max_{m=1,\dots,M} \Big| \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \Big| \tag{2.9}$$

Equal-mass ECE ($\mathrm{ECE}_{\mathrm{em}}$) [26] applies the same formula as eq. 2.7 but with bins chosen to contain an equal number of predictions rather than an equal width of $[0, 1]$.

Beyond varying only the binning scheme applied to the single top-label confidence, a separate line of work evaluates calibration jointly across all K class probabilities rather than only the confidence of the predicted class. Nixon *et al.* [26] proposed two such class-conditional extensions of ECE.

The *Static Calibration Error* (SCE) bins each class's one-vs-rest confidence separately using fixed-width bins, computes the calibration error within each bin, and averages the result across all K classes to evaluate marginal calibration (table 2.1) rather than only confidence calibration:

$$\mathrm{SCE} = \frac{1}{K} \sum_{k=1}^{K} \sum_{m=1}^{M} \frac{n_{m,k}}{N} \Big| \mathrm{acc}(m,k) - \mathrm{conf}(m,k) \Big|, \tag{2.10}$$

where M is the number of bins, $\mathrm{acc}(m,k)$ and $\mathrm{conf}(m,k)$ are the accuracy and confidence of bin m restricted to class k, $n_{m,k}$ is the number of predictions of class k falling in bin m, and N is the total number of predictions.

The *Adaptive Calibration Error* (ACE) applies the same construction but with R adaptively chosen ranges, each containing an equal number of predictions, in place of fixed-width bins:

$$\mathrm{ACE} = \frac{1}{KR} \sum_{k=1}^{K} \sum_{r=1}^{R} \Big| \mathrm{acc}(r,k) - \mathrm{conf}(r,k) \Big|. \tag{2.11}$$

A further refinement, the *Thresholded Adaptive Calibration Error (TACE)*, applies the same formula as ACE but first discards all predicted class probabilities below a fixed threshold t before adaptively binning the remainder into R ranges [26]:

$$\mathrm{TACE}(t) = \frac{1}{KR} \sum_{k=1}^{K} \sum_{r=1}^{R} \Big| \mathrm{acc}(r,k) - \mathrm{conf}(r,k) \Big|, \qquad \text{computed only over } \{(i,k) : f_k(x_i) > t\}. \tag{2.12}$$

Thresholding was introduced specifically to make adaptive binning usable in high-class-count settings, where the vast majority of the $K - 1$ non-predicted class probabilities are close to zero and would otherwise dominate the adaptive ranges [26]. We note, however, that a later reproduction study finds TACE to be sensitive to both of its hyperparameters (the number of ranges and the threshold) and, in practice, not to yield a consistent ranking of models [34].

The *Root-Mean-Squared Calibration Error* (RMSCE) replaces the L_1 per-bin gap of eq. 2.7 with an L_2 (squared) gap, which upweighs bins with large miscalibration relative to standard ECE [35]. Another binning-free alternative is the *Kolmogorov–Smirnov Calibration Error* (KSCE), which compares empirical cumulative distributions of confidence directly rather than via bins [36].

Table 2.2 compares ECE and all of these variants in a single place, listing down the underlying mechanism and whether it is class-wise or top-label.

Table 2.2.: Comparison of Expected Calibration Error (ECE) and other common calibration metrics. "Binning" indicates whether a choice of bins (fixed-width or adaptive) is required. "Class-wise" indicates whether the metric evaluates per-class (marginal) rather than only top-label (confidence) calibration.

Metric	**Binning**	**Class-wise**	**Consistent**
ECE [7]	Fixed-width	×	×[1]
MCE [7]	Fixed-width	×	Not verified
ECE_{em} [26]	Adaptive (equal-mass)	×	Not verified
SCE[2] [26]	Fixed-width	✓	Not verified
ACE[2] [26]	Adaptive (equal-mass)	✓	Not verified
TACE [26]	Adaptive (equal-mass, thresholded)	✓	Not verified[3]
RMSCE [35]	Fixed-width	×	Not verified
KSCE [36]	Binning-free	×	Not verified

[1] ECE is biased and not a consistent estimator of the underlying calibration error [25]; as mentioned in section 2.4.2 and [25].
[2] SCE and ACE are, respectively, equivalent to what some other sources term the class-wise calibration error (cwCE) and class-wise (equal-mass) calibration error (cwCE_{em}), differing only in the choice of fixed-width versus adaptive binning [37].
[3] Ashukha *et al.* [34] report that TACE is sensitive to its threshold and bin-count hyperparameters and does not consistently rank models in practice.

Throughout this thesis, we maintain a deliberate distinction between two roles a calibration function can play. A calibration metric (ECE from subsection 2.4.1 along with other alternatives like MCE, ACE, TACE, RMSCE, KSCE introduced in this subsection 2.4.3) is used only to evaluate a trained model, is typically non-differentiable (built from hard binning or empirical CDFs), and is never optimized directly. A calibration estimator (introduced later in section 2.6.1: Soft-ECE, MMCE, ECE_{KDE}) is a differentiable surrogate designed specifically for the term $\widehat{\mathcal{C}}$ inside the composite training objective (eq. 3.1), following calibration-refinement decomposition (eq. 2.14). These two categories are held to different standards: metrics are chosen for interpretability and evaluation-time reliability, estimators are chosen for differentiability and gradient behavior during training.

2.4.4. Proper Losses

A principled route to calibration runs through *proper scoring rules*, the standard for evaluating probabilistic forecasts systematized by Gneiting and Raftery [38]. A scoring rule $\mathcal{S}(f(x), y)$ is *proper* if the expected score is optimized by reporting the true conditional distribution, and *strictly proper* if that optimum is unique:

$$\mathbb{E}_{Y\sim q}\big[\mathcal{S}(q, Y)\big] \;\geq\; \mathbb{E}_{Y\sim q}\big[\mathcal{S}(p, Y)\big] \quad \text{for all } p, \text{ with equality iff } p = q. \tag{2.13}$$

Berta *et al.* [8] evaluated predictions with a loss function $\ell : \Delta_K \times \mathcal{Y}_K \to \mathbb{R}_+$. ℓ is a nonnegative

function of p and y such that $\ell(p, y)$ assesses the quality of prediction p for label y. Two of the most widely popular such examples include the *cross-entropy* [39] or negative log-loss (NLL), $\ell(p, y) = -\sum_{i=1}^{k} y_i \log(p_i)$ and *Brier score* [40], $\ell(p, y) = \|y - p\|_2^2$.

If label Y follows a categorical distribution $q \in \Delta_K$, then the expected loss can be denoted as: $\ell(p, q) = \mathbb{E}_{Y\sim q}[\ell(p, Y)]$. Berta *et al.* [8] also identifies that a requirement for the loss function ℓ is that it should be minimized when the predicted distribution p matches the true target distribution q. Losses that verify this property are categorized as *proper*.

Following the work of Gneiting and Raftery [38] and Berta *et al.* [8], we formally introduce the definition of *proper loss function*.

Definition 2.1 (Proper loss function)**.** We consider losses $\ell : \Delta_K \times \mathcal{Y}_K \to \mathbb{R} \cup \{\infty\}$ such that for any $p \in \Delta_K$, $\ell(p, \cdot)$ is measurable and quasi-integrable with respect to any $q \in \Delta_K$. This allows to define $\ell(p, q) := \mathbb{E}_{Y\sim q}[\ell(p, Y)]$, taking values in $\mathbb{R} \cup \{\infty\}$. If $\ell(q, q) \leq \ell(p, q)$ for all p, q, ℓ is said to be *proper*. Equality holds iff $p = q$ and ℓ is then said to be strictly proper.

Both *cross-entropy* and *Brier score* that were introduced earlier, are strictly proper and hence minimized in expectation by the true posterior $f^\star$ or η. They differ in a way that is central to this thesis: the log score is *unbounded*, whereas the Brier score is *bounded*. Boundedness is precisely the property exploited by the Reinforcement Learning with Calibration Rewards (RLCR) framework [12], which augments a correctness reward with a bounded proper scoring rule and proves that the resulting optimum is both accurate and calibrated. This furnishes a principled, theory-backed reason to prefer bounded proper scoring rules as calibration objectives at training time.

However, not all loss in practice are proper. The *hinge loss*, for instance, is defined directly on margins rather than on a probability report and therefore has no guarantee to minimize expectation in the form of eq. 2.13. More relevantly to this thesis, focal loss [41], which we revisit in section 2.6.2 as an implicit calibration regularizer, is also not a proper scoring rule, since it reweighs the per-example log-loss by a confidence-dependent factor that need not vanish at the true posterior. Nonetheless, focal loss tends to improve calibration empirically; that is why it is treated as an implicit calibration mechanism (section 2.6.2) in this thesis.

2.4.5. Calibration-Refinement Decomposition

Let $d_\ell(p, q) := \ell(p, q) - \ell(q, q)$ and $e_\ell(q) := \ell(q, q)$ be the divergence and entropy associated with loss function ℓ (we refer to these as ℓ-divergence and ℓ-entropy). Berta *et al.* [8] showed that for different choices of the loss function ℓ, we can recover well-known entropy and divergence functions (table 2.3).

We define $\gamma(p) := \mathbb{E}_{\mathcal{D}}[Y \mid P = p]$ as the *calibration function* of a stochastic classifier f. Now, since Y is one-hot ($Y = e_k$ when the true class is k), the conditional expectation $\mathbb{E}_{\mathcal{D}}[Y \mid P = p]$ is a vector whose k-th entry is $\mathbb{E}[Y_k \mid P = p] = \mathbb{E}[\mathbb{1}(Y = e_k) \mid P = p] = \mathbb{P}[Y = e_k \mid P =$

Table 2.3.: Controlling the choice of loss function ℓ can recover well-known entropy and divergence functions.

Loss ℓ	**Divergence** d_ℓ	**Entropy** e_ℓ
Log loss	KL divergence	Shannon entropy
$-\sum_i y_i \log(p_i)$	$\sum_i q_i \log \frac{q_i}{p_i}$	$-\sum_i q_i \log q_i$
Brier score	Squared distance	Gini index
$\|y-p\|_2^2$	$\|p-q\|_2^2$	$\sum_i q_i(1-q_i)$

p]. Since, the expectation of an indicator equals the probability of the event. So, $\gamma(p)$, the "calibration function" is, entry-by-entry, the vector $\big(\mathbb{P}(Y = e_1 \mid P = p), \ldots, \mathbb{P}(Y = e_K \mid P = p)\big)$, i.e., *the true class distribution among inputs that receive prediction* p.

Following the work of DeGroot and Fienberg [31], Bröcker [42], and Berta *et al.* [8], the population risk with a proper loss function ℓ can be decomposed as:

$$\mathcal{E}(f) = \mathbb{E}\big[\ell(f(X), Y)\big] = \mathcal{C}_\ell(f) + \mathcal{R}_\ell(f),$$
$$\underbrace{\mathcal{C}_\ell(f) := \mathbb{E}\big[d_\ell(P, \gamma(P))\big]}_{\text{Calibration error}}, \quad \underbrace{\mathcal{R}_\ell(f) := \mathbb{E}\big[e_\ell(\gamma(P))\big]}_{\text{Refinement error}}. \tag{2.14}$$

The first term, *Calibration error* measures the deviation of predictions from their conditional label distribution and vanishes if and only if f is calibrated. The second term, *Refinement error* measures the intrinsic uncertainty remaining after conditioning on predictions.

Bröcker [42] showed that the refinement term can be further decomposed into the uncertainty $\mathcal{U}_\ell(Y)$ (it is independent of f) and a sharpness term $\mathcal{S}_\ell(f)$, yielding the "*calibration-sharpness decomposition*":

$$\mathcal{E}(f) = \mathbb{E}\big[\ell(f(X), Y)\big] = \underbrace{\mathbb{E}\big[d_\ell(P, \gamma(P))\big]}_{\text{Calibration error } (\mathcal{C}_\ell(f))} + \underbrace{\underbrace{e_\ell(\mathbb{E}[Y])}_{\text{Uncertainty}(\mathcal{U}_\ell(Y))} - \underbrace{\mathbb{E}\big[d_\ell(\mathbb{E}[Y], \gamma(P))\big]}_{\text{Resolution / Sharpness } (\mathcal{S}_\ell(f))}}_{\text{Refinement error } (\mathcal{R}_\ell(f))} \tag{2.15}$$

The *sharpness principle* of Gneiting *et al.* [43] restates the goal of probabilistic forecasting as "maximize sharpness subject to calibration." However, for this thesis, we followed the calibration-refinement decomposition mentioned in eq. 2.14. Berta *et al.* [8] showed, both theoretically and empirically, that the calibration and refinement terms are *not* minimized at the same point during training. So a model chosen by validation loss is optimal for neither. Their "refine, then calibrate" approach (optimize refinement during training, fix calibration afterwards) is one possible response. The subject of this thesis is the converse question of folding calibration *into* training.

It is worth noting explicitly that the expected calibration error (ECE) of section 2.4.1 is itself not a proper loss. This can be seen in two ways. First, structurally: definition 2.1 requires a proper loss to be a per-example function $\ell(p, y)$ whose population expectation is minimized by truthful reporting; ECE is instead a population-level statistic computed by binning across an entire evaluation set and has no such per-example form. Second, conceptually: a classifier that outputs the constant marginal base rate for every input achieves zero ECE (it is perfectly calibrated in the sense of section 2.2) while discarding all input-dependent information. This distinction matters for the thesis. Since ECE is not proper, and hence not directly optimizable as a standalone training signal without also controlling sharpness, chapter 3 regularizes with the calibration component of a proper loss eq. 3.1 rather than with ECE directly.

Total proper losses such as the Brier score or cross-entropy are sometimes reported alongside ECE as an auxiliary, indirect check on calibration. However this might not be a good substitute for ECE: since we have already seen that these losses decompose into a calibration and a refinement term. A raw Brier score or NLL value conflates the two and cannot properly distinguish a well-calibrated-but-unsharp model from a sharp-but-miscalibrated one. Instead of the total loss, the calibration term of the eq. 2.14 is the correct comparison to ECE.

2.5. Post-hoc Calibration methods

When a trained model is miscalibrated, the common approach is to correct the outputs without retraining the entire model, we call this *recalibrating* a trained model. *Post-hoc calibration* holds the model's parameters fixed, fits a map g on a small held-out calibration set, and applies g to the frozen model's outputs at test time. Different post-hoc calibration methods share a common template: fit a map $g : [0, 1] \to [0, 1]$ (or on the logits) on a held-out set and apply it to the frozen model's outputs. They differ only in the family of maps they fit and in whether they act per-class or on the whole vector. These methods are cheap, model-agnostic, and treat the predictor as a black box. The empirical study of which supervised learners yield well-calibrated probabilities is itself old and associated with the findings by Niculescu-Mizil and Caruana [44].

We describe each principal method briefly, giving its calibration map explicitly, and compare them in table 2.4. Throughout, $s \in [0, 1]$ denotes a binary score, $\mathbf{z} \in \mathbb{R}^K$ a logit vector, $\mathbf{p} \in \Delta_K$ a predicted probability vector where $\Delta_K := \{p \in [0, 1]^K \ : \ \mathbf{1}^\top p = 1\}$, σ the logistic function, and $\{(s_i, y_i)\}_{i=1}^n$ the held-out calibration data with $y_i \in \{0, 1\}$.

2.5.1. Platt scaling

Platt scaling [5] is a parametric method which fits a logistic transformation to the scores by passing the model's real-valued score (or logit) z through a logistic sigmoid,

$$g(z) \ = \ \sigma(az + b) \ = \ \frac{1}{1 + \exp(az + b)}, \tag{2.16}$$

and estimates the two parameters a, b by minimizing the negative log-likelihood on the calibration set. The map is smooth and data-efficient, but it assumes the miscalibration is sigmoidal, which might not be true in practice.

2.5.2. Temperature scaling

Temperature Scaling, the popular standard for deep networks [1] is the one-parameter variation of Platt scaling that divides the logit vector $\mathbf{z}$ by a single scalar temperature $T > 0$ before the softmax,

$$g(\mathbf{z})_k = \frac{\exp(z_k/T)}{\sum_{j=1}^{K} \exp(z_j/T)}, \qquad T > 0, \tag{2.17}$$

where T is fit by minimizing the validation negative log-likelihood. Because the same T rescales every logit, the arg-max and the top-1 accuracy is left unchanged. $T > 1$ softens the distribution (reducing confidence) and $T < 1$ sharpens it. The generalizations *vector* and *matrix scaling* replace $\mathbf{z}/T$ by $\mathbf{W}\mathbf{z} + \mathbf{b}$ where $\mathbf{W}$ is a diagonal or full matrix; they are more expressive but introduce many parameters and can degrade accuracy [1].

2.5.3. Beta calibration

Beta calibration is a parametric method designed for scores that are already probabilities in $[0, 1]$: the map is derived from the Beta distribution and has three parameters [45]. Its calibrated output has a logit that is linear in $\ln s$ and $\ln(1 - s)$,

$$\ln \frac{g(s)}{1 - g(s)} = c + a \ln s - b \ln(1 - s), \qquad a, b \geq 0, \tag{2.18}$$

which strictly generalizes logistic calibration. Beta calibration family contains the identity map (at $a = b = 1$, $c = 0$; eq. 2.18 reduces to $g(s) = s$), so it cannot worsen an already well-calibrated classifier in the population limit. Platt scaling does not guarantee this.

2.5.4. Histogram binning

This is the simplest non-parametric method that partitions the score range into M bins $B_1, \ldots, B_M$ and maps any score lying in bin B_m to the fraction of positives among held-out points in that bin [46],

$$g(s) = \frac{1}{|B_m|} \sum_{i:\, s_i \in B_m} \mathbb{1}[\, y_i = 1 \,], \qquad s \in B_m. \tag{2.19}$$

It is assumption-free and easy to fit, but its output is piecewise constant, it discards the ordering of scores within a bin, and its quality is sensitive to the number and placement of the bins.

2.5.5. Isotonic regression

This non-parametric method removes the dependence on a fixed grid by fitting a monotonically non-decreasing, piecewise-constant map that minimises squared error subject only to a monotonicity constraint,

$$\hat{g} \;=\; \arg\min_{g \text{ non-decreasing}} \sum_{i=1}^{n} \big(g(s_i) - y_i\big)^2, \tag{2.20}$$

solved efficiently by the pool-adjacent-violators algorithm [6]. It is more flexible than a parametric fit like Platt Scaling and assumes only monotonicity, at a higher risk of overfitting when the calibration set is small.

NOTE: It is important to note that both *histogram binning* and *isotonic regression* map scores to empirical frequencies non-parametrically.

2.5.6. Bayesian Binning into Quantiles (BBQ)

It is a non-parametric refinement of histogram binning that modifies the arbitrariness of a single histogram by performing Bayesian model averaging over a collection of equal-frequency binning models $\{M_1, \dots, M_R\}$, weighting each by a Bayesian score [7].

$$g(s) \;=\; \sum_{r=1}^{R} P(M_r \mid \mathcal{D})\, P\big(y = 1 \mid s, M_r\big), \tag{2.21}$$

where $P(y = 1 \mid s, M_r)$ is the bin frequency under model M_r and the posterior weight $P(M_r \mid \mathcal{D})$ is proportional to that model's marginal likelihood (a Bayesian score) times its prior. The averaging removes the dependence on a single arbitrary binning and yields smoother estimates than a fixed histogram.

2.5.7. Dirichlet calibration

The natively multi-class generalization of beta calibration applies an affine map to the log-probability vector followed by a softmax,

$$g(\mathbf{p}) \;=\; \mathrm{softmax}\big(\mathbf{W} \ln \mathbf{p} + \mathbf{b}\big), \qquad \mathbf{W} \in \mathbb{R}^{K \times K},\ \mathbf{b} \in \mathbb{R}^{K}, \tag{2.22}$$

equivalently a log-transform, a single linear layer, and a softmax. Unlike temperature scaling, which only rescales confidence, Dirichlet Calibration, with its $K(K+1)$ parameters, can correct the class-wise (marginal) miscalibration of the entire probability vector [24].

Table 2.4 summarizes these methods along with the comparison between the map family, parameter count, whether the method is natively multiclass, and whether it provably preserves top-1 accuracy.

Table 2.4.: Comparison of common post-hoc calibration methods, ordered as in the text. "Multiclass-native" indicates whether the method calibrates the full vector directly or is applied one-vs-rest; "Preserves acc." indicates whether the map provably leaves the top-1 prediction unchanged.

Method	**Map family**	**#params**	**Multiclass-native**	**Preserves acc.**
Platt scaling [5]	Logistic (sigmoid)	2	×	∼
Temperature scaling [1]	Single-parameter logit	1	✓	✓
Beta calibration [45]	Beta family	3	×	∼
Histogram binning [46]	Non-param. (binning)	M bins	×	∼
Isotonic regression [6]	Non-param. monotone	data-dep.	×	∼
BBQ [7]	Non-param. Bayesian bins	—	×	∼
Dirichlet calibration [24]	Linear-on-log + softmax	$K(K+1)$	✓	×

Legend: ✓ yes × no ∼ depends on setting / not guaranteed. Temperature scaling preserves the arg-max because it scales all logits by the same factor; per-class and binning maps may alter it after renormalisation.

2.5.8. A shared limitation

The strengths of these methods, simplicity and model-agnosticism, come with structural limitations already noted in chapter 1. They operate on a *frozen* predictor and therefore cannot influence the learned representation of the model or its decision boundary; the correction is confined to the output layer. The simplest variants (temperature scaling) apply only a global, monotone transformation; and because the map is fit on one distribution, the resulting calibration is not guaranteed to survive a distribution shift [3]. These limitations are the direct motivation for studying the alternative: folding calibration into training itself.

2.6. Calibration During Training (End-to-End)

To be usable *inside* training, a calibration term must satisfy three requirements. (1) *A well-defined target*: it should estimate a genuine population calibration function, ideally one that is zero if and only if the model is calibrated, such as the canonical calibration error of section 2.2. (2) *Statistical soundness*: it should admit a finite-sample estimator, preferably consistent and low-bias, so that driving the term down corresponds to genuinely improving calibration. (3) *Differentiability*: the estimator must be (sub)differentiable with respect to the model parameters θ, so that it can be minimized by the same gradient-based optimization used for the predictive loss.

The standard binned ECE of section 2.4 fails (3). Assigning each prediction to a bin uses indicator functions, so the estimator is piecewise constant in θ and its gradient is zero almost everywhere [10]. Because this prevents backpropagation, the binned ECE cannot be optimized directly by gradient descent. This is precisely why an explicit calibration penalty requires a differentiable *surrogate*.

2.6.1. Auxiliary calibration losses

The most direct approach adds a differentiable calibration penalty to the predictive loss, optimizing $\widehat{\mathcal{E}}_\ell(f) + \lambda\,\Omega(f)$ for a penalty Ω and strength λ. Instances include the Difference of Confidence and Accuracy (DCA) term [2], MMCE [9], the soft-binned SB-ECE and S-AvUC objectives [10], and the $\mathrm{ECE}_{\mathrm{KDE}}$-regularised risk [11]. Karandikar *et. al* [10] reported that such penalties can reduce ECE substantially with little loss of accuracy and can improve robustness under dataset shift relative to cross-entropy plus post-hoc recalibration.

In this thesis, we will focus only on "auxiliary calibration loss". In particular, three estimators for the calibration error, of increasing statistical strength, are central to this thesis.

Soft-ECE (kernel-smoothed ECE). The most direct solution is to soften the binning. Hard bin membership is replaced by smooth weights derived from a kernel placed at fixed bin centers, a Gaussian softmax over distances to the centers, or a Laplacian kernel. So every prediction contributes fractionally to every bin and the per-bin confidence and accuracy become differentiable functions of θ [10].

Let $c_1, \dots, c_M$ be M fixed bin centers uniformly placed on $[0, 1]$. For a prediction p_i (the model's estimated probability for the input i), define the *soft membership weight* to bin j as

$$w_{j,i} = \frac{\exp\left(-\dfrac{(p_i - c_j)^2}{2\sigma^2}\right)}{\displaystyle\sum_{k=1}^{M} \exp\left(-\dfrac{(p_i - c_k)^2}{2\sigma^2}\right)}, \tag{2.23}$$

where $\sigma > 0$ is the kernel bandwidth. This is a Gaussian softmax over distances to bin centers, so every sample contributes fractionally to every bin. The *Differentiable Expected Calibration Error (DECE)* is then

$$\mathcal{L}_{\mathrm{DECE}} = \sum_{j=1}^{M} \frac{\sum_i w_{j,i}}{n} \left(\underbrace{\frac{\sum_i w_{j,i}\, p_i}{\sum_i w_{j,i}}}_{\text{Soft Confidence of bin } j} - \underbrace{\frac{\sum_i w_{j,i}\, y_i}{\sum_i w_{j,i}}}_{\text{Soft Accuracy of bin } j} \right)^2. \tag{2.24}$$

A precision-aware analogue replaces the soft accuracy term with a soft precision term:

$$\text{Soft Precision of bin } j \;=\; \frac{\sum_i w_{j,i}\, \mathbf{1}[\hat{y}_i = 1]\, y_i}{\sum_i w_{j,i}\, \mathbf{1}[\hat{y}_i = 1]}, \tag{2.25}$$

yielding $\mathcal{L}_{\text{DECE-prec}}$ by substituting the soft-accuracy part of eq. 2.24 with soft precision of eq. 2.25.

The resulting soft-binned calibration error is easy to implement and to add to the loss. Its weaknesses are inherited from binning: it depends on the number and placement of bin centers and on the kernel bandwidth, and it is neither a *consistent* nor a *de-biased* estimator of the underlying calibration error.

Remark 2.2. A Laplacian kernel $\exp(-|p_i - c_k|/\sigma)$ is often preferred over the Gaussian kernel in calibration settings, as it is less sensitive to outliers in the predicted probability distribution and yields sparser effective bin assignments. The hyperparameter σ for bandwidth is selected via cross-validation.

Maximum Mean Calibration Error (MMCE). *Kernel mean embeddings* give the Maximum Mean Calibration Error (MMCE), a trainable reproducing-kernel Hilbert space (RKHS) statistic that is zero iff the model is calibrated in top-label sense[9]. It embeds the per-sample calibration residuals in a RKHS through a differentiable kernel (typically Laplacian) and measures calibration as a maximum mean discrepancy [9]. Since MMCE removes binning entirely and needs no choice of evaluation points, it is mathematically clean for optimization. However, it targets only top-label (scalar-confidence) calibration and has been shown *not* to be a consistent estimator of the L_p canonical calibration error [11].

While sounds similar and abbreviated almost the same, the *Maximum Mean Calibration Error* (MMCE) is not to be confused with the (unrelated) *Maximum Calibration Error* MCE of subsection 2.4.3.

Let $r_i = \mathbf{1}[\hat{y}_i = y_i]$ be the indicator of correct prediction for sample i, and $c_i \in [0,1]$ its predicted confidence. The *squared MMCE* over a batch of n samples is

$$\mathcal{L}^2_{\text{MMCE}} = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} (r_i - c_i)(r_j - c_j)\, \kappa(c_i, c_j), \tag{2.26}$$

where $\kappa(c_i, c_j)$ is a differentiable kernel function, usually the Laplacian kernel $\kappa(c_i, c_j) = \exp(-|c_i - c_j|/\sigma)$.

Intuitively, MMCE tells us *whether the calibration residuals* $(r_i - c_i)$ *of two samples are correlated when their predicted confidences are similar?*

A perfectly calibrated model satisfies $\mathcal{L}_{\text{MMCE}} = 0$ if and only if $\text{CE}_p = 0$ [9], i.e., the model is calibrated. In imbalanced settings such as GRN edge prediction (the specific application case study for this thesis), one may additionally replace r_i with a precision indicator $r_i^{\text{prec}} = \mathbf{1}[\hat{y}_i = y_i = 1] / \max(\mathbf{1}[\hat{y}_i = 1], \epsilon)$, giving a precision-aware MMCE loss that penalizes overconfident false-positive predictions more heavily.

Canonical L_p error via kernel density estimation ($\mathrm{ECE}_{\mathrm{KDE}}$). The two preceding estimators operate at the level of top-label (scalar confidence) calibration. Popordanoska *et al.* [11] provides a stronger alternative which requires the entire probability vector to be calibrated simultaneously. It is also a consistent, low-bias, and differentiable estimator of the canonical calibration error. It forms a kernel density estimate of the conditional label distribution over the probability simplex using a Dirichlet kernel, and plugs that estimate into the L_p canonical calibration error [11]. Dirichlet Kernel is a natural choice on the simplex, since unlike Gaussian kernel, it places no mass outside it. The estimator is consistent (under mild regularity), differentiable, and admits a geometric-series debiasing scheme that reduces its bias from $\mathcal{O}(n^{-1})$ to $\mathcal{O}(n^{-2})$, at $\mathcal{O}(n^2)$ cost per mini-batch.

Formally, the L_p canonical calibration error is [11]:

$$\mathrm{CE}_p(f) = \mathbb{E}\left[\left\|\mathbb{E}[Y \mid f(X)] - f(X)\right\|_p^p\right]^{1/p}. \tag{2.27}$$

A classifier f is perfectly calibrated if and only if $\mathrm{CE}_p(f) = 0$.

Estimating the conditional expectation $\mathbb{E}[Y \mid f(X)]$ requires density estimation over the probability simplex Δ_K. Popordanoska *et al.* [11] uses a *Dirichlet kernel density estimate (KDE)*:

$$\widehat{\mathbb{E}}[Y \mid f(X)]\Big|_{f(x_j)} = \frac{\sum_{i\neq j} \kappa_{\mathrm{Dir}}\big(f(x_j);\, f(x_i)\big)\, y_i}{\sum_{i\neq j} \kappa_{\mathrm{Dir}}\big(f(x_j);\, f(x_i)\big)}, \tag{2.28}$$

where the *Dirichlet kernel* is defined as

$$\kappa_{\mathrm{Dir}}\big(f(x_j),\, f(x_i)\big) = \frac{\Gamma\left(\sum_{k=1}^K \alpha_{ik}\right)}{\prod_{k=1}^K \Gamma(\alpha_{ik})} \prod_{k=1}^K f(x_j)_k^{\alpha_{ik}-1}, \qquad \alpha_i = \frac{f(x_i)}{h} + \mathbf{1}, \tag{2.29}$$

with $h > 0$ a bandwidth parameter satisfying $h \to 0$ as $n \to \infty$, and Γ the Gamma function. The Dirichlet kernel is the natural choice for density estimation over the simplex Δ_K, as it respects the boundary structure of the simplex — unlike Gaussian kernels, which assign positive mass outside Δ_K.

The $\mathrm{ECE}^{\mathrm{KDE}}$ estimator of the L_p canonical calibration error is then

$$\widehat{\mathrm{CE}}_p(f)^p = \frac{1}{n}\sum_{j=1}^n \left\|\widehat{\mathbb{E}}[Y \mid f(X)]\Big|_{f(x_j)} - f(x_j)\right\|_p^p. \tag{2.30}$$

Remark 2.3 (Binary specialization). In a binary ($K = 2$) settings (like the edge prediction between two nodes in a graph), the Dirichlet kernel in Eq. 2.29 specializes to a *Beta kernel*:

$$\kappa_B\big(f(x_j),\, f(x_i)\big) = f(x_j)^{\alpha_i - 1}(1 - f(x_j))^{\beta_i - 1}\frac{\Gamma(\alpha_i + \beta_i)}{\Gamma(\alpha_i)\,\Gamma(\beta_i)}, \tag{2.31}$$

with $\alpha_i = f(x_i)/h + 1$ and $\beta_i = (1 - f(x_i))/h + 1$. Eq. 2.28— 2.30 then define a consistent, differentiable estimator of the L_p binary calibration error directly applicable to binary settings.

Table 2.5.: List of differentiable calibration estimators considered in this thesis. "Consistent" and "De-biased" refer to estimation of the underlying calibration error; "Canonical" indicates whether the full probability vector (rather than only top-label confidence) is calibrated.

Estimator	Consistent	Differentiable	De-biased	Canonical
Soft-ECE (kernel-smoothed) [10]	×	✓	×	×
MMCE [9]	×	✓	×	×
$\mathrm{ECE}_{\mathrm{KDE}}$ [11]	✓	✓	✓	✓

These three estimators, summarized in table 2.5, are used for the calibration term $\widehat{\mathcal{C}}$ that the end-to-end objective of chapter 3 minimizes. The choice of estimator is an empirical question, *which surrogate most reduces miscalibration while preserving predictive accuracy*, that the experimental chapters investigate.

2.6.2. Other methods to improve calibration during training

Implicit regularizers that affect calibration. Several techniques improve calibration as a side effect rather than by an explicit calibration term. *Label smoothing* replaces hard targets with softened ones and is known to improve calibration while changing the geometry of learned representations [47]. *Focal loss* down-weights well-classified examples and has been analysed as an implicit regularizer of predictive confidence, improving calibration [48, 41]. *Deep ensembles* improve both accuracy and uncertainty by averaging independently trained models [49]. These methods are evidence that along with post-hoc correction, the training procedure also controls calibration, and that calibration interacts with the inductive bias of the learner.

Reward-based calibration. Beyond supervised classification, the Reinforcement Learning with Calibration Rewards (RLCR) framework trains models with a reward that combines correctness and a bounded proper scoring rule, yielding provably accurate and calibrated predictions and demonstrating that explicit optimisation for calibration need not cost accuracy [12]. Although developed for reasoning language models, its core argument of a bounded proper scoring rule integrated into the objective shapes the solution toward calibration, is exactly the mechanism this thesis studies.

2.6.3. Calibration, accuracy, and architecture

Empirically, the relationship between calibration and accuracy is neither adversarial nor automatic. Minderer *et al.* [3] find that recent architectures can be both more accurate and better calibrated, and that calibration can improve with model size under distribution shift; large-

scale benchmark studies catalog how calibration varies with architecture, training, and metric choice across many thousands of models [4]. Together with the calibration–refinement analysis [8], these results frame end-to-end calibration not as a fixed trade-off but as a problem of *shaping* the learner, which leads naturally to a learning-theoretic treatment.

2.7. Statistical Learning Theory Foundations

This section assembles the theoretical foundation used in chapter 1's hypothesis that calibration acts as a structural constraint on the hypothesis space. We review the generalization gap, complexity-based bounds (Rademacher), PAC-Bayesian bounds, and how a calibration constraint can be situated within them.

2.7.1. Empirical risk minimization and the generalization gap

The object of interest is the *generalization gap* $\mathcal{E}_\ell(f) - \widehat{\mathcal{E}}_\ell(f)$ between population and empirical risk. Classical theory bounds the worst-case gap over a hypothesis class $\mathcal{F}$ in terms of a complexity measure of $\mathcal{F}$ [50, 51]. A central modern observation is that the classical capacity measures are far too loose for overparameterized deep networks, which can fit random labels yet still generalize [52]; this makes *data-* and *algorithm-dependent* notions of complexity, and the regularizing effect of training choices, the right lens for modern models.

2.7.2. Rademacher complexity bounds

The empirical Rademacher complexity of a function class $\mathcal{G}$ on a sample of size n is

$$\widehat{\mathfrak{R}}_m(\mathcal{G}) \;=\; \mathbb{E}_\sigma\left[\sup_{g\in\mathcal{G}} \frac{1}{n}\sum_{i=1}^{n} \sigma_i\, g(z_i)\right], \tag{2.32}$$

where σ_i are independent Rademacher (± 1) variables. It measures the ability of $\mathcal{G}$ to correlate with random noise, i.e. its effective capacity [53]. The standard bound states that, with probability at least $1-\delta$, every $g \in \mathcal{G}$ (taking values in $[0,1]$) satisfies

$$\mathbb{E}[g(Z)] \;\le\; \frac{1}{n}\sum_{i=1}^{n} g(z_i) \;+\; 2\,\widehat{\mathfrak{R}}_m(\mathcal{G}) \;+\; 3\sqrt{\frac{\ln(2/\delta)}{2n}}. \tag{2.33}$$

The relevance to this thesis is structural: if augmenting the objective with a calibration penalty effectively restricts the learner to a subset $\mathcal{F}_{\text{cal}} \subseteq \mathcal{F}$ of better-calibrated hypotheses, then $\widehat{\mathfrak{R}}_m(\mathcal{F}_{\text{cal}}) \le \widehat{\mathfrak{R}}_m(\mathcal{F})$ and the bound tightens.

2.7.3. PAC-Bayesian bounds

PAC-Bayes bounds are apt for modern, overparameterized models because they bound the risk of *randomized* predictors and have yielded the tightest (sometimes non-vacuous) guarantees for neural networks [54, 55, 56]. For a prior P over hypotheses chosen before seeing the data and any posterior Q (possibly data-dependent), McAllester's bound states that, with probability at least $1-\delta$,

$$\mathbb{E}_{f\sim Q}\big[\mathcal{E}_\ell(f)\big] \;\le\; \mathbb{E}_{f\sim Q}\big[\widehat{\mathcal{E}}_\ell(f)\big] \;+\; \sqrt{\frac{\mathrm{KL}(Q\,\|\,P) + \ln\frac{2\sqrt{n}}{\delta}}{2n}} \tag{2.34}$$

The bound improves when the posterior Q stays close (in KL divergence) to the prior P. This suggests a clean way to formalize calibration as regularization: a calibration objective can be viewed as biasing the posterior Q toward a calibration-respecting region, and if that region can be encoded in (or kept close to) the prior P, the complexity term $\mathrm{KL}(Q\|P)$ shrinks. Establishing such a connection precisely is a theoretical aim of chapter 1's RQ2.

2.7.4. Calibration as a constraint on the hypothesis space

Two threads make the "structural constraint" framing concrete. First, the calibration–refinement decomposition (section 2.4.5) shows that a proper-loss risk already *contains* a calibration term; penalizing calibration explicitly re-weights this existing component rather than adding an unrelated objective [8, 42]. Second, results show that calibration genuinely *constrains* the attainable set of predictors: Pleiss *et al.*[57] prove that calibration is compatible with only a restricted set of group error-rate constraints, an explicit example of calibration acting as a hard structural restriction on the hypothesis space. Taken together, these results support that the effect of end-to-end calibration on generalization is analyzable with the tools above.

2.8. Gene Regulatory Network Inference

2.8.1. Biological background

A gene regulatory network is a directed graph where edge $a \to b$ encodes that product of gene a (typically a transcription factor) influences the expression of gene b. Reconstructing GRNs from expression measurements is foundational for understanding differentiation, cellular response, and disease [13, 14]. Two features make the problem a stringent testbed for probabilistic prediction. The edges of interest are *directed* and ideally *causal*, so a method that merely captures statistical association can score well on association-based metrics while misrepresenting the regulatory logic. Secondly, the practical output is a *ranked, uncertain* set of candidate edges, whose usefulness depends on whether the attached confidences are trustworthy.

2.8.2. Data and its pathologies

GRN inference uses two broad data modalities. *Bulk* expression data, such as the microarray collection of the DREAM5 challenge, aggregate signal over many cells across many conditions [22]. *Single-cell* RNA-sequencing (scRNA-seq) resolves individual cells but introduces severe pathologies: high technical noise, extreme sparsity, and "dropout" events in which expressed transcripts are recorded as zero. Simulation studies reveal these consequences of technical noise by dominating the accuracy of network inference [15]. Many single-cell methods further rely on *pseudotime*, an inferred ordering of cells along a trajectory, whose errors propagate into the inferred network [14, 58]. These properties violate clean i.i.d. assumptions and elevated noise levels make calibrated uncertainty both harder to achieve and more valuable.

2.8.3. Classical and statistical approaches

Most early and still-competitive methods are unsupervised. *Correlation- and information-theoretic* methods score edges by pairwise dependence. For example, ARACNE uses mutual information with a data-processing-inequality pruning step [59], CLR uses a context-likelihood correction [60], and PIDC uses partial information decomposition for single-cell data [19]. *Probabilistic graphical models* provide the most rigorous paradigm: Bayesian networks model the joint distribution of expression and support reasoning about conditional independence and, under assumptions, causal structure and pseudo-temporal dynamics [13, 61]. There is a computational obstacle as learning Bayesian-network structure is NP-hard in general [16], and exact structure learning does not scale to the thousands of genes assayed at genome scale. This gap motivates faster, approximate, learning-based methods. *ODE- and dynamics-based* methods such as SCODE [58] and time-stamped approaches such as SINCERITIES [62] target temporal data specifically.

2.8.4. Machine-learning and deep-learning approaches

Scalability is supplied by feature-selection and regression formulations. GENIE3 decomposes network inference into p independent regression problems, predicting each gene from the others with tree ensembles and ranking edges by variable importance [17]. GRNBoost2 retains this scheme with gradient-boosted trees for far greater speed [18], and both serve as the regression backbone of the SCENIC pipeline [63]. More recently, *deep* models have been applied: CNNC casts gene-pair inference as image classification over co-expression histograms [20]; GENELink uses a graph attention network to predict regulatory links from single-cell data [21]; and DeepSEM embeds a structural equation model within a β-VAE to jointly infer the network and a denoised representation [64]. These supervised and generative deep methods are scalable compared to the classical structure learning methods. However, they are trained for, and evaluated by, predictive ranking, and often do not produce calibrated edge probabilities or guarantees causal validity.

Table 2.6 summarizes the principal method families along with the comparison of edge directionality, causal semantics, genome-scale scalability, output type, and whether calibrated uncertainty is an explicit training objective.

Table 2.6.: Families of gene regulatory network inference methods and the properties relevant to this thesis. The final column is the key observation: across all families, including deep methods that do emit probability-like scores, calibrated uncertainty is not an explicit objective, and edge confidences are neither trained for nor evaluated as calibrated quantities.

Family	**Representative methods**	**Dir.**	**Causal**	**Scale**	**Output**	**Calib.**
Correlation & mutual information	ARACNE [59], CLR [60], PIDC [19]	×	×	✓	Association score (ranking)	×
Bayesian networks / graphical models	Score-and-search BNs [13, 61]	✓	∼	×[a]	Structure / edge posterior	×
ODE & time-stamped dynamics	SCODE [58], SINCERITIES [62]	✓	∼	∼	Regression coeff. (ranking)	×
Tree-ensemble regression	GENIE3 [17], GRNBoost2 [18] (SCENIC [63])	✓	×	✓	Importance score (ranking)	×
Deep learning (supervised / generative)	CNNC [20], GENELink [21], DeepSEM [64]	✓	×[b]	✓	Edge probability / weighted adjacency	×

Legend: ✓ yes × no ∼ partial / only under modelling assumptions. *Dir.* = directed edges; *Causal* = explicit causal semantics; *Scale* = applicability at genome scale; *Calib.* = calibrated uncertainty is an explicit objective. [a] Exact Bayesian-network structure learning is NP-hard [16], limiting genome-scale use. [b] DeepSEM's structural-equation formulation carries a partial causal interpretation under its modeling assumptions.

2.8.5. Benchmarking, simulation, and evaluation

Progress in the field is organized around shared benchmarks. The DREAM5 challenge provides standardized datasets with gold-standard networks and established that aggregating predictions across methods is robust [22]; GeneNetWeaver supplies in-silico benchmark generation and profiling [65]; BEELINE systematizes single-cell GRN evaluation against curated and synthetic ground truth [14]; and SERGIO simulates single-cell expression from a user-specified GRN with realistic stochasticity and noise [15]. Evaluation almost always uses *ranking* metrics—area under the ROC (AuROC) and precision–recall curves and early precision [14, 22]. These quantify whether true edges are ranked above false ones but are *silent on calibration*: they do not test whether a predicted probability of 0.8 corresponds to an 80% empirical edge frequency, nor whether the recovered structure is causal. The combination of (i) controlled simulators that expose the true network and (ii) an evaluation culture that ig-

nores calibration is exactly what makes GRN inference a suitable and underexplored testbed for end-to-end calibrated learning.

2.9. Synthesis and Positioning

Three observations motivate the central focus of the thesis. *First*, calibration is a well-defined and hierarchically structured property (section 2.2), but its assessment is not straightforward. In particular, training objectives and evaluation estimators must be carefully distinguished (section 2.4), and both are rooted in the theory of proper scoring rules and the calibration–refinement decomposition (section 2.4.5). *Second*, although post-hoc recalibration methods (section 2.5) are well established, they remain fundamentally separate from the learning process itself. By contrast, in-training approaches (section 2.6), including differentiable calibration penalties, implicit regularization strategies, and reward-based formulations, modify the learner itself during optimization. *Third*, gene regulatory network (GRN) inference (Section 2.8) presents an especially compelling application domain. While scalable deep learning models have achieved strong predictive performance, they are typically designed and evaluated with ranking accuracy as the primary objective. As a result, calibrated and interpretable probabilistic inference remains largely unexplored. Moreover, the availability of simulation frameworks with controlled ground truth makes GRN inference suitable for studying calibration and generalization. The gap addressed by this thesis lies at the intersection of these three strands: an end-to-end treatment of calibration as an explicit learning constraint, analyzed through the lens of generalization theory and evaluated in the setting of simulation-supervised GRN inference.

3. End-to-End Calibration as a Structural Constraint

Chapter 1 advanced the hypothesis that integrating a calibration objective into empirical risk minimization (ERM) behaves like *implicit regularization*: it restricts the set of admissible predictors and thereby reshapes the bias–variance trade-off (RQ1, RQ2). This chapter makes that hypothesis precise. We (i) formalize the composite, calibration-regularized objective and relate its special cases to proper losses (section 3.1); (ii) recall the exact calibration–refinement decomposition that the objective re-weights (section 3.2); (iii) express penalized training as *constrained* ERM over a calibration-restricted hypothesis class (section 3.3); and (iv) analyze the generalization of that restricted class through Rademacher complexity (section 3.4) and a PAC-Bayesian lens (section 3.5). Section 3.6 interprets the results as capacity control, while section 3.8 states the assumptions and the questions that remain open.

Throughout the chapter, we adopt the notation of section 2.1: data $(X, Y) \sim \mathcal{D}$ where $X \in \mathcal{X}, Y \in \mathcal{Y} \subset \Delta_K$, a hypothesis class $\mathcal{F}$ of probabilistic predictors $f : \mathcal{X} \to \Delta_K$, a sample $S = \{(x_i, y_i)\}_{i=1}^n$, empirical and population risks $\widehat{\mathcal{E}_\ell}, \mathcal{E}_\ell$, and the Bayes predictor $f^\star(x) = \eta(x) = \pi(\cdot \mid x)$.

3.1. The Composite Calibration-Regularized Objective

Motivated by the calibration–refinement decomposition, introduced in section 2.4.5 and the observation that optimization dynamics may yield poorly calibrated predictors, we consider an *empirical, differentiable surrogate* $\widehat{\mathcal{C}}(f) : \mathcal{F} \to \mathbb{R}_{\geq 0}$ of the calibration error, for instance the kernel-density based canonical estimator $\mathrm{ECE}_{\mathrm{KDE}}$ [11], the maximum mean calibration error [9], or a soft-binned error [10]. We also consider $\widehat{\mathcal{E}} : \mathcal{F} \to \mathbb{R}_{\geq 0}$ to denote the total empirical proper loss (e.g. the Brier score).

Our objective is to solve the optimization problem

$$\min_{\theta \in \Theta} \mathcal{L}_\lambda(f_\theta) = \min_{\theta \in \Theta} \widehat{\mathcal{C}}(f_\theta) + \lambda \widehat{\mathcal{R}}(f_\theta) = \min_{\theta \in \Theta} (1 - \lambda)\widehat{\mathcal{C}}(f_\theta) + \lambda \widehat{\mathcal{E}}(f_\theta), \tag{3.1}$$

with refinement error, $\widehat{\mathcal{R}}(f_\theta) := \widehat{\mathcal{E}}(f_\theta) - \widehat{\mathcal{C}}(f_\theta)$ and with $\lambda \in (0, 1]$ being a regularization parameter. Following the calibration–refinement decomposition, we do not specify the refinement term independently; we *define* it as the residual. The goal is to shape the training trajectory $\{f_{\theta_t}\}$ such that the final predictor is both accurate and calibrated.

Definition 3.1 (Calibration-regularised ERM)**.** For weight λ, the calibration-regularised estimator is $\hat{f}_\lambda \in \arg\min_{\theta\in\Theta} \mathcal{L}_\lambda(f_\theta)$.

We note two things specifically. First, by eq. 3.1 the proper-loss ERM is recovered *exactly* at the *proper-score point* $\lambda = 1$, where the calibration penalty vanishes and $\mathcal{L}_1 = \widehat{\mathcal{E}}$; decreasing λ toward 0 places increasing weight on calibration. Second, the surrogate $\widehat{\mathcal{C}}$ should be *bounded* and built from a bounded proper scoring rule such as the Brier score [40, 12], rather than the unbounded log-loss, so that the objective is not dominated by a few confidently wrong examples; this is the same boundedness argument used by the RLCR framework [12] and is one reason this thesis prefers Brier-based surrogates.

Choosing the weights. The weight λ may be set in several ways, which the experimental chapters compare: *(i)* a fixed grid sweep over λ with selection by a validation criterion; *(ii) homoscedastic uncertainty weighting*, in which the weights are learned jointly with f by treating each term as a Gaussian log-likelihood with a learned observation-noise scale, after Kendall *et al.* [66]; and *(iii)* validation-based selection of a single trade-off parameter. Each induces a different effective constraint (section 3.3); the uncertainty-weighting scheme in particular tends to drive the calibration weight to large values, which must be anticipated when interpreting its solutions. A core empirical comparison task of this thesis is to determine which estimator most effectively reduces miscalibration while preserving predictive accuracy.

3.2. The Calibration–Refinement Decomposition, Made Precise

The decomposition that eq. 3.1 re-weights is classical. We have already seen in subsection 2.4.5 that for a proper loss (e.g., Brier Score) ℓ, the expected score admits the population decomposition [31, 42, 38]

$$\mathcal{E}(f) = \mathbb{E}\big[\ell(f(X), Y)\big] = \underbrace{\mathbb{E}\big[d_\ell\big(f(X), \gamma(f(X))\big)\big]}_{\mathcal{C}_\ell(f)\ \geq 0} + \underbrace{\mathbb{E}\big[e_\ell\big(\bar{\gamma}(f(X))\big)\big]}_{\mathcal{R}_\ell(f)} \tag{3.2}$$

where $\gamma(\mathbf{p}) = \mathbb{P}(Y \mid f(X) = \mathbf{p})$ is the true class distribution among inputs that receive prediction $\mathbf{p}$, d_ℓ is the Bregman divergence (reduces to KL divergence when we use log-loss) associated with ℓ, and $\mathcal{R}_\ell$ collects the resolution (sharpness) and the irreducible uncertainty $H(\gamma(Y))$. The calibration term $\mathcal{C}_\ell(f)$ is zero exactly when f is canonically calibrated (section 2.2).

Berta *et al.* [8] give a variational form of eq. 3.2 for cross-entropy and show that its two components are not minimized at the same iterate during training. In our composite objective, we estimate the calibration component by the surrogate $\widehat{\mathcal{C}} \approx \mathcal{C}_\ell$ and obtain the refinement component as the residual $\widehat{\mathcal{R}}_\ell = \widehat{\mathcal{E}} - \widehat{\mathcal{C}}_\ell$ (eq. 2.14), the empirical counterpart of $\mathcal{R}_\ell$, so the two estimated parts sum to the measured proper loss by construction. At $\lambda = 1$, the proper loss, and hence the consistent estimation of $f^\star$, is recovered; as λ decreases toward 0, the learner is pushed increasingly toward calibration at a controlled cost in refinement.

Remark 3.2 (Surrogate gap). $\widehat{\mathcal{C}}$ is a finite-sample surrogate for $\mathcal{C}_\ell$ and is in general *biased* (section 2.4.2) [25, 33]. Because refinement is defined as the residual $\widehat{\mathcal{E}} - \widehat{\mathcal{C}}_\ell$, any bias in $\widehat{\mathcal{C}}_\ell$ enters $\widehat{\mathcal{R}}_\ell$ with the opposite sign; only the total $\widehat{\mathcal{E}}$ is estimator-independent. The theory below treats $\widehat{\mathcal{C}}_\ell$ as the quantity actually optimized; relating the optimized surrogate to the population calibration $\mathcal{C}_\ell$ is itself part of the analysis and is revisited in section 3.8.

3.3. From Penalized to Constrained ERM

The claim that calibration "constrains the hypothesis space" is made literal by the standard correspondence between penalized and constrained optimization.

Proposition 3.3 (Penalized–constrained correspondence)**.** *Let $\lambda \in (0, 1]$. By eq. 3.1, minimizing the composite objective is equivalent to the penalized proper-loss problem $\min_{f \in \mathcal{F}} \widehat{\mathcal{E}}(f) + \mu\, \widehat{\mathcal{C}}_\ell(f)$ with $\mu = (1-\lambda)/\lambda \geq 0$. For every $\mu \geq 0$ there exists a level $c = c(\mu) \geq 0$ such that any minimizer $\hat{f}_\mu$ of the penalized problem is also a minimizer of the constrained problem*

$$\min_{f \in \mathcal{F}_c} \widehat{\mathcal{E}}(f), \qquad \mathcal{F}_c := \{ f \in \mathcal{F} : \widehat{\mathcal{C}}_\ell(f) \leq c \}, \tag{3.3}$$

and $c(\mu)$ is non-increasing in μ (equivalently, non-decreasingly in λ).

Proof. The equivalence to the penalized proper-loss problem is the algebraic identity in eq. 3.1 divided by λ. For the constrained form, take $c = \widehat{\mathcal{C}}(\hat{f}_\mu)$. Any f feasible for eq. 3.3 satisfies $\widehat{\mathcal{C}}_\ell(f) \leq c$, and by optimality of $\hat{f}_\mu$, $\widehat{\mathcal{E}}(\hat{f}_\mu) + \mu c \leq \widehat{\mathcal{E}}(f) + \mu \widehat{\mathcal{C}}_\ell(f) \leq \widehat{\mathcal{E}}(f) + \mu c$, whence $\widehat{\mathcal{E}}(\hat{f}_\mu) \leq \widehat{\mathcal{E}}(f)$. Monotonicity of $c(\cdot)$ follows from the standard envelope argument for the Lagrangian. A formal statement requires mild regularity (e.g. lower semicontinuity of $\widehat{\mathcal{C}}$); see [51] for the convex case. □

Proposition 3.3 licenses the central object of the analysis: the *calibration-restricted class* $\mathcal{F}_c \subseteq \mathcal{F}$. Increasing the calibration weight shrinks c and hence $\mathcal{F}_c$. The remaining question is whether this shrinkage improves generalization.

3.4. Generalization via Rademacher Complexity

We measure the capacity of a class by its empirical Rademacher complexity $\widehat{\mathfrak{R}}_m(\cdot)$ (section 2.7) [53]. Let $\ell \circ \mathcal{F} = \{(x, y) \mapsto \ell(f(x), y) : f \in \mathcal{F}\}$ denote the induced loss class for a $[0, 1]$-bounded loss ℓ.

Proposition 3.4 (Capacity is monotone in the calibration constraint)**.** *For any $c \geq 0$, the calibration-restricted class $\mathcal{F}_c \subseteq \mathcal{F}$ of eq. 3.3 satisfies*

$$\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}_c) \leq \widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}). \tag{3.4}$$

Consequently, with probability at least $1-\delta$, every $f \in \mathcal{F}_c$ satisfies

$$\mathcal{E}_\ell(f) \leq \widehat{\mathcal{E}}_\ell(f) + 2\,\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}_c) + 3\sqrt{\frac{\ln(2/\delta)}{2n}}, \tag{3.5}$$

so the uniform generalisation bound for the calibration-restricted class is no looser than that for $\mathcal{F}$, and is strictly tighter whenever the constraint removes hypotheses that attain the supremum in $\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F})$.

Proof sketch. $\mathcal{F}_c \subseteq \mathcal{F}$ implies $\ell \circ \mathcal{F}_c \subseteq \ell \circ \mathcal{F}$, and the Rademacher complexity is a supremum over the class, hence monotone under set inclusion; the bound eq. 3.5 is the standard Rademacher generalisation bound [53, 51] applied to $\mathcal{F}_c$. □

Proposition 3.4 is deliberately modest: monotonicity is an immediate consequence of inclusion. Its value is conceptual—it identifies the precise mechanism by which calibration can act as capacity control—and it isolates the genuinely substantive question, namely *how much* the constraint reduces complexity. Quantifying $\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}) - \widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}_c)$ for concrete classes (e.g. norm-bounded networks) is the non-trivial step and is taken up empirically in chapter 4; obtaining a closed-form lower bound on the reduction is left as an open problem (Section 3.8).

3.5. A PAC-Bayesian View

PAC-Bayes bounds are well suited to the overparameterised models used for GRN inference, since they bound the risk of randomised predictors and have given the tightest known guarantees for neural networks [54, 55, 56]. McAllester's bound states that for a data-independent prior P and any posterior Q, with probability at least $1-\delta$,

$$\mathbb{E}_{f\sim Q}\big[\mathcal{E}_\ell(f)\big] \leq \mathbb{E}_{f\sim Q}\big[\widehat{\mathcal{E}}_\ell(f)\big] + \sqrt{\frac{\mathrm{KL}(Q\,\|\,P) + \ln\frac{2\sqrt{n}}{\delta}}{2n}}. \tag{3.6}$$

We propose to read calibration-regularized training through Eq. 3.6 as follows. The composite objective Eq. 3.1 biases the learned posterior Q_λ toward the calibration-respecting region $\{f : \widehat{C}(f) \leq c\}$. If the prior P encodes a preference for the same region—for example a prior concentrated on well-calibrated predictors, or a data-dependent prior in the sense of [55]—then the posterior need not travel far from the prior to fit the data, so the complexity term $\mathrm{KL}(Q_\lambda \| P)$ is small and the bound tightens. Informally, calibration and a calibration-aware prior are "aligned," and alignment is what PAC-Bayes rewards.

Remark 3.5. The reading above is a *framework*, not a theorem. Turning it into a guarantee requires (a) constructing a valid prior P that provably charges the calibrated region without peeking at the test data, and (b) bounding $\mathrm{KL}(Q_\lambda \| P)$ in terms of $\boldsymbol{\lambda}$. Both are non-trivial and are stated here as objectives of the thesis rather than as established results.

3.6. Interpretation: Calibration as Capacity Control

Sections 3.3–3.5 support a single picture. The calibration penalty does not add an objective unrelated to generalization; by eq. 3.2 it re-weights a component that the proper risk already contains, and by proposition 3.3 it is equivalent to optimizing within a shrunken class $\mathcal{F}_c$. Reducing the class lowers variance (proposition 3.4) at the possible cost of increased bias if $f^\star \notin \mathcal{F}_c$, which is precisely the behavior of a regularizer. The proper-score point $\boldsymbol{\lambda} = 1$ is the bias-free setting (it targets $f^\star$); decreasing λ towards 0 trades a controlled amount of bias for reduced variance and better calibration. This is the formal content of the thesis's hypothesis that end-to-end calibration shapes the bias–variance trade-off, and it yields a concrete, testable prediction: as λ decreases, calibration error and the generalization gap should fall together until bias from an over-restricted $\mathcal{F}_c$ dominates. We examine this hypothesis in the controlled synthetic experiments in chapter 4.

3.7. Robustness Under Covariate Shift

The generalization guarantees of sections 3.4-3.5 are stated for the in-distribution setting, $\mathcal{D}_{\text{test}} = \mathcal{D}_{\text{train}}$. Under covariate shift (section 2.3), a natural question is whether the calibration-restricted class $\mathcal{F}_c$ (section 3.3) that improves in-distribution generalization also yields predictors whose calibration is more stable as $\mathbb{P}(X)$ shifts or whether the restriction only helps in absence of the shift. We distinguish three notions of robustness relevant to this question, following the empirical framework of Ovadia *et al.* [27], who benchmark both accuracy and calibration degradation under dataset shift and find that post-hoc recalibration in particular does not transfer reliably.

1. *Predictive Robustness*: degradation in a ranking-based metric (e.g. AUROC, AUPR) from $\mathcal{D}_{\text{train}}$ to $\mathcal{D}_{\text{test}}$, the standard notion of robustness in the broader shift literature.

2. *Calibration Robustness*: degradation in a calibration-specific metric (e.g. ECE, RMSCE, KSCE, or the calibration component $\mathcal{C}_\ell(f)$ of the decomposition eq. 2.14) from $\mathcal{D}_{\text{train}}$ to $\mathcal{D}_{\text{test}}$, isolated from predictive performance. This is the notion most directly relevant to whether calibration-regularized training (eq. 3.1) confers a benefit specific to calibration, rather than accuracy in general.

3. *Recalibration-map Transfer Robustness*: whether a post-hoc map g (section 2.5), fit on $\mathcal{D}_{\text{train}}$-distributed validation data, remains effective on $\mathcal{D}_{\text{test}}$ without refitting. Ovadia *et*

al. [27] find that this typically fails to hold for standard post-hoc methods, which is the shift-specific instance of the general limitation already noted in section 2.5.8.

For a metric M, we define the *robustness gap* as below:

$$\text{RobustnessGap}_{\text{M}}(\text{method}) = \text{M}(\text{method}; \mathcal{D}_{\text{test}}) - \text{M}(\text{method}; \mathcal{D}_{\text{train}}) \tag{3.7}$$

For metrics where lower is better (e.g., ECE, RMSCE, KSCE etc.), a positive gap indicates degradation under shift; for metrics where higher is better (AUROC, AUPR), a negative gap indicates the same. Comparing $|\texttt{RobustnessGap}|$ across methods puts both metric orientations on the same footing. Under this framework, end-to-end calibration is considered more robust than a predictive-only baseline if $|\texttt{RobustnessGap}|$ is smaller for the calibration-specific metrics of notion (2), independent of whether the predictive-robustness gap of notion (1) is comparable or worse. It is precisely this separation, calibration robustness without necessarily improved predictive robustness, that constitutes evidence for the mechanism this thesis studies (RQ1–RQ2), rather than a generic accuracy-under-shift effect already documented elsewhere.

3.8. Assumptions

For transparency we collect the assumptions on which the analysis rests:

- The sample S is drawn i.i.d. from $\mathcal{D}$
- The loss ℓ is bounded (e.g. Brier), so the Rademacher bound applies
- The calibration surrogate $\widehat{\mathcal{C}}$ is lower semicontinuous, so proposition 3.3 holds.

4. Simulation-Supervised Experiments

4.1. Overview and Objectives

The central question of this thesis is whether calibration can be improved by embedding it into the training objective as a structural component, that is, by solving

$$\min_{\theta\in\Theta} \widehat{\mathcal{C}}(f_\theta) + \lambda\,\widehat{\mathcal{R}}(f_\theta) \;=\; \min_{\theta\in\Theta} (1-\lambda)\,\widehat{\mathcal{C}}(f_\theta) + \lambda\,\widehat{\mathcal{E}}(f_\theta), \tag{4.1}$$

while statistical accuracy is preserved through an appropriate choice of $\lambda > 0$. Here $\widehat{\mathcal{C}}$ is an empirical calibration term, $\widehat{\mathcal{E}}$ the empirical total proper loss and $\widehat{\mathcal{R}} = \widehat{\mathcal{E}} - \widehat{\mathcal{C}}$ the refinement term, as introduced in chapter 3.

This chapter examines the question in the one setting where every quantity involved can be compared with the truth. We use a one-dimensional binary classification problem whose conditional class probability $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$ is known in closed form. On real data, calibration can only be estimated from observed labels, typically through binned statistics whose own bias and variance blur the comparison. Here, in addition, we can measure how far a trained model is from η itself. That distance, MSE_η, is the reference quantity of the chapter: it separates a model that is genuinely closer to the true posterior from one that merely produces a flatter reliability diagram.

The chapter is primarily an answer to RQ3, which asks how end-to-end calibrated models compare with baselines on exact measures of calibration, refinement and generalisation across model capacity and regularisation strength. It also supplies the empirical evidence for RQ1 (does the calibration term act as an implicit regulariser, and how does end-to-end training compare with predictive-only training followed by post-hoc calibration?) and it constrains how RQ2 can be answered, since the theoretical arguments of Chapter 3 make predictions that can be checked here. RQ4, which concerns gene regulatory network inference, is addressed in the following chapters.

A controlled setting is necessary for four reasons. First, MSE_η can be computed only when η is known. Second, capacity can be varied in a way whose approximation error is known exactly (Section 4.2.2), so that any remaining miscalibration can be attributed to estimation rather than to a hypothesis class that cannot represent the truth. Third, training sets can be redrawn from the same distribution as often as needed, which makes variance across training samples a measurable quantity rather than a thought experiment. Fourth, samples are independent and identically distributed by construction, so the assumptions under which the results of

Table 4.1.: Experiment questions of Track 1, the experiment that answers each of them, and where the result is reported. "1 seed" refers to data seed 1 and initialization seed 42; "30 seeds" to data seeds $s = 1, \dots, 30$ with initialization seed $10000 + s$.

	Question	Seeds	Reported in
Q1.1	Does the calibration–refinement decomposition hold during training, and do the two components move in opposite directions?	1	Sec. 4.3.1
Q1.2	Is there a phase transition during training that the penalty can prevent?	1	Sec. 4.3.1
Q1.3	How do post-hoc methods compare with each other and with end-to-end training across capacity?	1	Sec. 4.3.5
Q1.4	Bias and variance of TS-based and isotonic refinement estimators	30	Sec. 4.3.5
Q1.5	Does post-hoc calibration degrade for overparameterised models?	1	Sec. 4.3.5
Q1.6	Brier score or cross-entropy as base loss?	1	Sec. 4.3.2
Q1.7	Does focal loss match explicit end-to-end calibration?	30	Sec. 4.3.6
Q1.8	Does overparameterisation help under end-to-end training?	1 and 30	Sec. 4.3.4
Q1.9	Does the penalty help more when $m \gg n$?	1 and 30	Sec. 4.3.4
Q1.10	Does the penalty reduce bias and variance simultaneously?	30	Sec. 4.3.3
Q1.11	Does the penalty move the model towards η, or only improve the reliability diagram?	1	Sec. 4.3.2
Q1.12	Is there an optimal λ, and does it depend on m?	1	Sec. 4.3.2
A	Is the gain caused by the penalty or by the reduced step size on $\widetilde{E}$?	30	Sec. 4.3.7
B	Additive penalty at a fixed learning rate	30	Sec. 4.3.7
C	Choosing λ on validation data versus validation early stopping	30	Sec. 4.3.7
D	Learned loss weights (uncertainty weighting)	30	Sec. 4.3.7

Chapter 3 are derived hold exactly. This last point is the main difference to the real-data case study later in the thesis, where i.i.d. sampling cannot be assumed.

Table 4.1 lists the experiment questions of this track, the experiment that answers each of them and where the answer is reported. The table also records how many seeds each experiment uses. The distinction matters: the single-seed experiments are used to inspect mechanisms, and only the 30-seed experiments carry significance tests.

4.2. Experimental Setup

4.2.1. Data-Generating Process

Inputs and labels are generated as

$$\begin{aligned} X &\sim \mathrm{Uniform}[-1,1], \\ Y \mid X = x &\sim \mathrm{Bernoulli}\big(\eta(x)\big), \\ \eta(x) &= 0.1 + 0.8\,\sigma\big(4\sin(\pi x)\big). \end{aligned} \tag{4.2}$$

with $\sigma(t) = 1/(1+e^{-t})$. The Bayes-optimal predictor is $f^* = \eta$. Because $\eta(x) \in (0.1, 0.9)$ for every x, every label is genuinely random: even f^* misclassifies with probability $\min\{\eta(x), 1-\eta(x)\} \geq 0.1$. This Bernoulli draw is the only source of noise; no additional noise term is introduced. Two consequences are used repeatedly below.

- The irreducible Brier risk is $\mathbb{E}[\eta(X)(1-\eta(X))] \approx 0.147$ (estimated on the test set). For any predictor f that does not depend on the test labels, $\mathbb{E}[(f(X)-Y)^2] = \mathrm{MSE}_\eta(f) + \mathbb{E}[\eta(X)(1-\eta(X))]$. The Brier score and MSE_η therefore rank models almost identically, and they differ mainly by this constant.
- The Bayes classifier is correct with probability $\max\{\eta(x), 1-\eta(x)\} \leq 0.9$. A top-label confidence above 0.9 is therefore always overconfident in this problem, whichever model produces it.

Each training set has $n_{\text{train}} = 100$ points and is redrawn for every data seed. The test set has $n_{\text{test}} = 50{,}000$ points, is drawn once with a fixed seed and is shared by all runs, so that differences between runs are not confounded by test-set noise. The training set is deliberately small so that the larger models can interpolate it.

4.2.2. Model and Hypothesis Class

The learner is a logistic model on a Legendre polynomial basis,

$$p_\theta(x) = \sigma\Big(\sum_{k=0}^{m} \theta_k\, \phi_k(x)\Big), \tag{4.3}$$

where $\phi_0, \ldots, \phi_m$ are the Legendre polynomials on $[-1,1]$ and the degree m controls capacity (the model has $m+1$ parameters). Following the supervisor's recommendation, we use this model instead of a neural network because it isolates the effect of capacity from confounders such as depth, normalisation layers and optimiser heuristics. The Legendre basis spans the same space as the monomials but is numerically well conditioned.

The capacity grid is $m \in \{5, 10, 100, 120, 200, 300, 1000\}$. Using the rule of thumb $m \approx n^r$, it covers an underparameterised regime ($m \in \{5, 10\}$, $r \approx 0.35$–0.5), the interpolation threshold

($m = 100 = n_{\text{train}}$, $r = 1$), a near-threshold overparameterised regime ($m \in \{120, 200, 300\}$, $r \approx 1.04$–1.24) and a strongly overparameterised regime ($m = 1000$, $r = 1.5$). The three primary capacities are $m \in \{10, 100, 1000\}$; the full grid is used where the question concerns the shape of a curve in m. Parameters are initialised as $\theta \sim \mathcal{N}(0, 0.01\, I)$, so all models start with predictions close to 0.5. The initialisation seed is set independently of the data seed.

Can the hypothesis class represent η? Since η is bounded away from 0 and 1, the function $g = \sigma^{-1} \circ \eta$ is continuous and bounded on $[-1, 1]$. By the Weierstrass approximation theorem it can be approximated uniformly by polynomials, and because σ is uniformly continuous, $\sigma(P_m) \to \eta$ uniformly as $m \to \infty$. Figure 4.1 shows how fast this happens. The figure uses no data at all: g is projected onto the first $m + 1$ Legendre polynomials by Gauss–Legendre quadrature, and σ of the projection is compared with η.

The convergence is spectral, which is expected because g is real-analytic. At $m = 10$ the approximation error is small but not zero (sup-norm $2.6{\times}10^{-2}$, L_2 error $6.6{\times}10^{-3}$). At $m = 100$ it has fallen to about 10^{-12}, which is at the level of floating-point noise, and it stays there up to $m = 1000$. Both $m = 100$ and $m = 1000$ can therefore represent η essentially exactly, and every deviation from η observed at these capacities is estimation error: it comes from the finite sample, the optimisation and overfitting, not from the function class.

The error at $m = 10$ is also small in absolute terms. Its squared L_2 error, about 4×10^{-5}, is a lower bound on the MSE_η any model of degree 10 can reach. The trained models in this chapter have $\text{MSE}_\eta \approx 10^{-2}$ at best, so approximation bias accounts for well under one percent of their error even at $m = 10$.

4.2.3. Training Objective and Weight Schemes

End-to-end objective. All end-to-end models minimize the convex combination

$$\mathcal{L}_\lambda(\theta) = (1 - \lambda)\, \widetilde{C}_{\text{soft}}(\theta) + \lambda\, \widetilde{E}(\theta), \tag{4.4}$$

where $\widetilde{E}$ is either the cross-entropy (CE) or the Brier score on the training set. The calibration term is the soft-binned estimator

$$\widetilde{C}_{\text{soft}}(\theta) = \sum_{b=1}^{B} \Big(\frac{1}{n} \sum_{i=1}^{n} w_{ib} \Big) \big(\bar{y}_b - \bar{p}_b \big)^2, \qquad w_{ib} = \text{softmax}_b\Big(-\frac{(p_i - c_b)^2}{2h^2} \Big), \tag{4.5}$$

with bin centres $c_b = (b - \frac{1}{2})/B$, $B = 15$ bins and bandwidth $h = 0.08$. Here $\bar{p}_b$ and $\bar{y}_b$ are the $w_{\cdot b}$-weighted means of predictions and labels in bin b. The term is differentiable, and it is a smoothed version of a binned squared calibration error on the training predictions.

In computing its gradient, the bin weights w_{ib} are treated as constants (a straight-through approximation). The gradient therefore flows into the penalty through the predictions inside $\bar{p}_b$, but not through the assignment of predictions to bins. This is an approximation of the true gradient of (4.5), and all results below refer to it.

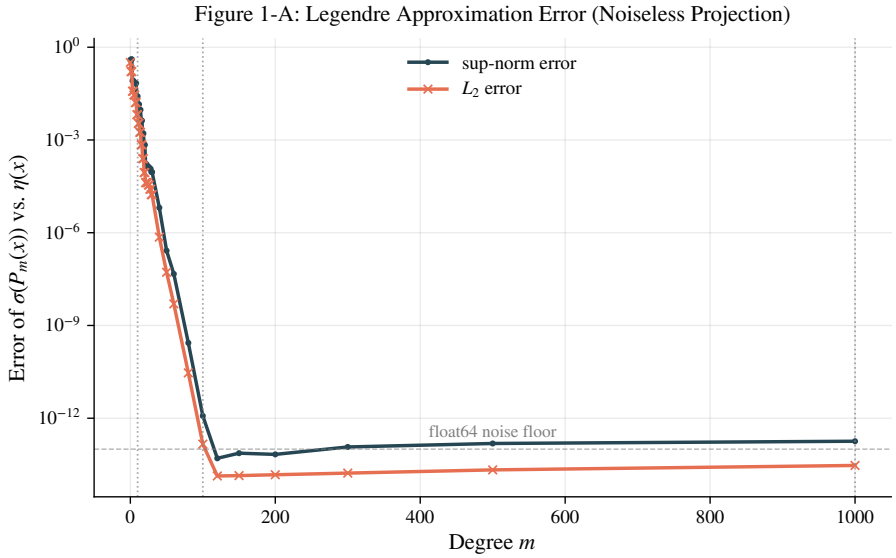


Figure 4.1.: Approximation error of $\sigma(P_m)$ relative to η as a function of the degree m, for the noiseless Legendre projection of $g = \sigma^{-1} \circ \eta$ (no training data involved). Dotted vertical lines mark the primary capacities $m \in \{10, 100, 1000\}$.

Role of λ and the limiting cases. At $\lambda = 1$, (4.4) reduces to the plain proper loss, which is the predictive-only baseline. The objective is formally defined for $\lambda \in (0, 1]$. We nevertheless include $\lambda = 0$ as a limiting case, because it shows what the calibration term does on its own. Unlike $\widetilde{E}$, the calibration term is not a proper scoring rule: any calibrated predictor minimises it, including the constant predictor that outputs the base rate everywhere. *Decreasing* λ increases the weight on calibration. This is the opposite of the direction suggested by the wording of Q1.10 in the experiment plan; we follow the direction implied by (4.4) throughout.

An identity that matters for interpretation. All models are trained with full-batch gradient descent at a fixed learning rate η_{lr}. One step on (4.4) can be rewritten as

$$\theta \leftarrow \theta - \eta_{\text{lr}}\big[(1-\lambda)\nabla\widetilde{C}_{\text{soft}} + \lambda\nabla\widetilde{E}\big] = \theta - (\eta_{\text{lr}}\lambda)\Big[\nabla\widetilde{E} + \tfrac{1-\lambda}{\lambda}\nabla\widetilde{C}_{\text{soft}}\Big]. \tag{4.6}$$

Training on the convex objective with learning rate η_{lr} is therefore *exactly* the same as training on the additive objective $\widetilde{E} + \beta\,\widetilde{C}_{\text{soft}}$ with $\beta = (1-\lambda)/\lambda$ and learning rate $\eta_{\text{lr}}\lambda$. We confirmed this numerically: after 1,200 epochs the two parameter vectors agree to within 3×10^{-16}.

The consequence is that lowering λ changes two things at once. It strengthens the calibration term relative to the proper loss, and it shrinks the effective step size on the proper loss by a factor of λ. With a fixed budget of epochs and no early stopping, the second effect alone behaves like stopping plain training early. Berta et al. [8] study exactly that mechanism. Separating the two effects is the purpose of ablation A in Section 4.3.7.

Weight schemes. Four ways of setting the relative weight of the two terms are studied.

1. *Fixed grid.* The main experiments use $\lambda \in \{0, 0.1, 0.5, 1\}$. The single-seed sweep behind Figure 4.5 uses a finer 21-point grid $\lambda \in \{0, 0.05, \dots, 1\}$; the finer grid changes where the optimum is located (Section 4.3.2).

2. *Additive penalty.* The objective $\widetilde{E} + \beta\, \widetilde{C}_{\text{soft}}$ is trained at the unchanged learning rate 0.5, with $\beta \in \{0, 0.5, 1, 2\}$ from the experiment plan. We add $\beta \in \{9, 19\}$, the values that correspond to $\lambda = 0.1$ and $\lambda = 0.05$ via (4.6). Because the step size on $\widetilde{E}$ is the same in every run, this form isolates the effect of the penalty.

3. *Validation-based selection.* λ is chosen from $\{0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1\}$ by the Brier score on a held-out validation set of 20 points (an 80/20 split of the training data). This is the realistic counterpart of the fixed-grid experiments, whose "best" λ is read off the ground-truth MSE_η on the test set. That quantity is not available in practice, so selecting on it is an oracle choice.

4. *Learned weights (uncertainty weighting).* Following Kendall et al. [66], the loss weights are learned jointly with θ by minimizing

$$\mathcal{L}_{\text{UW}}(\theta, s) = \tfrac{1}{2} e^{-s_C}\, \widetilde{C}_{\text{soft}}(\theta) + \tfrac{1}{2} e^{-s_E}\, \widetilde{E}(\theta) + \tfrac{1}{2}(s_C + s_E), \tag{4.7}$$

   with log-variances $s = (s_C, s_E)$ initialized at zero, updated with the same learning rate as θ, and clipped to $[-8, 8]$. For fixed θ the minimizer is $s_i^* = \log L_i$, which gives each loss the effective weight $1/(2L_i)$. The alternative parameterization $e^{-s}L + \frac{1}{2}s$ leads to the same effective weight. The exact form of the scheme was not fixed in the experiment plan, so this experiment is exploratory.

4.2.4. Baselines

Predictive-only training. The model is trained on the plain proper loss ($\lambda = 1$) with CE or Brier as the base loss. The CE baseline is the reference in all significance tests.

Predictive-only training followed by post-hoc calibration. The 100 training points are split 80/20. The baseline model is trained on the 80 points, and four recalibration maps are fitted on the 20 held-out points:

- temperature scaling (one parameter, fitted by NLL) [1];
- Platt scaling (two parameters) [5];
- isotonic regression [6];
- histogram binning with 15 equal-width bins [6].

The end-to-end model it is compared with is trained on all 100 points. This is deliberate: not having to give up training data for a calibration hold-out is one of the practical arguments for end-to-end calibration, and equalising the data would remove that trade-off from the comparison.

Focal loss. Focal loss [41, 48] is included as an implicit calibration method, with $\gamma \in \{0, 1, 2, 5\}$. We use the standard asymmetric form

$$\mathcal{L}_{\text{Focal}}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\Big[y_i(1-p_i)^{\gamma}\log p_i + (1-y_i)\,p_i^{\gamma}\log(1-p_i)\Big], \tag{4.8}$$

which down-weights easy examples of both classes. The experiment plan wrote a single modulating factor $(1-p_i)^{\gamma}$ for both terms. That version does not down-weight confidently correct negative examples (for $y = 0$ and $p = 0.05$ its factor is 0.90 instead of 0.0025 at $\gamma = 2$), so it does not implement the mechanism of the cited works. Both forms coincide at $\gamma = 0$, where they reduce to CE; we verified that $\gamma = 0$ reproduces the CE training trajectory exactly.

Controls introduced for the ablations. Two further baselines are used in Section 4.3.7. The *step-size-matched control* is plain CE trained with learning rate 0.5λ. By (4.6), it differs from end-to-end training at λ only in the absence of the penalty. *Validation early stopping* trains plain CE on 80 points and keeps the checkpoint with the lowest Brier score on the 20 validation points among all 1,200 epochs. This is checkpoint selection rather than stopping with a patience rule, and it is the natural competitor to validation-based selection of λ.

4.2.5. Evaluation Metrics

All metrics are computed on the fixed test set.

Metrics exact up to Monte Carlo error. These are computed against η rather than the labels:

- $\text{MSE}_{\eta} = \frac{1}{N}\sum_j \big(p_\theta(x_j) - \eta(x_j)\big)^2$, the primary metric of this chapter.
- The *ground-truth refinement* used for Q1.4: $\sum_b \frac{n_b}{N}\bar{\eta}_b(1-\bar{\eta}_b)$, where $\bar{\eta}_b$ is the mean of the true $\eta(x)$ over test points in bin b of p_θ (30 bins). It approximates the refinement term $\mathbb{E}[\,C(1-C)\,]$ with $C = \mathbb{E}[Y \mid f(X)]$, following Bröcker [42] and Berta et al. [8], but uses the true η instead of noisy labels.

Metrics estimated from test labels.

- Test CE and Brier score.
- The hard-binned Brier decomposition with 10 equal-width bins in p: $\widehat{C}_{\text{Brier}} = \sum_b \frac{n_b}{N}(\bar{p}_b - \bar{y}_b)^2$, with refinement defined as the residual so that total = calibration + refinement holds exactly.
- For the CE training curves in Figure 4.2 only, the binned analogue $\sum_b \frac{n_b}{N} \mathrm{KL}(\bar{y}_b \,\|\, \bar{p}_b)$. It follows the correspondence between log-loss and KL divergence in [8].
- Top-label ECE with 10 bins of confidence $\max(p, 1-p) \in [0.5, 1]$, compared with the accuracy of the thresholded prediction.
- Its worst-bin (MCE) and root-mean-square (RMSCE) [26] variants.
- The binning-free KS calibration error (KSCE) [36].

The two calibration notions differ in an important way. The Brier calibration term compares the predicted probability with the empirical frequency of $Y = 1$, which is a canonical (distribution-level) notion. Top-label ECE compares confidence with the accuracy of the decision and ignores which class the probability refers to. As Section 4.3.6 shows, the two can rank methods differently.

4.2.6. Implementation and Reproducibility

Optimisation. All models are trained by full-batch gradient descent with learning rate 0.5 for 1,200 epochs, without early stopping unless stated otherwise, in double precision. The soft-ECE penalty uses $B = 15$ and $h = 0.08$; the evaluation decomposition uses 10 bins. Experiments Q1.1–Q1.12 were implemented in PyTorch with automatic differentiation. The ablations A–D were implemented in NumPy with the analytic gradients of the same objectives. The NumPy implementation reproduces a value from the PyTorch runs (test CE 0.5854648820627176 for one fixed configuration) to within 10^{-15}, and it reproduces the corresponding rows of Table 4.3 to all reported decimal places.

Seeds. Mechanism-level experiments (Figures 4.2–4.8) use one data seed (1) and one initialisation seed (42). Multi-seed experiments use data seeds $s = 1, \ldots, 30$ with initialisation seed $10000 + s$ and, where a split is needed, split seed $20000 + s$. The single-seed post-hoc experiment uses split seed 123.

Statistical testing. Paired comparisons over the 30 seeds use the Wilcoxon signed-rank test with a Bonferroni correction over the number of comparisons stated with each table. Markers are * for $p_{\text{Bonf}} < 0.05$ and ** for $p_{\text{Bonf}} < 0.01$. Variances are reported descriptively; no test on variances was specified in the protocol.

Compute and code. The largest sweep (360 training runs) took about eight minutes on Google Colab; the ablations took of the order of ten minutes in total. All notebooks and the full-resolution figures are available at the project repository.

4.3. Results

The results are presented in the order in which the experiments build on each other.

1. How calibration and refinement evolve during a single training run (Section 4.3.1).
2. What happens at the end of training as λ varies (Section 4.3.2).
3. Generalisation and variance across 30 seeds (Section 4.3.3).
4. The role of capacity (Section 4.3.4).
5. The comparison with post-hoc calibration and focal loss (Sections 4.3.5 and 4.3.6).
6. The ablations of Section 4.3.7.

The ablations change the interpretation of several earlier results, so the first sections describe what was observed and point forward where the ablations qualify it. The interpretation itself is collected in Section 4.4.

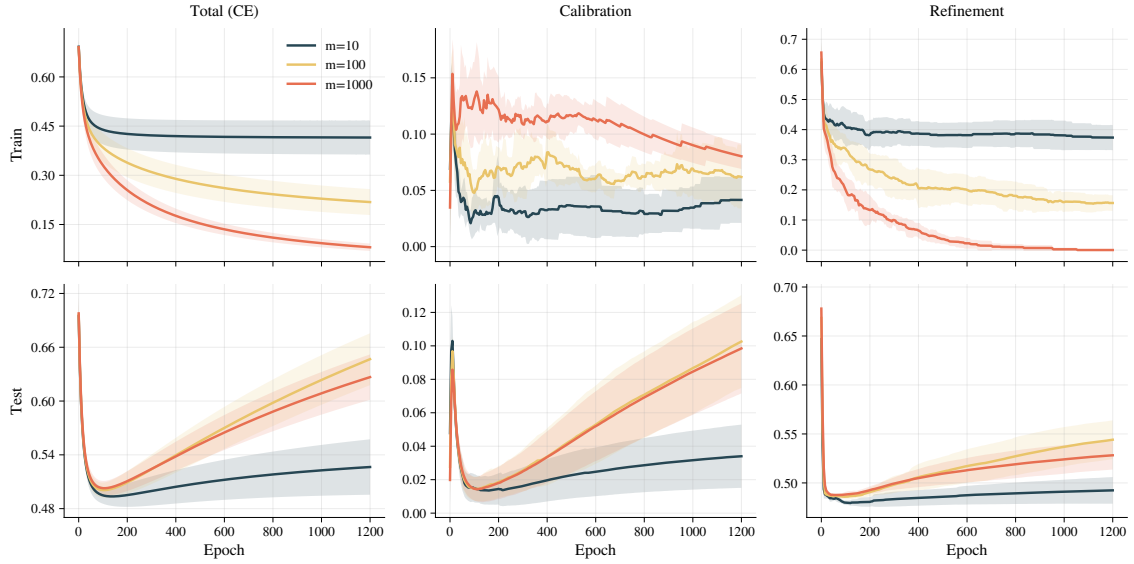
4.3.1. Training Dynamics of the Calibration–Refinement Decomposition

Figure 4.2 decomposes the cross-entropy into its calibration and refinement components over the course of training, on the training set (upper rows) and the test set (lower rows). It shows two configurations: the predictive-only baseline ($\lambda = 1$) and end-to-end training with $\lambda = 0.1$.

Q1.1: do calibration and refinement move in opposite directions? In (4.4) the decomposition holds by construction, because refinement is defined as the residual of the total loss. The substantive question is how the two components behave during training.

Under the baseline, their behaviour on the test set diverges clearly once the model is large enough to fit the training labels. Between initialisation and the final epoch, test refinement at $m = 100$ falls from 0.688 to 0.525, while test calibration rises from 0.014 to 0.138. The values at $m = 1000$ are nearly identical ($0.689 \to 0.519$ and $0.015 \to 0.134$). At $m = 10$ the calibration term also increases, but much less ($0.020 \to 0.053$).

The curves show the structure behind these endpoints. During roughly the first 80 to 100 epochs, both components decrease together. After that, test calibration increases almost linearly, while test refinement stays nearly flat and even rises slightly. On the training set the



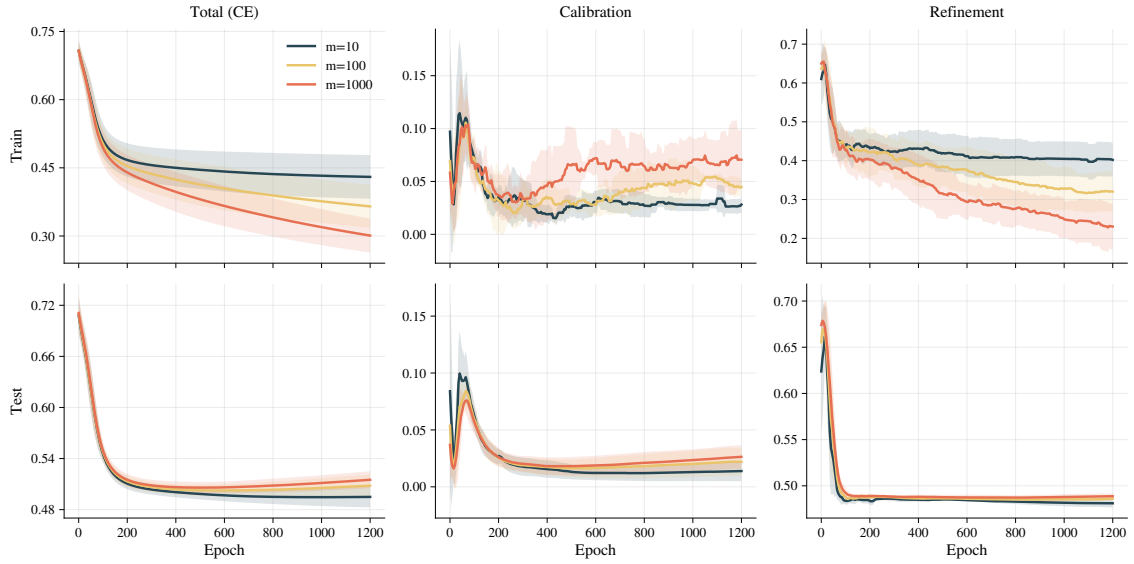


Figure 4.2.: Calibration–refinement decomposition of the cross-entropy over 1,200 epochs, for $m \in \{10, 100, 1000\}$ (5 seeds). Top block: predictive-only baseline ($\lambda = 1$). Bottom block: end-to-end training with $\lambda = 0.1$. Within each block, the upper row is the training set and the lower row the test set. The full grid, covering $\lambda \in \{0, 0.1, 0.5, 1\}$ for both the CE and the Brier base loss is available on GitHub in high resolution.

picture is different: refinement keeps falling and reaches almost zero at $m = 1000$, which reflects the model memorising the training labels.

The two components are therefore not minimised at the same time, which is the pattern Berta et al. [8] describe. A precise summary of what we observe is this: after an initial phase of joint improvement, test calibration degrades while test refinement has essentially saturated. The late increase in test loss is almost entirely a calibration effect.

Q1.2: is there a phase transition, and does the penalty prevent it? For $m \geq 100$ the transition between the two phases is sharp and occurs early, at roughly epoch 80. At $m = 10$ it is present but mild.

With $\lambda = 0.1$ the transition is delayed and much weaker. Test calibration reaches its minimum at around epoch 250 to 400 and ends at 0.036 ($m = 100$) and 0.038 ($m = 1000$) instead of 0.138 and 0.134. Test refinement ends at 0.486–0.487, slightly lower than under the baseline. The penalty therefore postpones and damps the calibration-degradation phase; it does not remove it.

The identity eq. 4.6 gives a second reading of the same curves. At $\lambda = 0.1$ the proper-loss part of the objective moves ten times more slowly than under the baseline. After 1,200 epochs it has progressed about as far as 120 epochs of plain CE, which is close to where the baseline's test loss is lowest. Whether the damping comes from the penalty or from this slower time scale cannot be decided from Figure 4.2; ablation A (Section 4.3.7) addresses exactly this.

Reliability diagrams. The end-to-end calibration objective improves calibration across all three model capacities. As shown in Figure 4.3, the final ECE decreases from 0.062 to 0.043 at $m = 10$, from 0.112 to 0.061 at $m = 100$, and from 0.113 to 0.065 at $m = 1000$ when λ decreases from 1 to 0.1. The improvement is particularly pronounced at the larger capacities, where the reliability diagrams show a smaller discrepancy between confidence and empirical accuracy. These results suggest that incorporating the calibration penalty can improve the calibration of the model's predictions across a range of capacities.

4.3.2. Calibration–Refinement Trade-off Across Weights

This section looks at the final epoch as a function of λ. Figure 4.4 shows the learned functions. Figure 4.5 shows MSE_η, ECE and the Brier score on the 21-point grid, and Figure 4.6 splits the Brier score into its calibration and refinement terms. All three figures use a single seed.

Q1.11: does the penalty move the model towards η? The limiting case $\lambda = 0$ answers the question for the calibration term on its own, and the answer is no. Without the proper loss, the model collapses to a nearly constant prediction around 0.5. At $m = 1000$ the test

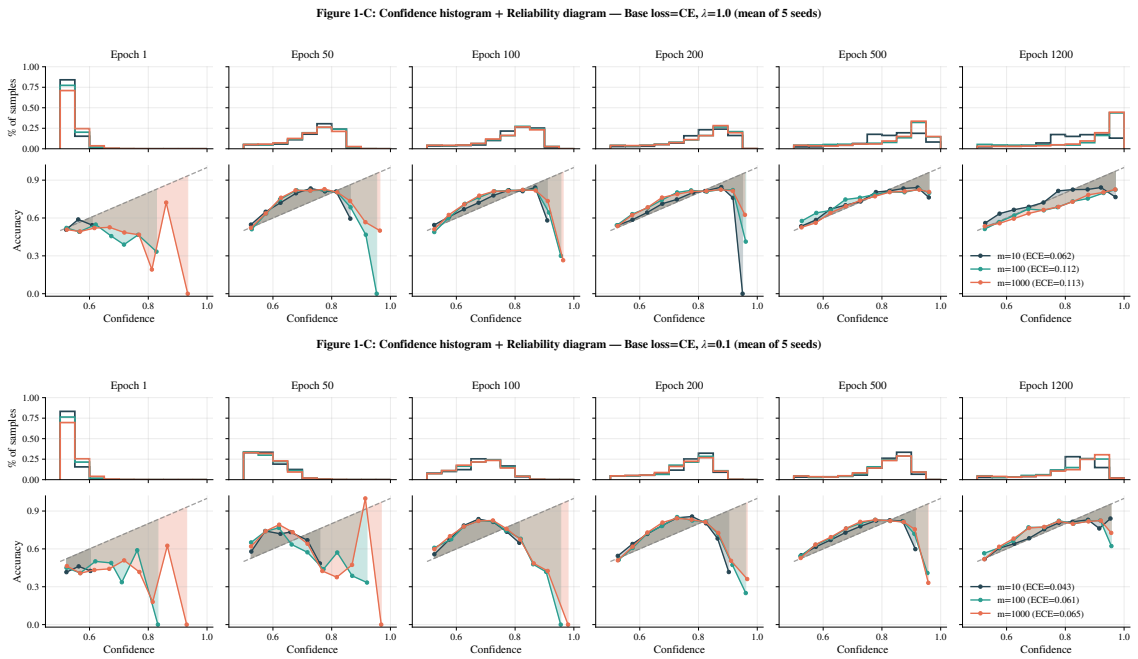


Figure 4.3.: Confidence histograms (upper rows) and reliability diagrams (lower rows) at epochs 1, 50, 100, 200, 500 and 1,200 for the CE base loss (5 seeds). Top block: $\lambda = 1$. Bottom block: $\lambda = 0.1$. Shaded areas show the gap to the diagonal. The legend in the last panel of each block gives the final ECE. The full grid, covering $\lambda \in \{0, 0.1, 0.5, 1\}$ and both base losses is available on GitHub in high resolution.

predictions have mean 0.4995 and standard deviation 0.032, and the CE and Brier families coincide exactly because the base loss no longer enters the objective.

Measured against η, this solution is useless: $\mathrm{MSE}_\eta \approx 0.105$ at every capacity, which is essentially the 0.103 of the constant predictor 0.5. It is nevertheless well calibrated in the canonical sense. Its Brier calibration term is the lowest of all runs at $m = 1000$ (0.0036), and all of its loss sits in refinement (0.243–0.250, close to the 0.25 of a coin flip; Figure 4.6). This is the calibration–refinement trade-off in its purest form.

Its top-label ECE is not particularly low (0.087, 0.078 and 0.059 for the three capacities). In the 30-seed experiment of Section 4.3.3, however, its ECE at $m \geq 100$ is lower than at $\lambda = 0.1$, although its MSE_η is roughly four to five times higher. A calibration metric alone would not have identified this model as degenerate.

For $\lambda > 0$ the picture changes. At every capacity and for both base losses, MSE_η decreases monotonically as λ is lowered from 1 towards the optimum. Figure 4.4 shows what this looks like:

- At $\lambda = 1$, the models with $m \geq 100$ produce narrow spikes towards 0 and 1 (for example near $x \approx -0.7$ and $x \approx -0.6$), where they fit individual noisy training labels. They also

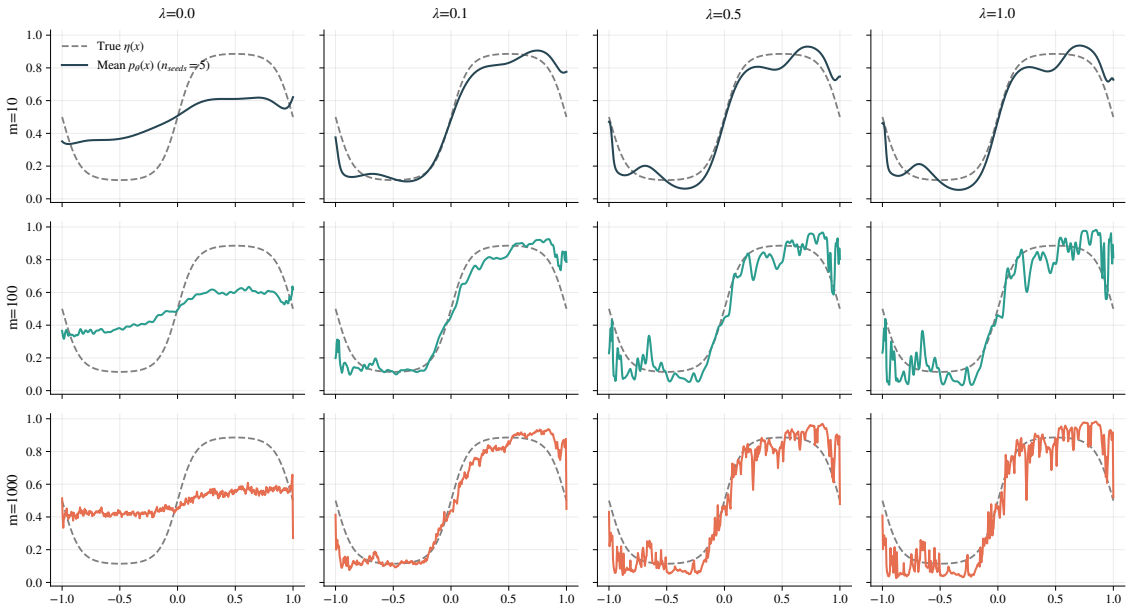


Figure 4.4.: Learned function $p_\theta(x)$ (solid) against the true posterior $\eta(x)$ (dashed) at the final epoch, CE base loss, for $m \in \{10, 100, 1000\}$ (rows) and $\lambda \in \{0, 0.1, 0.5, 1\}$ (columns), single seed. The corresponding figure for the Brier base loss is available on GitHub in high resolution.

produce a plateau at 0.95–1 on the interval where $\eta \approx 0.88$.

- At $\lambda = 0.1$, most spikes disappear and the function follows η over most of the domain.
- Three kinds of deviation remain at $\lambda = 0.1$: a moderate overshoot on the upper plateau, high-frequency ripple at $m = 1000$, and boundary effects at $x = \pm 1$. The boundary effects are most visible for $m = 10$ at $x = -1$, where the prediction approaches 0.9 although $\eta(-1) = 0.5$.

Within the interior of the λ range, then, the penalised objective does move the model towards η and not only towards a better-looking reliability diagram.

Q1.12: is there an optimal λ, and does it depend on m? On the fine grid, the λ that minimises MSE_η is:

- 0.05 for the CE base loss at all three capacities, with $\mathrm{MSE}_\eta = 0.0097, 0.0118$ and 0.0123;
- 0.15, 0.10 and 0.10 for the Brier base loss, with $\mathrm{MSE}_\eta = 0.0091, 0.0112$ and 0.0112.

The optimum is thus interior in both families and nearly independent of capacity, but it de-

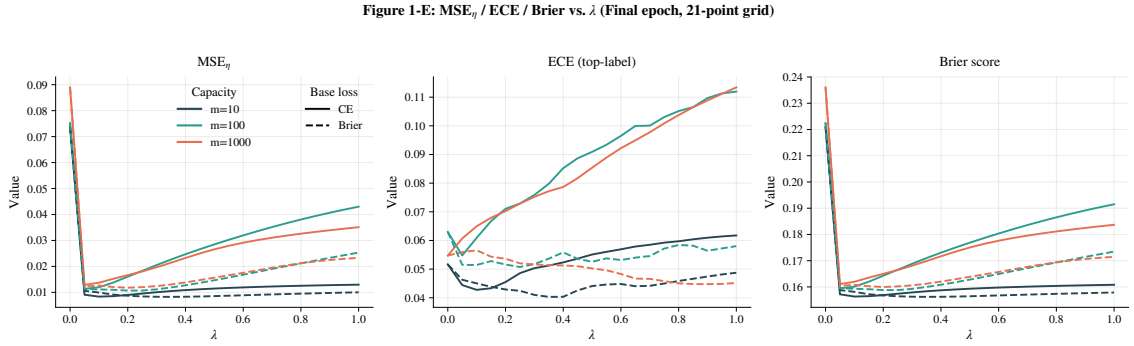


Figure 4.5.: MSE$_\eta$, top-label ECE and Brier score at the final epoch as functions of λ (21-point grid, 5 seeds). Color encodes capacity and line style the base loss.

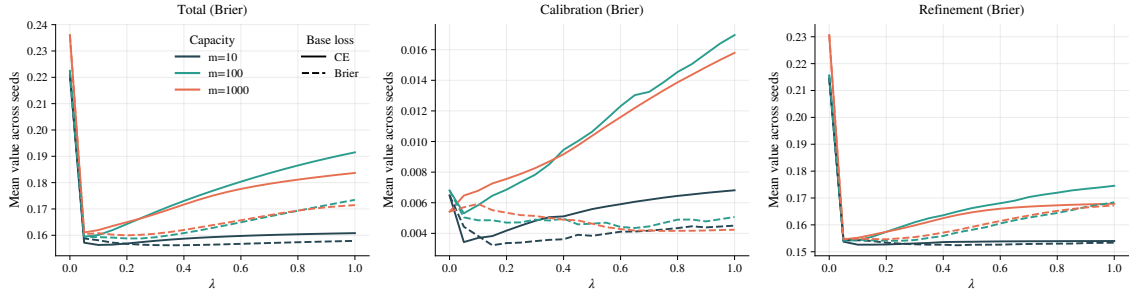


Figure 4.6.: Hard-binned Brier decomposition (total = calibration + refinement) at the final epoch as a function of λ, for the same runs as figure 4.5. Every model is scored with the Brier decomposition, whichever loss it was trained with.

pends on the base loss.

The CE optimum lies at the second-smallest grid point. We cannot exclude that a grid refined near zero would place it lower still; the only thing established is that it lies between 0 and 0.1. The coarse grid $\{0, 0.1, 0.5, 1\}$ used in the multi-seed experiments places the same optimum at 0.1, simply because 0.05 is not in that grid. This is why the experiments below use $\lambda = 0.1$ in some places and $\lambda = 0.05$ in others: the difference reflects grid resolution, not a different optimum.

At the optimum, sharpness is not sacrificed. The Brier refinement term in Figure 4.6 is lowest for $\lambda \in [0.05, 0.1]$ (for example 0.154 against 0.163 at $\lambda = 1$ for CE and $m = 1000$), so both components improve together.

The ECE-optimal and MSE$_\eta$-optimal values of λ do not always agree. For the Brier base loss at $m \geq 100$, ECE is lowest at $\lambda = 0.7$–0.9, while MSE$_\eta$ is lowest at 0.10. A λ chosen to minimise ECE would therefore not select the model closest to η.

All values in this paragraph come from a single seed and are selected with MSE$_\eta$ on the test

set, which is an oracle criterion. The realistic counterpart is ablation C.

Q1.6: Brier score or cross-entropy as base loss? At $\lambda = 1$, the Brier base loss is better than CE on both metrics at every capacity, and the gap grows with capacity:

- at $m = 1000$: MSE_η 0.0195 against 0.0325, ECE 0.056 against 0.127;
- at $m = 10$: MSE_η 0.0109 against 0.0138, ECE 0.062 against 0.071.

The Brier family is also much less sensitive to λ. Its calibration term stays around 0.004–0.007 over nearly the whole range, while the CE calibration term at $m \geq 100$ grows steadily with λ, from about 0.006 to 0.018 (Figure 4.6). At the respective optima, however, the difference largely vanishes (at most 0.0011 in MSE_η).

This agrees with the boundedness argument for the Brier score made in Chapter 3. We add a hypothesis that we have not tested separately. The gradient of the Brier score with respect to the logit, $2(p - y)\, p(1 - p)$, is at most half that of CE and vanishes for confident predictions. It therefore slows the fitting of individual noisy labels, which is a step-size effect of the same kind that ablation A isolates for λ.

These comparisons rest on a single seed. The 30-seed experiments were run for the CE base loss only.

4.3.3. Effect on Generalisation and on Variance Across Seeds

Table 4.3 and Figure 4.7 report the 30-seed results for the CE base loss on the coarse grid.

End-to-end ($\lambda = 0.1$) against the baseline ($\lambda = 1$). With $\lambda = 0.1$ the model is better than the baseline on every metric at every capacity, and the difference is significant in 25 of 27 metric–capacity combinations (paired Wilcoxon test, Bonferroni over three capacities). The exceptions are MCE at $m = 100$ and $m = 1000$. The average calibration error (ECE, RMSCE, KSCE and the Brier calibration term) improves reliably, but the gap in the single worst bin does not.

The size of the effect depends strongly on capacity:

- MSE_η falls from 0.0189 to 0.0101 at $m = 10$, from 0.0482 to 0.0127 at $m = 100$ and from 0.0378 to 0.0129 at $m = 1000$;
- ECE falls from 0.069 to 0.046, from 0.122 to 0.055 and from 0.114 to 0.055;
- the Brier refinement term also falls significantly (for example 0.177 $\rightarrow$ 0.155 at $m = 100$), so the gain in calibration is not paid for with sharpness.

Table 4.2.: Mean and across-seed variance of MSE_η at $m = 1000$ (CE base loss, 30 seeds). Decreasing λ means increasing weight on the calibration term.

λ	mean MSE_η	std	variance
1.0	0.0378	0.0083	7.0×10^{-5}
0.5	0.0289	0.0075	5.6×10^{-5}
0.1	0.0129	0.0052	2.7×10^{-5}
0.0	0.0580	0.0456	2.1×10^{-3}

Generalisation gap. In the single-seed runs of Figure 4.2, the gap between training and test cross-entropy at the final epoch shrinks under the penalty. At $m = 1000$ it goes from approximately 0.58 (training loss about 0.07, test loss 0.65) to approximately 0.26 (about 0.26 against 0.52). At $m = 10$ it goes from approximately 0.22 to approximately 0.15. These values are read from the curves and are approximate.

Q1.10: are bias and variance reduced simultaneously? Table 4.2 shows the mean and the across-seed variance of MSE_η at $m = 1000$ as the weight on calibration increases, that is, as λ decreases.

From $\lambda = 1$ over 0.5 to 0.1, both quantities fall together: the mean by about two thirds, the variance from 7.0×10^{-5} to 2.7×10^{-5}. The same pattern holds at $m = 100$ (standard deviation $0.0128 \to 0.0090 \to 0.0054$) and at $m = 10$ ($0.0078 \to 0.0072 \to 0.0054$).

At $\lambda = 0$ the relationship breaks down completely. The mean rises to 0.058, worse than the baseline, and the variance to 2.1×10^{-3}, about eighty times its value at $\lambda = 0.1$. Once the proper loss no longer anchors the model to the data, which degenerate solution it reaches depends heavily on the seed.

The answer to Q1.10 is therefore yes, over the interior range $\lambda \in [0.1, 1]$ that was tested. The effect is not monotone down to $\lambda = 0$.

Two qualifications apply. First, "bias" and "variance" here are the mean and the across-seed variance of MSE_η. They are not the squared-bias and variance terms of the bias–variance decomposition of the predictor itself, which would require the pointwise mean and variance of $p_\theta(x)$ across seeds; we did not compute these. Second, ablation A shows that the step-size control reproduces this pattern as well (Section 4.3.7).

4.3.4. Effect of Model Capacity

Capacity under predictive-only training. Under the baseline, the error is not monotone in capacity. Table 4.3 gives $\mathrm{MSE}_\eta = 0.0189$, 0.0482 and 0.0378 for $m = 10$, 100 and 1000. ECE (0.069, 0.122, 0.114) and test CE (0.551, 0.657, 0.625) show the same ordering. The worst model is the one at the interpolation threshold $m = n_{\text{train}}$, and the strongly overparameterised

Table 4.3.: Final-epoch test metrics, CE base loss, mean ± standard deviation over 30 seeds. Markers on the $\lambda = 0.1$ rows give the result of a paired Wilcoxon test against the baseline $\lambda = 1$, Bonferroni-corrected over the three capacities (* $p < 0.05$, ** $p < 0.01$, n.s. not significant). All marked differences favor $\lambda = 0.1$.

m	λ	Test CE	Brier	ECE	MSE_η	Cal (Brier)	Ref (Brier)	MCE	RMSCE	KSCE
10	0.0	0.5969 ± 0.0905	0.2032 ± 0.0436	0.0528 ± 0.0269	0.0559 ± 0.0443	0.0060 ± 0.0052	0.1973 ± 0.0447	0.1628 ± 0.1489	0.0621 ± 0.0307	0.0478 ± 0.0236
	0.1	$0.4981 \pm 0.0170^{**}$	$0.1577 \pm 0.0054^{**}$	$0.0459 \pm 0.0179^{**}$	$0.0101 \pm 0.0054^{**}$	$0.0051 \pm 0.0038^{**}$	$0.1525 \pm 0.0029^{**}$	$0.1594 \pm 0.1486^{**}$	$0.0552 \pm 0.0181^{**}$	$0.0337 \pm 0.0160^{*}$
	0.5	0.5293 ± 0.0292	0.1637 ± 0.0071	0.0630 ± 0.0184	0.0163 ± 0.0072	0.0075 ± 0.0036	0.1562 ± 0.0050	0.1991 ± 0.1324	0.0775 ± 0.0193	0.0368 ± 0.0167
	1.0	0.5512 ± 0.0424	0.1663 ± 0.0077	0.0694 ± 0.0195	0.0189 ± 0.0078	0.0090 ± 0.0039	0.1573 ± 0.0053	0.2080 ± 0.1265	0.0870 ± 0.0204	0.0388 ± 0.0164
100	0.0	0.6233 ± 0.0938	0.2163 ± 0.0450	0.0497 ± 0.0177	0.0692 ± 0.0458	0.0048 ± 0.0041	0.2115 ± 0.0471	0.3999 ± 0.2429	0.0624 ± 0.0206	0.0399 ± 0.0201
	0.1	$0.5077 \pm 0.0181^{**}$	$0.1604 \pm 0.0055^{**}$	$0.0545 \pm 0.0212^{**}$	$0.0127 \pm 0.0054^{**}$	$0.0060 \pm 0.0042^{**}$	$0.1545 \pm 0.0026^{**}$	0.1821 ± 0.1359 n.s.	$0.0633 \pm 0.0219^{**}$	$0.0347 \pm 0.0166^{**}$
	0.5	0.5841 ± 0.0268	0.1805 ± 0.0088	0.0917 ± 0.0175	0.0330 ± 0.0090	0.0123 ± 0.0046	0.1682 ± 0.0072	0.1533 ± 0.0433	0.1004 ± 0.0183	0.0496 ± 0.0180
	1.0	0.6573 ± 0.0363	0.1958 ± 0.0125	0.1217 ± 0.0178	0.0482 ± 0.0128	0.0190 ± 0.0054	0.1768 ± 0.0103	0.1930 ± 0.0513	0.1316 ± 0.0187	0.0632 ± 0.0189
1000	0.0	0.6001 ± 0.0939	0.2054 ± 0.0449	0.0506 ± 0.0302	0.0580 ± 0.0456	0.0058 ± 0.0070	0.1995 ± 0.0468	0.5076 ± 0.2773	0.0628 ± 0.0298	0.0367 ± 0.0222
	0.1	$0.5099 \pm 0.0176^{**}$	$0.1607 \pm 0.0053^{**}$	$0.0546 \pm 0.0202^{**}$	$0.0129 \pm 0.0052^{**}$	$0.0061 \pm 0.0042^{**}$	$0.1546 \pm 0.0026^{**}$	0.1590 ± 0.1398 n.s.	$0.0620 \pm 0.0201^{**}$	$0.0355 \pm 0.0178^{**}$
	0.5	0.5724 ± 0.0274	0.1766 ± 0.0074	0.0854 ± 0.0212	0.0289 ± 0.0075	0.0105 ± 0.0047	0.1661 ± 0.0069	0.1361 ± 0.0362	0.0925 ± 0.0207	0.0479 ± 0.0168
	1.0	0.6252 ± 0.0368	0.1855 ± 0.0082	0.1143 ± 0.0184	0.0378 ± 0.0083	0.0162 ± 0.0048	0.1694 ± 0.0076	0.1580 ± 0.0330	0.1198 ± 0.0177	0.0594 ± 0.0173

Figure 1-G: Multi-seed box plots by base loss (30 seeds)

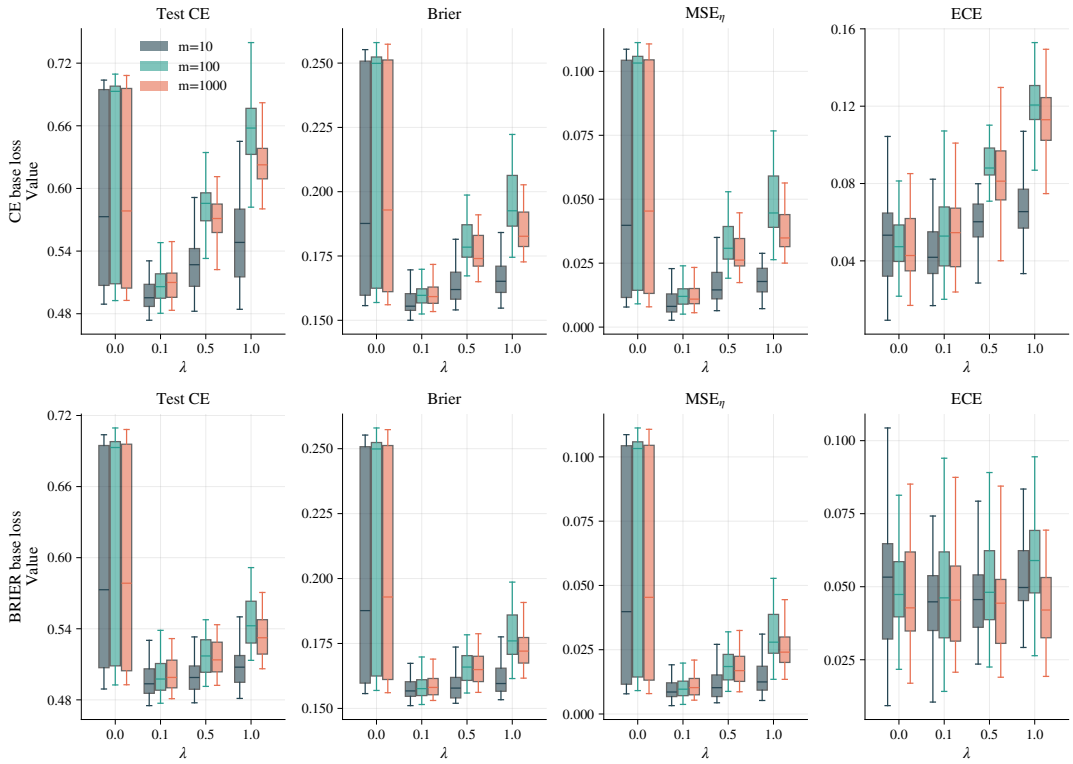


Figure 4.7.: Distribution over 30 seeds of test CE, Brier score, MSE$_\eta$ and ECE for the CE base loss, grouped by λ, with one box per capacity (median, interquartile range, whiskers without outliers).

model partly recovers.

This is consistent with a double-descent-type curve, but three capacities are not enough to establish one. The single-seed post-hoc sweep over the full grid points in the same direction (Figure 4.8, trained on 80 points). There, the raw ECE is 0.03–0.06 for $m \leq 10$, 0.136–0.146 for $m \in [100, 300]$ and 0.131 at $m = 1000$.

Either way, the degradation starts at the threshold itself, not only deep in the over-parameterised regime. Since $m = 100$ and $m = 1000$ can both represent η essentially exactly (Figure 4.1), the degradation is estimation error.

Q1.8: does over-parameterisation help under end-to-end training? No. At $\lambda = 0.1$ the three capacities reach MSE$_\eta = 0.0101$, 0.0127 and 0.0129, and at the fine-grid optimum of the single-seed sweep 0.0097, 0.0118 and 0.0123 (CE). In both cases the smallest model is numerically the best. The larger models are only slightly worse, and the differences between capacities are smaller than one standard deviation across seeds.

The calibration penalty thus removes most of the damage done by over-parameterisation, but it does not turn capacity into an advantage. In this problem there is little for extra capacity to gain, because the approximation error at $m = 10$ is already negligible (Section 4.2.2). A target function that the small model cannot represent would be needed to test whether over-parameterisation can help under end-to-end training.

Q1.9: does the penalty help more when $m \gg n$? In absolute terms, yes. Moving from $\lambda = 1$ to $\lambda = 0.1$ reduces MSE_η by 0.0088 at $m = 10$, by 0.0355 at $m = 100$ and by 0.0249 at $m = 1000$. These are relative reductions of 47%, 74% and 66%. The benefit is largest at the interpolation threshold and remains large in the strongly overparameterised regime, because that is where the baseline does most damage. Ablation A shows that the same capacity dependence appears for the step-size control.

4.3.5. Comparison with Post-hoc Calibration

Q1.3 and Q1.5: post-hoc methods across capacity. Figure 4.8 and Table 4.4 compare four post-hoc methods, fitted on 20 held-out points, with end-to-end training at $\lambda = 0.1$ over the full capacity grid (single seed). The ranking depends mainly on how miscalibrated the raw model already is, and that in turn depends on capacity.

For $m \leq 10$ the raw CE model is already reasonably calibrated (ECE 0.027 at $m = 5$, 0.059 at $m = 10$), and *every* post-hoc method makes it worse. Temperature scaling, for example, doubles the ECE at $m = 5$ (0.027 $\rightarrow$ 0.057). With little real miscalibration to correct and only 20 noisy calibration points, the fitted maps mostly fit noise.

From $m = 100$ onwards, the raw model is clearly miscalibrated (ECE 0.131–0.146), and the methods separate:

- *Temperature scaling* is the best method at every capacity in this range, including end-to-end training. It reaches 0.035 at $m = 100$, but its ECE grows steadily with capacity to 0.075 at $m = 1000$.
- *Platt scaling* is somewhat worse (0.075–0.090).
- *Isotonic regression* leaves the ECE essentially unchanged (0.129–0.136).
- *Histogram binning* is often worse than no calibration at all (up to 0.214 at $m = 300$).
- *End-to-end training* stays in a narrow band across the whole grid: 0.049–0.083 for the CE family and 0.061–0.069 for the Brier family.

For the Brier base loss, the raw models are already fairly well calibrated at every capacity (ECE 0.040–0.070). Temperature scaling helps only for $m \in \{100, 120, 200, 300\}$, and at $m = 1000$ the raw Brier model (0.056) is better than both temperature scaling (0.076) and end-to-end

Table 4.4.: Test ECE before and after post-hoc calibration, and for end-to-end training ($\lambda = 0.1$), by base loss and capacity (single seed, single 80/20 split).

Loss	m	Raw	Temp. scaling	Platt	Isotonic	Histogram	End-to-end
CE	5	0.0271	0.0566	0.0602	0.1025	0.2288	0.0487
	10	0.0585	0.0656	0.0646	0.0908	0.2417	0.0612
	100	0.1358	0.0345	0.0779	0.1331	0.1353	0.0822
	120	0.1403	0.0356	0.0758	0.1341	0.1322	0.0830
	200	0.1464	0.0561	0.0749	0.1363	0.1694	0.0807
	300	0.1435	0.0591	0.0745	0.1355	0.2137	0.0834
	1000	0.1311	0.0746	0.0901	0.1294	0.1592	0.0816
Brier	5	0.0401	0.0611	0.0624	0.1049	0.2109	0.0690
	10	0.0398	0.0749	0.0712	0.1111	0.2072	0.0636
	100	0.0699	0.0481	0.0794	0.1418	0.2382	0.0663
	120	0.0695	0.0484	0.0790	0.1450	0.2353	0.0654
	200	0.0665	0.0595	0.0768	0.1340	0.2392	0.0618
	300	0.0654	0.0627	0.0789	0.1381	0.2551	0.0614
	1000	0.0564	0.0759	0.0919	0.1506	0.1777	0.0628

training (0.063).

These single-split numbers have to be read against their variability. Repeating the 80/20 split ten times at $m = 1000$ (CE) gives the following mean $\pm$ standard deviation of ECE:

- raw model: 0.130 ± 0.006;
- temperature scaling: 0.073 ± 0.041 (range 0.032–0.134);
- Platt scaling: 0.099 ± 0.036;
- isotonic regression: 0.133 ± 0.029;
- histogram binning: 0.205 ± 0.048.

The advantage of temperature scaling over end-to-end training at a single split (0.075 against 0.082) is therefore well within the split-to-split variation of temperature scaling itself. The failure of isotonic regression and histogram binning, on the other hand, is systematic at this calibration-set size.

In summary, post-hoc calibration does not degrade in a simple monotone way. Two effects combine as capacity grows. The raw model becomes badly miscalibrated from the interpolation threshold onwards. The best global correction (temperature scaling) becomes less effective beyond the threshold and much less stable. That is consistent with the hypothesis that a single global map cannot fully correct miscalibration that varies across the input space. It is not evidence that end-to-end training is better calibrated than temperature scaling: in this comparison it is not.

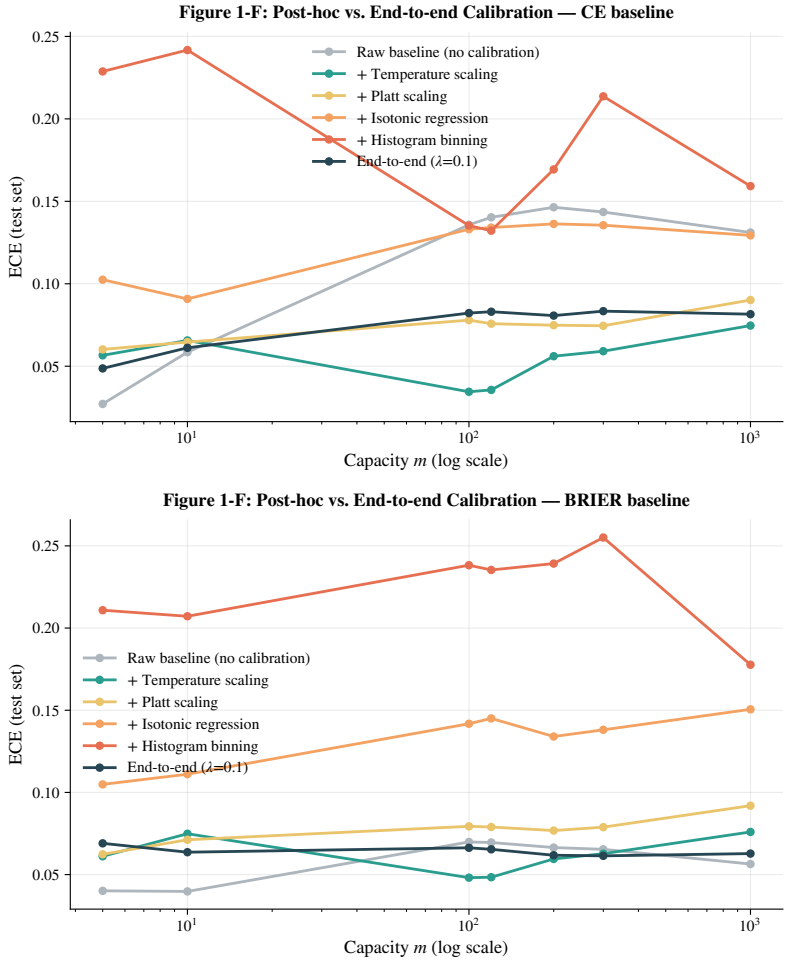


Figure 4.8.: Test ECE of the raw baseline, the four post-hoc methods and end-to-end training ($\lambda = 0.1$) across the full capacity grid, for the CE (top) and Brier (bottom) base losses (single seed, single 80/20 split). The baseline is trained on 80 points and the maps are fitted on the remaining 20; the end-to-end model uses all 100 points.

Q1.4: estimating refinement with a recalibration map. A post-hoc map can also be used to estimate the refinement of a model, since the loss after optimal recalibration equals the refinement term [8]. Figure 4.9 and Table 4.5 compare two such estimators against the ground-truth refinement of Section 4.2.5:

- a *TS-refinement* estimator, which uses temperature scaling as the recalibration class, as proposed by Berta et al.;
- an *isotonic-refinement* estimator.

Both are fitted and evaluated with the Brier score, on 20 held-out points, for CE baseline models over 30 seeds.

Figure 1-H: Refinement estimator bias-variance comparison

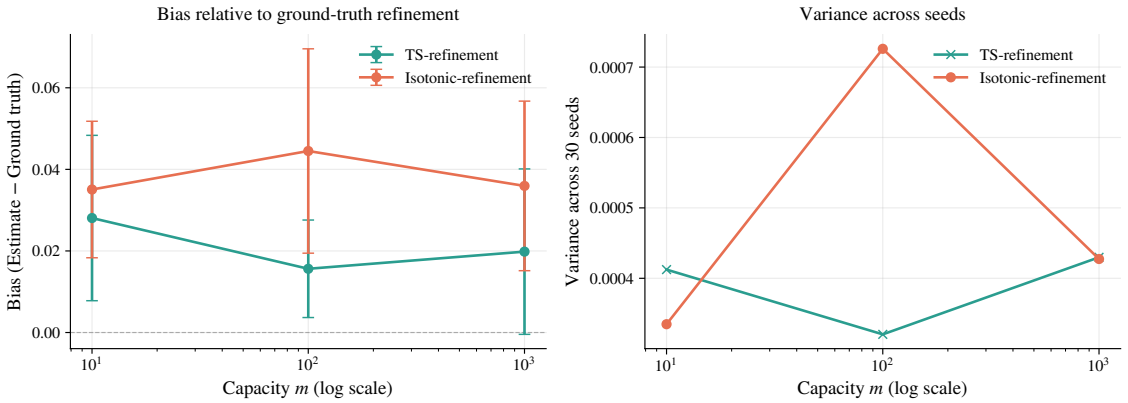


Figure 4.9.: Bias (left; mean ± standard deviation over 30 seeds) and across-seed variance (right) of TS-based and isotonic refinement estimators relative to the ground-truth refinement, for CE baseline models.

Table 4.5.: Refinement estimators relative to the ground-truth refinement (30 seeds, CE baseline, Brier-based estimators, 20 calibration points). Bias is mean ± standard deviation of (estimate − ground truth); variance is the across-seed variance of the estimate. The last column tests $|\text{bias}_{\text{TS}}|$ against $|\text{bias}_{\text{iso}}|$ (paired Wilcoxon, Bonferroni over three capacities).

	TS-refinement		Isotonic refinement		
m	bias	variance	bias	variance	|bias| test
10	$+0.0281 \pm 0.0203$	4.1×10^{-4}	$+0.0351 \pm 0.0167$	3.3×10^{-4}	n.s. ($p = 0.40$)
100	$+0.0156 \pm 0.0120$	3.2×10^{-4}	$+0.0445 \pm 0.0251$	7.3×10^{-4}	** ($p < 0.001$)
1000	$+0.0198 \pm 0.0203$	4.3×10^{-4}	$+0.0359 \pm 0.0208$	4.3×10^{-4}	* ($p = 0.017$)

Both estimators are positively biased at every capacity; they overestimate refinement. This is expected for two reasons. A restricted class such as temperature scaling cannot reach the ideal recalibration map, and a map fitted on 20 noisy points is further from it still.

TS-refinement has the smaller mean bias everywhere. The difference is significant at $m = 100$ (0.016 against 0.045, $p_{\text{Bonf}} < 0.01$) and at $m = 1000$ (0.020 against 0.036, $p_{\text{Bonf}} = 0.017$), but not at $m = 10$ (0.028 against 0.035).

The variance ranking does change with capacity. Isotonic regression is the more stable estimator at $m = 10$ (3.3×10^{-4} against 4.1×10^{-4}). TS-refinement is more than twice as stable at $m = 100$. The two are equal at $m = 1000$.

With small calibration sets, TS-refinement is therefore the safer default once the model is at or beyond the interpolation threshold. This supports its use in the refine-then-calibrate procedure of Berta et al. The variance comparison is descriptive only.

4.3.6. Comparison with Focal Loss

Figure 4.10 and table 4.6 compare focal loss at $\gamma \in \{0, 1, 2, 5\}$ with end-to-end training at $\lambda = 0.05$, the MSE_η optimum of the fine grid for CE (30 seeds). As a consistency check, focal loss at $\gamma = 0$ reproduces the $\lambda = 1$ rows of table 4.3 exactly.

Q1.7: does focal loss match explicit end-to-end calibration? The answer differs between the two metrics.

On MSE_η, end-to-end training is better than every γ at every capacity, and every difference is significant. Compared with the best γ, it is about 2.0, 3.3 and 2.3 times closer to η at $m = 10$, 100 and 1000 (0.0097 against 0.0189, 0.0111 against 0.0365, 0.0117 against 0.0268). It is also best on test CE and the Brier score at every capacity.

On top-label ECE, the ranking depends on capacity. At $m = 10$, end-to-end training is clearly best (0.045 against 0.069 for the best focal setting, $\gamma = 0$). At $m = 100$ and $m = 1000$, focal loss with $\gamma = 2$ is significantly better than end-to-end training (0.035 against 0.051, and 0.032 against 0.052).

The best γ also shifts with capacity. At $m = 10$, any $\gamma > 0$ makes both metrics worse, monotonically in γ. At $m \geq 100$, $\gamma = 2$ is best on both metrics and $\gamma = 5$ overshoots, ending worse than plain CE on ECE.

The gap on MSE_η therefore does not narrow with capacity; it is largest at the interpolation threshold. The ranking on ECE reverses. This is the clearest case in the chapter of the two metrics disagreeing: focal loss with $\gamma = 2$ aligns confidence with accuracy on average better than any other method, yet its predicted probabilities are two to three times further from η than those of end-to-end training.

Top-label ECE only asks whether confidence matches the accuracy of the decision. It is blind to how the probability mass is distributed within the confidence range, and to whether the probability of the class is correct. Two cautions apply to this comparison. The λ of the end-to-end model was selected for MSE_η and not for ECE. The best γ was selected after the fact for each capacity, so both methods are compared at an oracle choice of their hyperparameter.

4.3.7. Ablations and Sensitivity Analysis

The identity (4.6) shows that every result obtained so far confounds two effects of lowering λ: a stronger calibration term and a smaller step size on the proper loss. The four ablations below separate these effects and test weight schemes that do not rely on an oracle. All use the CE base loss and 30 seeds. Significance markers compare the stated pair of methods. Because an end-to-end model can be significantly *worse* than its control, we use $^*/^{**}$ for a significant improvement and $^\dagger/^{\dagger\dagger}$ for a significant deterioration, at $p_{\text{Bonf}} < 0.05$ and < 0.01 respectively.

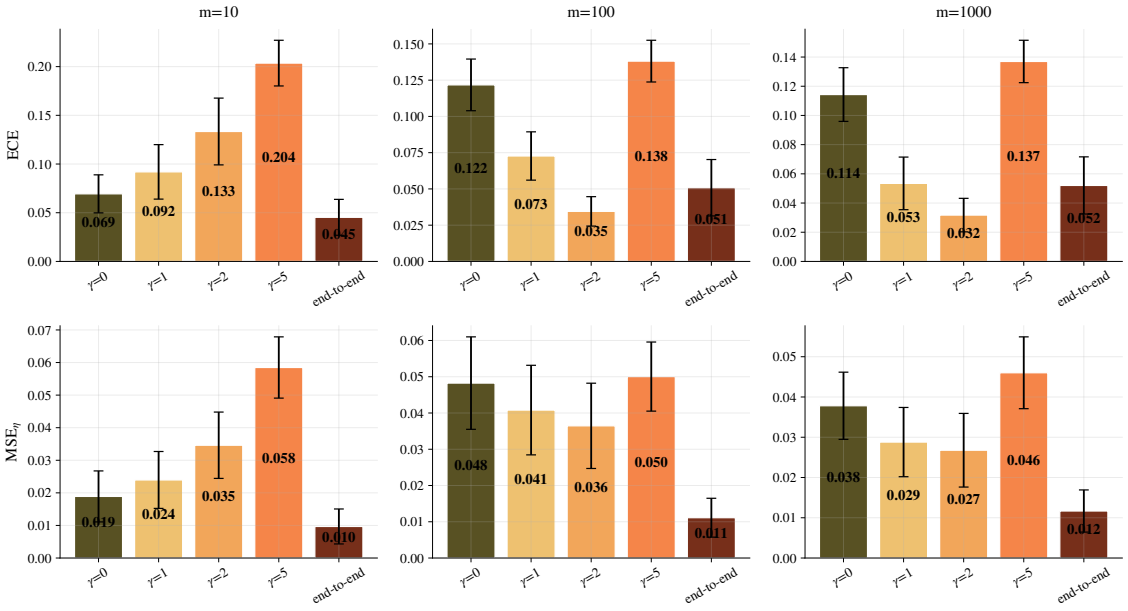


Figure 4.10.: ECE (top) and MSE_η (bottom) for focal loss with $\gamma \in \{0, 1, 2, 5\}$ and for end-to-end training ($\lambda = 0.05$, CE base loss), mean $\pm$ standard deviation over 30 seeds.

A: step-size-matched control. Figure 4.11 and Table 4.7 compare end-to-end training at $\lambda \in \{0.05, 0.1, 0.5\}$ with plain CE trained at the learning rate 0.5λ. By (4.6), the two differ only in the presence of the penalty. The results fall into three regimes.

- **At $\lambda = 0.1$, the penalty adds nothing measurable to MSE_η or calibration.** End-to-end training and its control reach the same MSE_η at every capacity (0.0101 against 0.0100, 0.0127 against 0.0127, 0.0129 against 0.0129). ECE and the Brier calibration term do not differ significantly; at $m \geq 100$ the control is even numerically better on ECE (0.048 against 0.055). The only significant effect of the penalty is a very small reduction of the refinement term at $m \geq 100$ (about 0.001). The step-size effect alone accounts for essentially all of the improvement over the baseline that Section 4.3.3 attributed to $\lambda = 0.1$ (between 99.8% and 101% of the MSE_η gain). The reduction in across-seed variance is reproduced as well: the control has standard deviation 0.0049 against 0.0052 at $m = 1000$.

- **At $\lambda = 0.5$, the penalty is harmful.** At $m \geq 100$ it makes MSE_η, ECE and the calibration term significantly *worse* than the control (for example 0.0289 against 0.0277 in MSE_η and 0.085 against 0.080 in ECE at $m = 1000$).

- **At $\lambda = 0.05$, the penalty contributes a genuine improvement.** End-to-end training is significantly better than its control on MSE_η at all three capacities (0.0097 against 0.0131, 0.0111 against 0.0146, 0.0117 against 0.0155) and on the Brier calibration term. ECE improves as well, significantly only at $m = 10$ after correction; at $m \geq 100$ the

Table 4.6.: Focal loss and end-to-end training ($\lambda = 0.05$), mean $\pm$ standard deviation over 30 seeds. The best mean per capacity and column is in bold. Paired Wilcoxon tests of end-to-end training against each γ (Bonferroni over 12 comparisons per metric) are significant at $p < 0.01$ for ECE and MSE_η in all cases except ECE at $m = 1000$, $\gamma = 1$ (n.s.).

m	Method	Test CE	Brier	ECE	MSE_η
10	focal, $\gamma = 0$	0.5512 ± 0.0424	0.1663 ± 0.0077	0.0694 ± 0.0195	0.0189 ± 0.0078
	focal, $\gamma = 1$	0.5341 ± 0.0247	0.1711 ± 0.0086	0.0919 ± 0.0280	0.0240 ± 0.0087
	focal, $\gamma = 2$	0.5524 ± 0.0238	0.1816 ± 0.0101	0.1334 ± 0.0343	0.0346 ± 0.0102
	focal, $\gamma = 5$	0.6019 ± 0.0199	0.2053 ± 0.0094	0.2036 ± 0.0234	0.0585 ± 0.0094
	end-to-end	$\mathbf{0.4963} \pm 0.0173$	$\mathbf{0.1574} \pm 0.0054$	$\mathbf{0.0453} \pm 0.0184$	$\mathbf{0.0097} \pm 0.0054$
100	focal, $\gamma = 0$	0.6573 ± 0.0363	0.1958 ± 0.0125	0.1217 ± 0.0178	0.0482 ± 0.0128
	focal, $\gamma = 1$	0.5734 ± 0.0262	0.1882 ± 0.0121	0.0727 ± 0.0167	0.0408 ± 0.0124
	focal, $\gamma = 2$	0.5513 ± 0.0258	0.1837 ± 0.0116	$\mathbf{0.0345} \pm 0.0101$	0.0365 ± 0.0118
	focal, $\gamma = 5$	0.5837 ± 0.0201	0.1970 ± 0.0094	0.1381 ± 0.0144	0.0500 ± 0.0095
	end-to-end	$\mathbf{0.5014} \pm 0.0172$	$\mathbf{0.1589} \pm 0.0054$	0.0508 ± 0.0194	$\mathbf{0.0111} \pm 0.0054$
1000	focal, $\gamma = 0$	0.6252 ± 0.0368	0.1855 ± 0.0082	0.1143 ± 0.0184	0.0378 ± 0.0083
	focal, $\gamma = 1$	0.5430 ± 0.0196	0.1763 ± 0.0085	0.0535 ± 0.0180	0.0288 ± 0.0086
	focal, $\gamma = 2$	0.5299 ± 0.0206	0.1741 ± 0.0090	$\mathbf{0.0317} \pm 0.0115$	0.0268 ± 0.0091
	focal, $\gamma = 5$	0.5754 ± 0.0189	0.1930 ± 0.0088	0.1370 ± 0.0145	0.0460 ± 0.0089
	end-to-end	$\mathbf{0.5039} \pm 0.0169$	$\mathbf{0.1595} \pm 0.0053$	0.0521 ± 0.0196	$\mathbf{0.0117} \pm 0.0052$

improvement is numerical (0.051 against 0.069). The step-size effect still explains most of the gain over the baseline (63%, 91% and 86% for MSE_η at $m = 10$, 100 and 1000), so the penalty's own share is roughly 10–40%.

The control at learning rate 0.025 underfits: it is worse than the control at 0.05. The penalty appears to compensate for this. As a result, end-to-end training at $\lambda = 0.05$ is numerically better at $m \geq 100$ than the best of the three controls (0.0111 against 0.0127 and 0.0117 against 0.0129). This last comparison is across different pairs and was not tested.

B: additive penalty at a fixed learning rate. Figure 4.12 and Table 4.8 add the penalty to plain CE without changing the learning rate. The result is unambiguous: at the full step size, the penalty never helps.

- At $m = 10$, no value of β changes any metric significantly.
- At $m = 100$ and $m = 1000$, MSE_η, ECE and the calibration term get worse monotonically as β increases, and significantly so for every $\beta \geq 0.5$. At $m = 1000$, $\beta = 19$ raises MSE_η from 0.0378 to 0.0474, ECE from 0.114 to 0.142 and the calibration term from 0.0162 to 0.0232.

The same $\beta = 19$ at learning rate 0.025, which is end-to-end training at $\lambda = 0.05$, gave the best results of ablation A. Whether the penalty helps or hurts therefore depends on how fast the

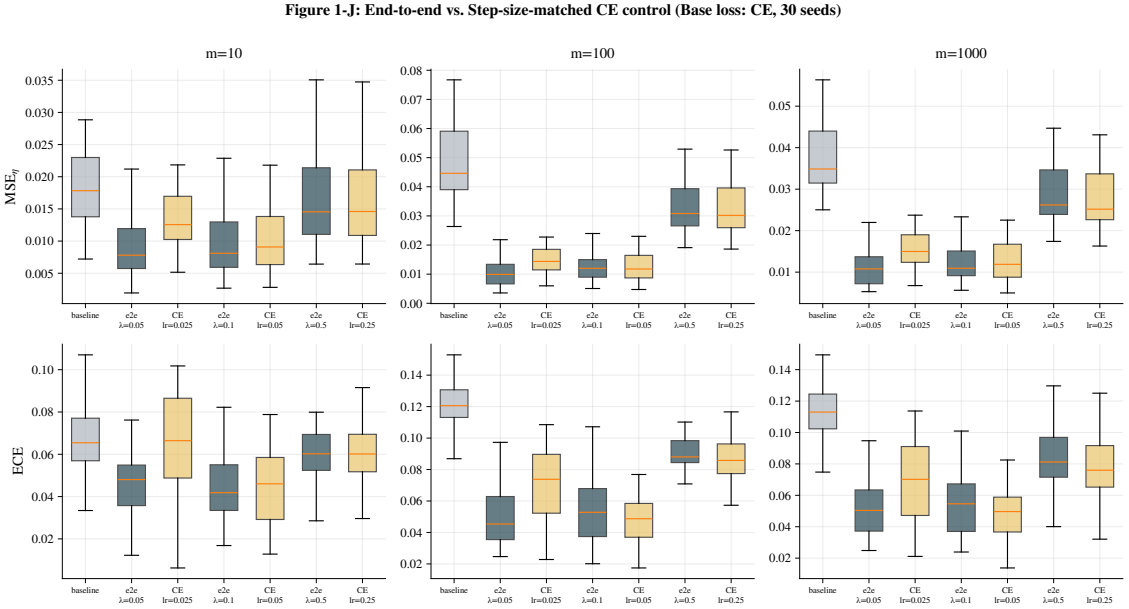


Figure 4.11.: Ablation A: MSE$_\eta$ (top) and ECE (bottom) for the baseline, end-to-end training at $\lambda \in \{0.05, 0.1, 0.5\}$ (dark) and plain CE at the matched learning rate 0.5λ (light), over 30 seeds.

proper loss is being fitted, not on its weight alone.

We offer a hypothesis for this dependence; it is consistent with ablations A and B but was not tested directly. The penalty is computed on the *training* predictions. When the proper loss is fitted quickly, those predictions soon approach the training labels. In the bins close to 0 and 1, the label mean $\bar{y}_b$ then lies further from 0.5 than the prediction mean $\bar{p}_b$, so the gradient of (4.5) pushes predictions further towards the extremes. The penalty then rewards training-set calibration of a model that is overfitting, and this increases its overconfidence on the test set. When the proper loss is fitted slowly, the training-set bin gaps still resemble the population gaps, and the penalty corrects genuine miscalibration.

C: validation-based selection of λ against validation early stopping. All "best" values of λ so far were read off the ground-truth MSE$_\eta$. Ablation C replaces this oracle with a 20-point validation set and compares the result with plain CE stopped early on the same validation set (Figure 4.13, Table 4.9).

Selecting λ by validation Brier score works reasonably well. The selected value is $\lambda = 0.05$ in 15, 19 and 19 of the 30 seeds for $m = 10$, 100 and 1000, and $\lambda = 1$ in 6, 1 and 3 seeds. Compared with the oracle choice $\lambda = 0.1$ on the same 80 training points, the selection costs about 0.002–0.003 in MSE$_\eta$.

Against validation early stopping, end-to-end training with a selected λ reaches a lower MSE$_\eta$

Table 4.7.: Ablation A: end-to-end training against plain CE at the matched learning rate 0.5λ (mean ± standard deviation, 30 seeds). Markers on end-to-end rows compare with the control directly below (paired Wilcoxon, Bonferroni over nine comparisons per metric). $^{*}/^{**}$: significantly better; $^{\dagger}/^{\dagger\dagger}$: significantly worse; unmarked: not significant.

m	Condition	MSE_η	ECE	Cal (Brier)	Ref (Brier)
10	baseline (lr 0.5)	0.0189 ± 0.0078	0.0694 ± 0.0195	0.0090 ± 0.0039	0.1573 ± 0.0053
	end-to-end, $\lambda = 0.05$	$0.0097 \pm 0.0054^{*}$	$0.0453 \pm 0.0184^{*}$	$0.0050 \pm 0.0037^{*}$	0.1524 ± 0.0031
	CE, lr 0.025	0.0131 ± 0.0045	0.0644 ± 0.0263	0.0083 ± 0.0040	0.1523 ± 0.0019
	end-to-end, $\lambda = 0.1$	0.0101 ± 0.0054	0.0459 ± 0.0179	0.0051 ± 0.0038	0.1525 ± 0.0029
	CE, lr 0.05	0.0100 ± 0.0047	0.0447 ± 0.0171	0.0052 ± 0.0035	0.1523 ± 0.0025
	end-to-end, $\lambda = 0.5$	0.0163 ± 0.0072	0.0630 ± 0.0184	0.0075 ± 0.0036	0.1562 ± 0.0050
	CE, lr 0.25	0.0161 ± 0.0071	0.0623 ± 0.0178	0.0074 ± 0.0036	0.1561 ± 0.0049
100	baseline (lr 0.5)	0.0482 ± 0.0128	0.1217 ± 0.0178	0.0190 ± 0.0054	0.1768 ± 0.0103
	end-to-end, $\lambda = 0.05$	$0.0111 \pm 0.0054^{**}$	0.0508 ± 0.0194	$0.0057 \pm 0.0041^{*}$	$0.1532 \pm 0.0021^{**}$
	CE, lr 0.025	0.0146 ± 0.0048	0.0685 ± 0.0242	0.0083 ± 0.0040	0.1539 ± 0.0020
	end-to-end, $\lambda = 0.1$	0.0127 ± 0.0054	0.0545 ± 0.0212	0.0060 ± 0.0042	$0.1545 \pm 0.0026^{**}$
	CE, lr 0.05	0.0127 ± 0.0048	0.0484 ± 0.0158	0.0051 ± 0.0033	0.1551 ± 0.0027
	end-to-end, $\lambda = 0.5$	$0.0330 \pm 0.0090^{\dagger\dagger}$	$0.0917 \pm 0.0175^{\dagger\dagger}$	$0.0123 \pm 0.0046^{\dagger\dagger}$	$0.1682 \pm 0.0072^{**}$
	CE, lr 0.25	0.0326 ± 0.0090	0.0877 ± 0.0178	0.0115 ± 0.0046	0.1686 ± 0.0075
1000	baseline (lr 0.5)	0.0378 ± 0.0083	0.1143 ± 0.0184	0.0162 ± 0.0048	0.1694 ± 0.0076
	end-to-end, $\lambda = 0.05$	$0.0117 \pm 0.0052^{**}$	0.0521 ± 0.0196	$0.0058 \pm 0.0043^{*}$	$0.1538 \pm 0.0018^{**}$
	CE, lr 0.025	0.0155 ± 0.0049	0.0683 ± 0.0245	0.0084 ± 0.0040	0.1546 ± 0.0020
	end-to-end, $\lambda = 0.1$	0.0129 ± 0.0052	0.0546 ± 0.0202	0.0061 ± 0.0042	$0.1546 \pm 0.0026^{**}$
	CE, lr 0.05	0.0129 ± 0.0049	0.0473 ± 0.0161	0.0050 ± 0.0033	0.1556 ± 0.0028
	end-to-end, $\lambda = 0.5$	$0.0289 \pm 0.0075^{\dagger\dagger}$	$0.0854 \pm 0.0212^{\dagger\dagger}$	$0.0105 \pm 0.0047^{\dagger\dagger}$	$0.1661 \pm 0.0069^{\dagger\dagger}$
	CE, lr 0.25	0.0277 ± 0.0072	0.0795 ± 0.0223	0.0097 ± 0.0047	0.1657 ± 0.0066

at all capacities:

- 0.0142 against 0.0210 at $m = 10$;
- 0.0169 against 0.0232 at $m = 100$;
- 0.0185 against 0.0250 at $m = 1000$.

The difference is significant at $m = 10$ and $m = 100$ ($p_{\mathrm{Bonf}} < 0.05$) but not at $m = 1000$ ($p_{\mathrm{Bonf}} = 0.07$). The same pattern holds when both pipelines are refitted on all 100 points. The ECE differences (0.059 against 0.076, and similar values at the other capacities) are not significant at any capacity.

This advantage has at least two possible sources. Ablation A offers one: the selected λ is mostly 0.05, the one value at which the penalty contributes on its own. The second concerns how noisy each selection rule is. Early stopping selects one of 1,201 checkpoints on 20 points, and the chosen epoch varies enormously between seeds (median 102–137 epochs, standard deviation 226–439, range from a few epochs to 1,200). Selecting one of seven values of λ may simply be less noisy. The experiment cannot separate the two explanations; selecting the learning rate

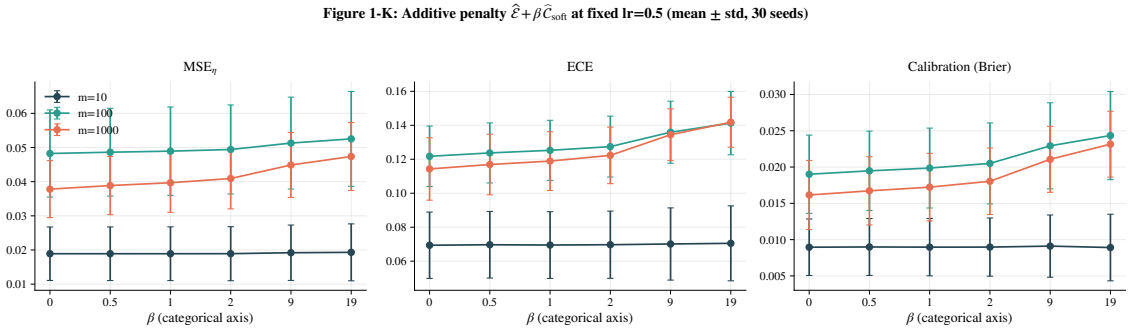


Figure 4.12.: Ablation B: MSE$_\eta$, ECE and Brier calibration term for the additive objective $\widetilde{E} + \beta\,\widetilde{C}_{\text{soft}}$ at the fixed learning rate 0.5 (mean ± standard deviation, 30 seeds). $\beta = 9$ and $\beta = 19$ correspond to $\lambda = 0.1$ and $\lambda = 0.05$ at a correspondingly reduced learning rate.

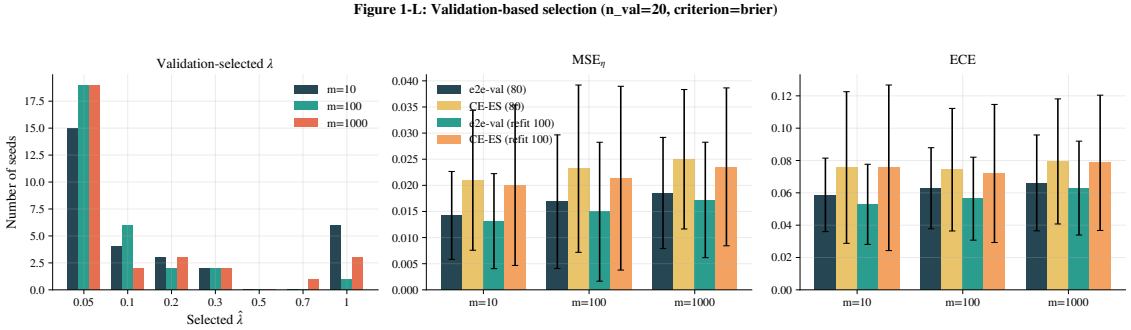


Figure 4.13.: Ablation C: distribution of the validation-selected λ over 30 seeds (left), and test MSE$_\eta$ (middle) and ECE (right) for end-to-end training with validation-selected λ and for CE with validation early stopping, trained on 80 points or refitted on all 100 (mean ± standard deviation).

of plain CE from the same seven-point grid would.

D: learned loss weights. Uncertainty weighting (4.7) fails in this setting (Table 4.10, Figure 4.14). It is significantly worse than both the baseline and end-to-end training at $\lambda = 0.1$, on MSE$_\eta$ and on ECE, at every capacity. At $m = 100$, for example, it reaches MSE$_\eta = 0.103$ against 0.048 for the baseline, and a test CE of 1.69. The learned effective weight $\lambda_{\text{eff}} = w_E/(w_C + w_E)$ fluctuates between roughly 0.01 and 0.2 throughout training rather than settling, and the clipping bound on the log-variances is reached in 70 of 90 runs.

The effective weights $1/(2L_i)$ explain the failure. As the model fits the training data, the training CE falls, so the weight on the proper loss *increases*. The effective step size on $\widetilde{E}$ therefore grows exactly when the model begins to overfit, which is the opposite of the mechanism that helped in ablation A. At the same time, the small training calibration error gives the penalty a large weight in a fast-fitting regime, which is where ablation B shows it to be harmful.

Table 4.8.: Ablation B: additive penalty at learning rate 0.5 (mean ± standard deviation, 30 seeds). $\beta = 0$ is the CE baseline. Markers compare each $\beta > 0$ with $\beta = 0$ (paired Wilcoxon, Bonferroni over 15 comparisons per metric); ††: significantly worse at $p < 0.01$; unmarked: not significant.

m	β	MSE_η	ECE	Cal (Brier)	Ref (Brier)
10	0	0.0189 ± 0.0078	0.0694 ± 0.0195	0.0090 ± 0.0039	0.1573 ± 0.0053
	0.5	0.0189 ± 0.0078	0.0697 ± 0.0196	0.0090 ± 0.0039	0.1573 ± 0.0053
	1	0.0189 ± 0.0079	0.0695 ± 0.0196	0.0090 ± 0.0040	0.1573 ± 0.0053
	2	0.0189 ± 0.0079	0.0697 ± 0.0198	0.0090 ± 0.0040	0.1573 ± 0.0054
	9	0.0192 ± 0.0081	0.0701 ± 0.0213	0.0091 ± 0.0043	0.1575 ± 0.0055
	19	0.0193 ± 0.0083	0.0705 ± 0.0221	0.0089 ± 0.0046	0.1578 ± 0.0055
100	0	0.0482 ± 0.0128	0.1217 ± 0.0178	0.0190 ± 0.0054	0.1768 ± 0.0103
	0.5	$0.0486 \pm 0.0129^{\dagger\dagger}$	$0.1237 \pm 0.0177^{\dagger\dagger}$	$0.0195 \pm 0.0055^{\dagger\dagger}$	0.1767 ± 0.0103
	1	$0.0489 \pm 0.0129^{\dagger\dagger}$	$0.1253 \pm 0.0177^{\dagger\dagger}$	$0.0199 \pm 0.0055^{\dagger\dagger}$	0.1766 ± 0.0102
	2	$0.0494 \pm 0.0130^{\dagger\dagger}$	$0.1275 \pm 0.0179^{\dagger\dagger}$	$0.0205 \pm 0.0056^{\dagger\dagger}$	0.1764 ± 0.0101
	9	$0.0513 \pm 0.0134^{\dagger\dagger}$	$0.1359 \pm 0.0183^{\dagger\dagger}$	$0.0229 \pm 0.0059^{\dagger\dagger}$	0.1759 ± 0.0097
	19	$0.0525 \pm 0.0139^{\dagger\dagger}$	$0.1413 \pm 0.0186^{\dagger\dagger}$	$0.0243 \pm 0.0061^{\dagger\dagger}$	0.1757 ± 0.0096
1000	0	0.0378 ± 0.0083	0.1143 ± 0.0184	0.0162 ± 0.0048	0.1694 ± 0.0076
	0.5	$0.0389 \pm 0.0085^{\dagger\dagger}$	$0.1169 \pm 0.0178^{\dagger\dagger}$	$0.0167 \pm 0.0047^{\dagger\dagger}$	$0.1699 \pm 0.0077^{\dagger\dagger}$
	1	$0.0397 \pm 0.0087^{\dagger\dagger}$	$0.1189 \pm 0.0173^{\dagger\dagger}$	$0.0172 \pm 0.0047^{\dagger\dagger}$	$0.1702 \pm 0.0078^{\dagger\dagger}$
	2	$0.0409 \pm 0.0089^{\dagger\dagger}$	$0.1223 \pm 0.0166^{\dagger\dagger}$	$0.0180 \pm 0.0046^{\dagger\dagger}$	$0.1706 \pm 0.0079^{\dagger\dagger}$
	9	$0.0449 \pm 0.0095^{\dagger\dagger}$	$0.1345 \pm 0.0152^{\dagger\dagger}$	$0.0211 \pm 0.0046^{\dagger\dagger}$	$0.1715 \pm 0.0082^{\dagger\dagger}$
	19	$0.0474 \pm 0.0099^{\dagger\dagger}$	$0.1418 \pm 0.0147^{\dagger\dagger}$	$0.0232 \pm 0.0045^{\dagger\dagger}$	$0.1719 \pm 0.0083^{\dagger\dagger}$

The experiment is exploratory. The learning rate of the log-variances, the clipping range and the choice to compute the weights on training rather than validation losses were fixed without tuning, and a different configuration may behave differently.

4.4. Discussion

4.4.1. The Central Question in the Light of the Controlled Experiments

The central question asks whether calibration can be improved by solving (4.1) while statistical accuracy is maintained through a suitable λ. Taken at face value, the experiments of this chapter answer it positively. Compared with predictive-only training under the same budget of 1,200 epochs, the convex objective with $\lambda \approx 0.05$–0.1 improves every calibration measure except the worst-bin gap. It also improves the distance to the true posterior, the test cross-entropy and the refinement term, significantly and at every capacity (Table 4.3). Nothing is traded away: at a well-chosen λ, calibration and statistical accuracy improve together, and an interior optimum exists for both base losses (Figure 4.5).

The ablations, however, change what this result means. The convex combination couples the

Table 4.9.: Ablation C: validation-based pipelines (20 validation points, criterion: Brier score; mean ± standard deviation, 30 seeds). Markers on end-to-end rows compare with the early-stopping row of the same version (80 points or refit; paired Wilcoxon, Bonferroni over six comparisons per metric). The last two rows per capacity are references: the oracle $\lambda = 0.1$, and plain CE trained for all 1,200 epochs, both on 80 points.

m	Pipeline	MSE_η	ECE	Brier
10	end-to-end, selected λ (80)	$0.0142 \pm 0.0084^*$	0.0587 ± 0.0227	$0.1618 \pm 0.0082^*$
	CE, early stopping (80)	0.0210 ± 0.0134	0.0757 ± 0.0469	0.1683 ± 0.0132
	end-to-end, selected λ (refit)	$0.0131 \pm 0.0091^*$	0.0529 ± 0.0248	0.1607 ± 0.0089
	CE, early stopping (refit)	0.0200 ± 0.0154	0.0755 ± 0.0512	0.1674 ± 0.0152
	end-to-end, $\lambda = 0.1$ (80)	0.0122 ± 0.0058	0.0535 ± 0.0198	0.1598 ± 0.0058
	CE, 1,200 epochs (80)	0.0240 ± 0.0097	0.0794 ± 0.0213	0.1714 ± 0.0095
100	end-to-end, selected λ (80)	$0.0169 \pm 0.0128^*$	0.0629 ± 0.0251	$0.1646 \pm 0.0125^*$
	CE, early stopping (80)	0.0232 ± 0.0160	0.0743 ± 0.0379	0.1706 ± 0.0157
	end-to-end, selected λ (refit)	$0.0150 \pm 0.0133^*$	0.0564 ± 0.0257	$0.1627 \pm 0.0130^*$
	CE, early stopping (refit)	0.0214 ± 0.0176	0.0720 ± 0.0427	0.1688 ± 0.0173
	end-to-end, $\lambda = 0.1$ (80)	0.0153 ± 0.0059	0.0622 ± 0.0226	0.1630 ± 0.0060
	CE, 1,200 epochs (80)	0.0528 ± 0.0140	0.1366 ± 0.0169	0.2003 ± 0.0137
1000	end-to-end, selected λ (80)	0.0185 ± 0.0106	0.0661 ± 0.0297	0.1663 ± 0.0105
	CE, early stopping (80)	0.0250 ± 0.0133	0.0794 ± 0.0387	0.1725 ± 0.0132
	end-to-end, selected λ (refit)	0.0172 ± 0.0110	0.0629 ± 0.0290	0.1650 ± 0.0109
	CE, early stopping (refit)	0.0235 ± 0.0151	0.0786 ± 0.0419	0.1711 ± 0.0149
	end-to-end, $\lambda = 0.1$ (80)	0.0151 ± 0.0053	0.0605 ± 0.0203	0.1629 ± 0.0054
	CE, 1,200 epochs (80)	0.0396 ± 0.0081	0.1200 ± 0.0183	0.1873 ± 0.0082

weight on calibration to the step size on the proper loss (4.6), and in our setting most of the benefit comes from the second channel:

- At $\lambda = 0.1$, plain cross-entropy trained with a ten times smaller learning rate reproduces the improvement completely: in MSE_η, in the calibration measures and in the across-seed variance.
- At $\lambda = 0.5$ the penalty makes things measurably worse than this control.
- At the full step size, adding the penalty never helps and, for $m \geq n$, always hurts.
- The calibration term contributes on its own in only one regime: at the smallest weight tested ($\lambda = 0.05$), where it accounts for roughly 10–40% of the gain in MSE_η and significantly lowers the calibration error.

A more precise answer to the central question is therefore this. The objective (4.1), optimised with a fixed learning rate and budget, does improve calibration without sacrificing accuracy. It does so mainly because it slows the fitting of the proper loss, which acts as an implicit form of early stopping, and only secondarily because the calibration term steers the model. In the

Figure 1-M: Learned loss weights over training (mean, 10–90% band across seeds)

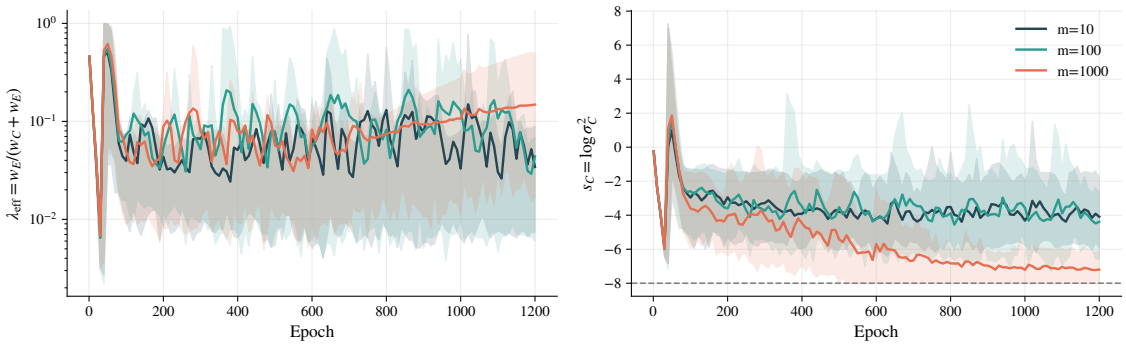


Figure 4.14.: Ablation D: implied convex weight $\lambda_{\text{eff}} = w_E/(w_C + w_E)$ (left, logarithmic scale) and log-variance s_C of the calibration term (right) during training under uncertainty weighting (mean and 10–90% band over 30 seeds). The dashed line marks the lower clipping bound.

realistic setting where λ has to be chosen on held-out data, end-to-end training still reaches a lower MSE_η than validation-based early stopping of plain cross-entropy, significantly so for $m \leq n$.

4.4.2. RQ1: Calibration as an Implicit Regulariser

Does the calibration objective act as an implicit regulariser? Theoretically, yes. Lowering λ from 1 to 0.1 produces every signature of regularisation one would look for:

- the mean of MSE_η falls, and so does its variance across training samples;
- the gap between training and test loss shrinks;
- the spikes with which the overparameterised baseline fits individual noisy labels disappear (Figure 4.4);
- the damage done at the interpolation threshold is largely undone.

What ablation A shows is where this regularisation comes from. For $\lambda \geq 0.1$ it is an effect of the optimisation trajectory: a smaller effective step on the proper loss within a fixed number of epochs, which is algorithmic regularisation of the kind studied by Berta et al. [8]. It is not an effect of the calibration term. The term itself regularises only when fitting is slow, and it works against generalisation when fitting is fast (ablation B).

Our hypothesis for this dependence is that the penalty is estimated on the training predictions. Once a model has begun to fit individual labels, low training-set calibration error is no longer evidence of calibration on new data, and minimising it pushes predictions further

Table 4.10.: Ablation D: uncertainty weighting compared with the baseline and end-to-end training at $\lambda = 0.1$ (mean $\pm$ standard deviation, 30 seeds). ††: significantly worse than both references at $p < 0.01$ (paired Wilcoxon, Bonferroni over six comparisons per metric).

m	Method	MSE_η	ECE	Test CE	final λ_{eff}
10	uncertainty weighting	$0.0670 \pm 0.0852^{\dagger\dagger}$	$0.1576 \pm 0.1197^{\dagger\dagger}$	1.0119 ± 0.7193	0.038 ± 0.041
	baseline ($\lambda = 1$)	0.0189 ± 0.0078	0.0694 ± 0.0195	0.5512 ± 0.0424	–
	end-to-end, $\lambda = 0.1$	0.0101 ± 0.0054	0.0459 ± 0.0179	0.4981 ± 0.0170	–
100	uncertainty weighting	$0.1028 \pm 0.0853^{\dagger\dagger}$	$0.2257 \pm 0.0963^{\dagger\dagger}$	1.6949 ± 1.1359	0.165 ± 0.306
	baseline ($\lambda = 1$)	0.0482 ± 0.0128	0.1217 ± 0.0178	0.6573 ± 0.0363	–
	end-to-end, $\lambda = 0.1$	0.0127 ± 0.0054	0.0545 ± 0.0212	0.5077 ± 0.0181	–
1000	uncertainty weighting	$0.0697 \pm 0.0135^{\dagger\dagger}$	$0.1946 \pm 0.0170^{\dagger\dagger}$	1.1628 ± 0.2438	0.148 ± 0.156
	baseline ($\lambda = 1$)	0.0378 ± 0.0083	0.1143 ± 0.0184	0.6252 ± 0.0368	–
	end-to-end, $\lambda = 0.1$	0.0129 ± 0.0052	0.0546 ± 0.0202	0.5099 ± 0.0176	–

towards the labels. If this hypothesis holds, a calibration penalty estimated on the same data as the proper loss can only act as a regulariser while the model is still far from interpolating. That would limit how far the idea of calibration as a structural constraint can be taken in the overparameterised regime.

How does it reshape the bias–variance trade-off relative to post-hoc calibration? Post-hoc calibration and training-time weighting act on different components of the loss. A recalibration map that is injective, such as temperature or Platt scaling, does not change which inputs receive which ranks, so it cannot reduce the refinement term [42, 8]; it can only remove calibration error. In the overparameterised baseline, however, the excess loss is partly a refinement problem. The Brier refinement term at $m = 100$ is 0.177 under the baseline and 0.155 at $\lambda = 0.1$ (Table 4.3), a gap larger than the whole calibration term. End-to-end training (or, equivalently here, a smaller step size) reduces both components. Post-hoc calibration can at best reduce one.

On the calibration measure alone, temperature scaling on a 20-point hold-out is competitive with end-to-end training and is better on the one split we examined (Table 4.4), but its result varies strongly with the choice of that split. Isotonic regression and histogram binning do not work at this calibration-set size. End-to-end training is the only method whose calibration is stable across the whole capacity range, and it does not give up training data for calibration.

These comparisons are single-split and single-seed, and they measure ECE only. They should be read as showing that the two approaches solve different problems, not as a ranking.

4.4.3. RQ2: What the Experiments Imply for the Theory

Chapter 3 approaches generalisation under end-to-end calibration through the complexity of a calibration-constrained hypothesis class and through PAC-Bayesian bounds. The experiments cannot confirm or refute a bound, but they do indicate which kind of argument fits the observed behaviour. Three observations are relevant.

First, the improvements in generalisation that we observe are algorithm-dependent. They depend on the product of learning rate and number of epochs, and they disappear or reverse when the penalty is added at the full step size. An argument that bounds generalisation uniformly over a hypothesis class, such as one based on Rademacher complexity, does not see the optimisation trajectory. It therefore cannot account for most of the effect measured here. Algorithm-dependent analyses, for instance of early-stopped gradient descent, or PAC-Bayesian bounds with a posterior that depends on the training procedure, are closer to the mechanism we observe.

Second, a soft penalty is not a hard constraint. The set of models with small empirical calibration error does not shrink the hypothesis class in a useful way when the model can interpolate: an interpolating model can be well calibrated on its own training data. This is consistent with ablation B.

Third, where the penalty does contribute on its own (ablation A at $\lambda = 0.05$), the effect is modest but significant. A theoretical account of it would need to explain why it appears only when the proper loss is fitted slowly.

These are interpretations of the experiments and are offered as directions for the analysis, not as results.

4.4.4. RQ3: Summary of the Controlled Comparison

Table 4.11 collects the answers to the experiment questions. Measured against the true posterior, three statements hold across capacities.

- End-to-end calibrated models at a small λ are closer to η than predictive-only models, better calibrated on average and no less sharp.
- The benefit is largest at and beyond the interpolation threshold, where predictive-only training does the most damage.
- Overparameterisation brings no advantage under end-to-end training in this problem, but it no longer brings a disadvantage either.

Two findings concern the evaluation itself rather than the methods. The first is that label-based calibration metrics can be misleading. The degenerate model at $\lambda = 0$ has a top-label ECE comparable to the best end-to-end model while being four to five times further from η.

Table 4.11.: Summary of the answers to the Track 1 experiment questions.

Q	Answer
Q1.1	Yes, the two components are not minimised at the same time. After an early phase of joint improvement, test calibration degrades while test refinement has saturated, clearly so for $m \geq n$.
Q1.2	Yes, with a transition at about epoch 80 for $m \geq n$. The penalty delays and damps it but does not prevent it; at $\lambda \geq 0.1$ this is mostly the slower effective time scale.
Q1.3	Temperature scaling is the strongest post-hoc method and competitive with end-to-end training on ECE for $m \geq n$, but unstable across calibration splits. Isotonic regression and histogram binning fail with 20 calibration points. All post-hoc methods hurt for $m \leq 10$.
Q1.4	TS-refinement has lower bias at every capacity (significant for $m \geq n$). The variance ranking flips: isotonic is lower at $m = 10$, TS at $m = 100$, equal at $m = 1000$.
Q1.5	The raw model degrades from the interpolation threshold on. Temperature scaling becomes less effective and less stable beyond it, but does not collapse.
Q1.6	Brier is better than CE at $\lambda = 1$ at every capacity (single seed), with the gap growing in m; at the respective optimal λ the difference is small.
Q1.7	Not on MSE_η: end-to-end training is two to three times closer to η. On ECE, focal loss with $\gamma = 2$ is better for $m \geq n$.
Q1.8	No: $m = 10$ is numerically best under end-to-end training, but capacity matters little.
Q1.9	Yes in absolute terms (largest reduction at $m = n$), but the step-size control shows the same dependence.
Q1.10	Yes for $\lambda \in [0.1, 1]$ (mean and across-seed variance of MSE_η fall together); not at $\lambda = 0$. At $\lambda = 0.1$ the step-size control reproduces it.
Q1.11	Towards η for $\lambda \in (0, 1)$; the calibration term alone ($\lambda = 0$) produces a useless, near-constant predictor.
Q1.12	Yes: interior optimum near 0.05 (CE) and 0.10–0.15 (Brier), almost independent of m.
A–D	The gain at $\lambda \geq 0.1$ is explained by the reduced step size. The penalty contributes on its own only at $\lambda = 0.05$ and hurts at full step size. Validation-selected λ beats validation early stopping on MSE_η for $m \leq n$. Uncertainty weighting fails in the configuration tested.

Focal loss with $\gamma = 2$ has the best ECE at $m \geq 100$ while being two to three times further from η than end-to-end training. A model should therefore never be judged by a calibration metric alone; a proper score or its refinement term has to be reported alongside it. The second finding is that selecting λ on the test-set ground truth is optimistic. Selection on 20 validation points costs about 0.002–0.003 in MSE_η.

4.4.5. Limitations

Scope of the simulation. All results come from a single one-dimensional problem with a very smooth η, a fixed training-set size of $n = 100$ and one model family. Because even $m = 10$ represents η almost exactly, the setting cannot show a benefit of capacity. The results should not be extrapolated to problems whose target cannot be represented by small models.

Optimisation. Every experiment uses full-batch gradient descent with one learning rate and one budget of epochs. The central mechanism identified here, the coupling between λ and the effective step size, is specific to optimisers whose step scales with the gradient. Adaptive optimisers such as Adam are approximately invariant to a rescaling of the whole loss. For them the step-size channel should be much weaker, and the relative weight of the two terms should matter more. The soft-ECE gradient uses a straight-through approximation, and its hyperparameters ($B = 15$, $h = 0.08$) were not varied.

Statistical basis. The following rest on a single seed:

- the training dynamics, the reliability diagrams and the curves over the fine λ grid;
- the comparison of base losses;
- the post-hoc comparison, which in addition uses a single split whose variability is large.

The 30-seed experiments use the CE base loss only. The "best" λ in all experiments except ablation C is an oracle choice based on the test-set MSE_η, and the optimum for CE lies at the edge of the grid. Variances were compared descriptively. The words bias and variance refer to the mean and the across-seed variance of MSE_η, not to a pointwise bias–variance decomposition of the predictor.

Metrics and definitions.

- The top-label ECE and MCE are binned estimators with their own bias; MCE in particular is dominated by sparsely populated bins.
- The calibration term of the cross-entropy used in figure 4.2 is a binned analogue of the Brier formula.
- The focal loss follows the standard asymmetric form rather than the formula written in the experiment plan.

Comparisons and explanations.

- The post-hoc comparison deliberately gives end-to-end training 100 training points and the post-hoc pipeline 80.
- The uncertainty-weighting experiment uses a single, untuned configuration.
- The explanations offered for ablation B and for the advantage in ablation C are hypotheses that the experiments are consistent with but do not isolate.

4.4.6. Optional Task: Calibration under Controlled Covariate Shift

The full experimental design, figures, numerical results, and discussion for this optional task are reported in Appendix A.1. The paired statistical tests are provided in Appendix A.2.

5. Calibration for Gene Regulatory Network Inference

5.1. Overview and Objectives

So far we have studied end-to-end calibration in a controlled binary-classification setting where the data-generating posterior is known. Now we move to gene regulatory network (GRN) inference, where the objective is to predict whether a directed regulatory edge exists between a transcription factor (TF) and a target gene. The transition is important: biological expression data exhibit dependence across genes and cells, the true posterior probability of an edge is not known, and the evaluation depends on how candidate edges and negative examples are constructed.

The central question is similar: whether incorporating a differentiable calibration penalty into a Transformer-based edge predictor can improve the reliability of its predicted edge probabilities without unacceptably damaging its ability to rank true regulatory edges above non-edges. The experiments compare three approaches: (i) predictive-only training with binary cross-entropy (BCE) or Brier loss; (ii) end-to-end calibration-aware training, which combines a proper predictive loss with a smooth differentiable calibration surrogate; and (iii) post-hoc calibration of a predictive-only model using temperature scaling or isotonic regression.

The empirical study has three parts. First, the main experiment compares these methods on a simulated GRN generated with SERGIO and on a DREAM5 expression/gold-standard dataset. Second, a controlled feature-shift experiment tests how sensitive predictions are to changes in the scale and offset of the expression features while the candidate-pair labels are held fixed. Third, an exploratory cross-network experiment trains on one DREAM5 network and evaluates on another. The latter two analyses are for evaluating its performance under covariate shift and is treated as stress tests. A biological shift doesn't always satisfy the formal pure-covariate-shift assumption.

The chapter therefore addresses the practical part of the thesis research question: whether calibration-aware training is useful when a model must assign probabilities to candidate regulatory edges. Since no known posterior is available for the GRN tasks, the results cannot establish exact conditional calibration. They instead combine ranking metrics with empirical calibration measures and proper scoring rules.

5.2. Experimental Setup

5.2.1. Task Formulation

Let $\mathcal{G} = (V, E)$ denote a directed gene regulatory network, where vertices represent genes and a directed edge (u, v) represents a regulatory relationship from regulator u to target v. For a candidate TF–target pair (u, v), the model receives the paired expression profiles of the TF and target across a set of cells or experimental conditions. The binary label is

$$y_{uv} = \begin{cases} 1, & \text{if the pair is present in the available gold-standard network,} \\ 0, & \text{otherwise.} \end{cases}$$

The model returns a probability $p_{uv} \in [0, 1]$ intended to represent the likelihood that the candidate pair is a regulatory edge.

For each pair, the input tensor has shape $(T, 2)$, where T is the number of retained cells or conditions and the two channels are regulator expression and target expression. The experiments cap T at 64 to keep the computation manageable and to ensure a consistent input shape across datasets. The Transformer operates on these paired profiles; gene identifiers are not directly supplied as learned identity features. The model therefore learns from expression-profile patterns, not from a separate gene-identity embedding.

This formulation should be interpreted as supervised edge prediction against a supplied reference network. A gold-standard label is an operational reference, not proof of causal regulation in every biological context. Likewise, a high attention weight is not, by itself, evidence that a predicted edge is causal.

5.2.2. Datasets

SERGIO simulation

SERGIO is a simulator for single-cell gene-expression data designed around a user-specified gene regulatory network and nonlinear regulatory dynamics [15]. It is useful here because the network structure used to generate the expression data is known, so simulated candidate edges can be labelled directly from the generating graph. This gives a controlled test of whether the model can recover edges in data generated from a known regulatory structure. It does not, however, make the resulting expression profiles independent and identically distributed in the broad biological sense; genes are connected through the simulated regulatory mechanism, and all cells share the same underlying network.

The notebook defines a three-layer network with 25 master regulators, 50 transcription factors, and 100 target genes, giving 175 genes in total. Master regulators regulate the TF layer, and TFs regulate the target-gene layer. Each TF receives two master-regulator inputs, while

each target gene receives between two and four TF inputs. Interaction strengths are sampled between 1 and 5 in magnitude, with a 20% chance of a negative sign; Hill coefficients are 2 with probability 0.9 and 1 otherwise. The simulation uses one bin containing 300 cells. The SERGIO expression matrix is standardised gene-wise, and at most 64 cells are retained for each pair profile.

The prediction task is restricted to TF-to-target edges. There are $50 \times 100 = 5{,}000$ candidate pairs, of which 303 are positive in the saved pair-level dataset, giving a positive prevalence of approximately 6.1%. The saved tensor has shape $(5000, 64, 2)$. This is a sparse-edge setting and its class prevalence is substantially lower than that of the sampled DREAM5 candidate set, so the absolute AUPR values across the two datasets should not be interpreted as directly comparable measures of biological difficulty.

DREAM5 expression data and gold standard

DREAM5 was a community challenge for gene-network inference that compared methods against reference regulatory networks and expression datasets [22]. The challenge is widely used as a benchmark because it provides expression measurements alongside network reference information. Unlike the SERGIO experiment, the expression data and labels are not generated by the local simulation code. The model's labels are derived from the corresponding DREAM5 gold-standard edge list, and expression measurements are paired with that network's own reference labels. Expression files and gold standards must not be mixed across networks.

For our experiment, we have used *"Network 1 - in silico"* from the official website. The prepared pair dataset is named `dream5_net1_pair_data.npz` in the saved results. It contains 12,000 candidate pairs. The main notebook's saved output reports a positive prevalence of approximately one third for this sampled set. The DREAM5 preparation procedure samples negative pairs rather than enumerating every possible non-edge. Consequently, its measured prevalence is a property of this candidate set, not the natural prevalence of regulatory edges in the full network. AUROC, AUPR, Brier score, and calibration metrics therefore describe the prepared evaluation set and should not be compared directly with official DREAM5 leaderboard results unless the candidate set and evaluation protocol match.

For the cross-network experiment, we used the *"Network 3 - E. coli"* as the secondary network. Therefore, `dream5_net1_pair_data.npz` acts as source and `dream5_net3_pair_data.npz` as target. The saved target preparation output reports 12,000 pairs and a positive prevalence of approximately 17.2%. This difference in prevalence is relevant to both AUPR and calibration. It also means that cross-network evaluation changes more than the marginal expression distribution: the reference network and label composition differ as well.

5.2.3. Transformer Architecture and Training Objectives

Each candidate TF–target pair is represented by a sequence $X_{uv} \in \mathbb{R}^{T\times 2}$, where each time / condition position contains the standardized expression values of the regulator and target. A shared linear projection maps the two input channels into a d_{model}-dimensional hidden representation. This sequence is processed by a Transformer encoder, the resulting hidden states are mean-pooled across the T positions, and a feed-forward prediction head maps the pooled representation to one logit. Applying the sigmoid function to the logit yields the predicted edge probability. The architecture therefore models patterns across the paired expression profiles; it does not include gene-identity embeddings or an explicit graph message-passing layer.

The *small* model uses $d_{\text{model}} = 16$, two attention heads, one Transformer encoder layer, and dropout 0.1. Its encoder feed-forward width is $2d_{\text{model}} = 32$. The *medium* model uses $d_{\text{model}} = 32$, four attention heads, two encoder layers, and dropout 0.1, with feed-forward width 64. Both models use the same input projection, mean pooling, and prediction head structure: layer normalisation, a linear layer of width d_{model}, GELU activation, dropout, and a final linear layer producing a scalar logit. The medium model thus increases both hidden width and encoder depth while preserving the same task representation. The full architecture and optimisation settings are listed in Appendix B.1.

Let $p_\theta(x)$ be the model's predicted probability and let $\mathcal{L}_{\text{base}}$ denote either binary cross-entropy (BCE) or the Brier loss. End-to-end calibration-aware training minimises

$$\mathcal{L}_\lambda(\theta) = (1-\lambda)\,\widetilde{\mathcal{C}}_{\text{soft}}(\theta) + \lambda\,\mathcal{L}_{\text{base}}(\theta), \tag{5.1}$$

where $\widetilde{\mathcal{C}}_{\text{soft}}$ is a differentiable soft-binned calibration surrogate. It uses 12 soft bin centres and a Gaussian-style bandwidth of 0.08; soft bin memberships are detached when calculating the penalty. This surrogate is an ECE-style training penalty, not the canonical binned ECE reported at evaluation time. The parameter λ controls the balance between the predictive loss and calibration surrogate. At $\lambda = 1$, the calibration term vanishes and training reduces to the predictive-only baseline. Values $\lambda = 0.5$ and $\lambda = 0.1$ increase the relative contribution of the calibration penalty.

For post-hoc calibration, temperature scaling learns a single temperature from validation predictions, while isotonic regression learns a non-parametric monotone mapping. Both calibrators are fitted on validation predictions and then applied to held-out test predictions. This separation avoids fitting the calibration map to test labels.

5.2.4. Splits, Reproducibility, and Hyperparameters

The main experiment uses three random seeds (42, 123, and 2026). For each seed, the candidate pairs are divided into training, validation, and test subsets using the utility module's stratified split procedure. The same seed-specific split is used to compare methods within a dataset. The small and medium architectures are both evaluated with BCE and Brier base losses, the same set of λ values, and the same post-hoc calibration procedures. Training uses

up to 30 epochs, batches of 32, learning rate 10^{-3}, and early stopping with patience 5. The full architecture, optimisation, calibration, splitting, and dataset settings are summarised in Appendix B.1. The complete three-seed result table, including both architectures, both base losses, all objective weights, and both post-hoc methods, is given in Appendix B.3. The shift and cross-network result tables are reproduced in Appendices B.4 and B.5.

The unit of splitting is the candidate pair, not the gene or cell. Consequently, a held-out pair may share a TF or target gene with pairs in the training set. The experiment evaluates generalisation to held-out pairs within the prepared candidate universe; it does not measure generalisation to entirely unseen genes. This limitation is particularly relevant because expression profiles from pairs sharing genes are dependent.

5.2.5. Evaluation Metrics

The evaluation reports four primary quantities:

- **AUROC**: the area under the receiver-operating-characteristic curve, measuring ranking across positive and negative candidate pairs.
- **AUPR**: the area under the precision–recall curve. It is informative under class imbalance but depends on positive prevalence and the negative-sampling protocol.
- **Brier score**: the mean squared error between predicted probabilities and binary labels,

$$\mathrm{BS} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2.$$

  Lower values indicate better probabilistic predictions, combining aspects of calibration and discrimination.
- **ECE10**: the expected calibration error estimated with ten probability bins. Lower values indicate a smaller weighted discrepancy between mean predicted confidence and empirical positive frequency within bins. Because ECE is bin-dependent and estimated from finite samples, it is interpreted together with the Brier score and reliability diagrams rather than as a definitive measure of calibration.

The shift experiment additionally reports calibration-in-the-large, defined as the absolute difference between mean predicted probability and observed positive prevalence. The cross-network experiment reports a score-conditional calibration gap computed by binning predictions and averaging the absolute within-bin difference between observed frequency and mean prediction, subject to a minimum bin count. The code's score-conditional gap is an empirical binned diagnostic and should not be interpreted as exact conditional calibration.

5.3. Main Results: Calibration and Edge Prediction

5.3.1. Aggregate Results Across Seeds

The main experiment reports means and standard deviations across the three seeds for AUROC, AUPR, Brier score, and ECE10. Figure 5.1 plots the AUPR–ECE trade-off for individual seed-level evaluations. Figures 5.2 and 5.3 show reliability diagrams and the distributions of held-out predictions, respectively. These views are complementary: AUPR describes ranking, reliability diagrams compare predicted probabilities with empirical event frequencies, and confidence distributions show how positive and negative candidate pairs are separated by the model.

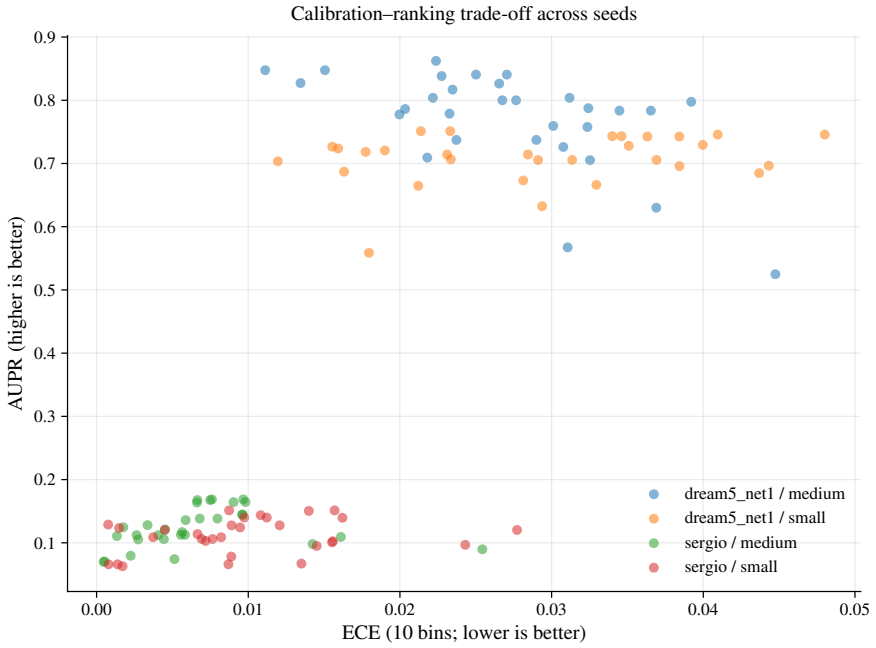


Figure 5.1.: Calibration–ranking trade-off for seed-level test evaluations. Each point represents one method/configuration on a prepared GRN candidate set. Lower ECE and higher AUPR are preferred, but AUPR depends on candidate-pair prevalence.

The saved aggregate results show that there is no single method that dominates every metric and dataset. On DREAM5 Network 1, the medium BCE model trained end-to-end with $\lambda = 0.5$ has the strongest mean AUPR among the displayed BCE configurations (0.833) and a Brier score of 0.120, compared with 0.787 AUPR and 0.136 Brier for the predictive-only medium model. Its mean ECE10 is 0.028, slightly lower than the predictive-only model's 0.030. The same configuration does not dominate all settings: for the small architecture, the predictive-only model has mean AUROC 0.833 and AUPR 0.734, while end-to-end training at $\lambda = 0.5$ gives AUROC 0.816 and AUPR 0.718 and has a higher ECE10 (0.038 versus 0.037). These values suggest that capacity and the calibration weight interact, rather than supporting a universal rule that a stronger calibration penalty improves both calibration and ranking.

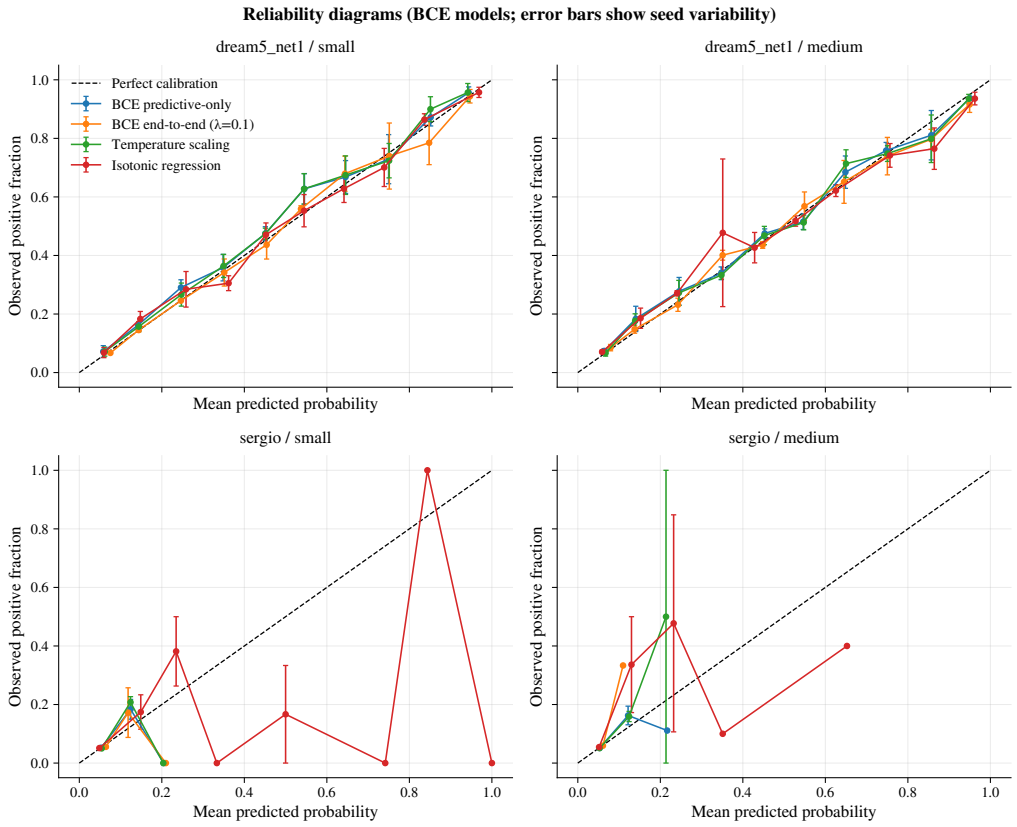


Figure 5.2.: Reliability diagrams for the BCE-based predictive-only, end-to-end, temperature-scaled, and isotonic models, separated by dataset and architecture. Points closer to the diagonal indicate closer agreement between predicted probability and empirical edge frequency in the corresponding bins.

For SERGIO, the predictive-only medium model has mean AUROC 0.629 and AUPR 0.142. End-to-end training at $\lambda = 0.5$ gives AUROC 0.611 and AUPR 0.137, while $\lambda = 0.1$ gives lower AUPR (0.118). The mean Brier scores are all close to 0.056, reflecting the low positive prevalence and the concentration of predicted probabilities near the majority-class range. The small SERGIO model shows a similar pattern: the predictive-only model reaches AUROC 0.620 and AUPR 0.132, and the $\lambda = 0.5$ end-to-end configuration has the same rounded AUPR (0.132) with AUROC 0.622. The calibration penalty therefore does not uniformly improve edge ranking on the simulated network.

Table 5.1 reports a compact selection of the saved mean metrics. The complete seed-aggregated comparison, including standard deviations for every metric and all method configurations, is provided in Appendix B.3.

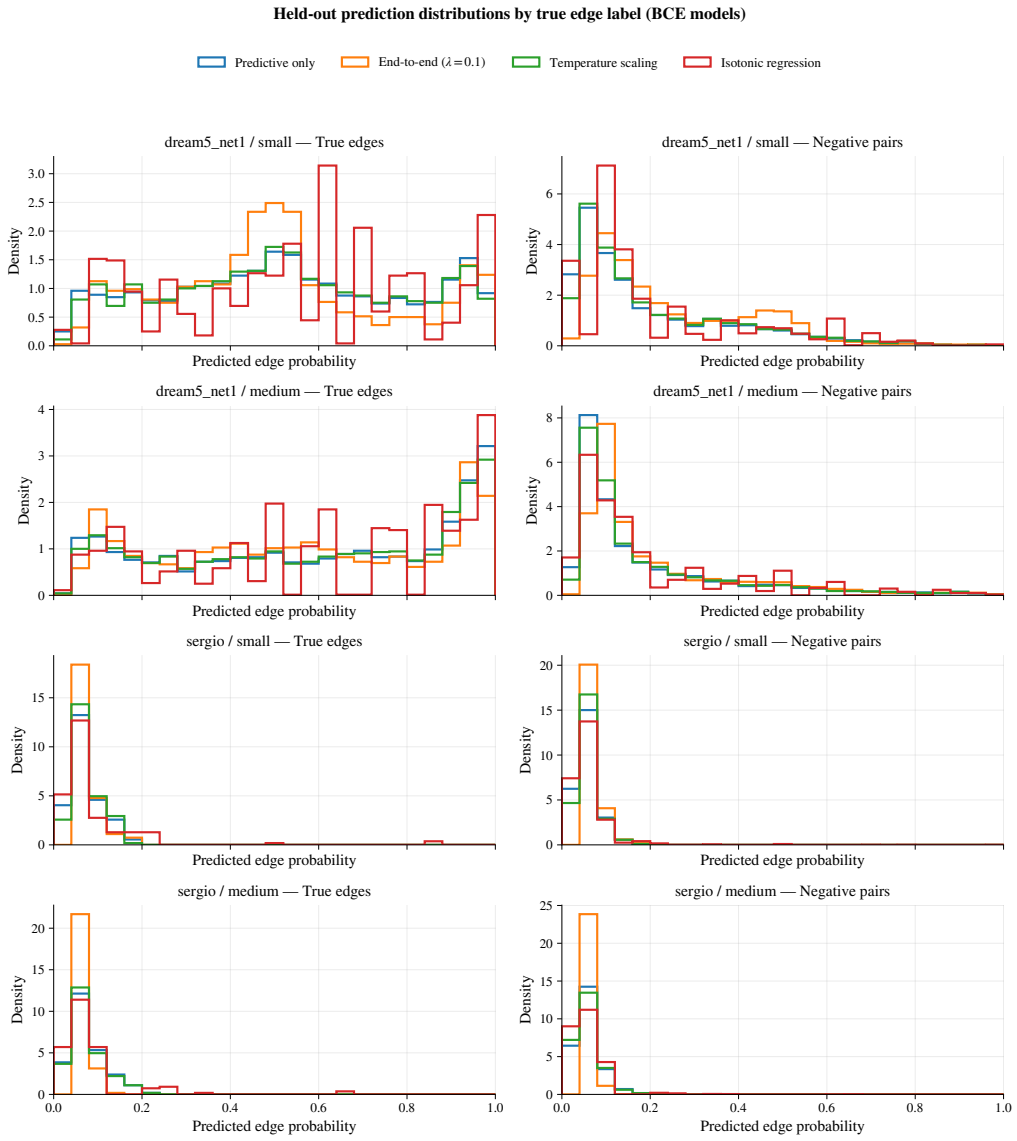


Figure 5.3.: Held-out predicted-probability distributions for positive edges and negative candidate pairs. The distributions illustrate separation and confidence concentration; they do not alone establish calibration.

Table 5.1.: Selected mean test metrics from the three-seed main experiment. ECE10 and Brier are lower-is-better; AUROC and AUPR are higher-is-better.

Dataset	Capacity	Method	AUROC	AUPR	Brier	ECE10
DREAM5 Net 1	Medium	Predictive-only BCE	0.857	0.787	0.136	0.030
DREAM5 Net 1	Medium	End-to-end BCE, $\lambda = 0.5$	0.882	0.833	0.120	0.028
DREAM5 Net 1	Medium	Temperature scaling	0.857	0.787	0.136	0.028
DREAM5 Net 1	Small	Predictive-only BCE	0.833	0.734	0.152	0.037
DREAM5 Net 1	Small	End-to-end BCE, $\lambda = 0.5$	0.816	0.718	0.157	0.038
SERGIO	Medium	Predictive-only BCE	0.629	0.142	0.056	0.008
SERGIO	Medium	End-to-end BCE, $\lambda = 0.5$	0.611	0.137	0.056	0.011
SERGIO	Small	Predictive-only BCE	0.620	0.132	0.056	0.009
SERGIO	Small	End-to-end BCE, $\lambda = 0.5$	0.622	0.132	0.056	0.011

5.3.2. Post-hoc Calibration

Temperature scaling preserves the ranking of a model's predictions because it applies a monotone transformation to the logits. Accordingly, the AUROC and AUPR values for temperature scaling are identical to those of the corresponding predictive-only model in the saved output. It can nevertheless change probability-sensitive metrics, including Brier score and ECE10. On DREAM5 Network 1, temperature scaling reduces mean ECE10 for the medium BCE model from 0.030 to 0.028, while the Brier score remains effectively unchanged after rounding (0.136). For the small model, the ECE10 decreases from 0.037 to 0.033, again with essentially unchanged Brier score.

Isotonic regression is more flexible because it estimates a non-parametric monotone mapping. In the saved results, the small DREAM5 BCE model's mean ECE10 decreases from 0.037 to 0.026, while its Brier score remains 0.152 after rounding. For the medium DREAM5 model, isotonic regression reduces ECE10 to 0.027, but its AUPR is lower than the predictive-only model's AUPR. This is consistent with the fact that a post-hoc monotone map can alter ties and probability calibration without improving the underlying ranking. The isotonic results should be interpreted cautiously because the calibration mapping is estimated from a finite validation set.

For SERGIO, post-hoc calibration yields smaller or similar changes. The mean ECE10 for the medium BCE predictive-only model is approximately 0.008, and temperature scaling gives a similar value; isotonic regression is also close. Since the positive prevalence is only about 6.1%, these low ECE values should not be read in isolation. A model that assigns probabilities close to the low base rate can have a small aggregate calibration error while offering only moderate edge-ranking performance. This motivates reporting AUROC, AUPR, Brier score, and reliability plots together.

5.3.3. BCE versus Brier as the Base Loss

The main notebook repeats the comparison with BCE and Brier base losses for both model capacities. The saved results indicate that the effect of the base loss depends on the dataset and calibration weight. On DREAM5 Network 1, medium BCE with $\lambda = 0.5$ achieves a higher mean AUPR (0.833) and lower Brier score (0.120) than the corresponding medium Brier-loss configuration, which has mean AUPR 0.776 and Brier score 0.140. On SERGIO, the predictive-only BCE and Brier baselines are similar, while the strongly calibration-weighted Brier configuration loses ranking performance. These comparisons do not establish that BCE is universally preferable; they show that the calibration penalty should be evaluated jointly with the predictive loss and model capacity.

5.4. Robustness to Controlled Expression-Feature Shift

5.4.1. Design

The controlled-shift experiment reuses the prepared SERGIO and DREAM5 Network 1 candidate pairs. Each model is trained on the original training inputs, and early stopping uses the original validation inputs. At test time, the same held-out pairs are evaluated twice: once with the original features and once after a fixed channel-wise affine transformation. The regulator channel is transformed as $x'_{\mathrm{TF}} = 1.5x_{\mathrm{TF}} + 0.5$, and the target channel as $x'_{\mathrm{target}} = 0.75x_{\mathrm{target}} - 0.25$. The binary labels are unchanged. This transformation changes the feature marginals while preserving the evaluated pairs and their positive prevalence.

The experiment uses three seeds and both model capacities. It compares predictive-only BCE ($\lambda = 1$), end-to-end BCE at $\lambda = 0.5$ and $\lambda = 0.1$, temperature scaling, and isotonic regression. The post-hoc calibrators are fitted on the original validation predictions and then applied to both test versions. This design asks how robust each fitted predictor and calibrator is to a specified expression-feature perturbation. It does not constitute a biological intervention and does not prove that the conditional distribution $P(Y \mid X)$ is invariant under the transformation.

5.4.2. Results

The feature transformation changes the channel means as intended. For SERGIO, the original pair tensor has channel means approximately $(0, 0)$ after gene-wise standardisation, whereas the transformed tensor has means $(0.5, -0.25)$. For DREAM5 Network 1, the corresponding means change from approximately $(0.004, 0.008)$ to $(0.505, -0.244)$.

The results show a clear reduction in predictive performance on shifted DREAM5 features (Figure 5.4). For the medium model, the mean AUROC of predictive-only BCE decreases from 0.859 on the original test inputs to 0.757 on the shifted inputs, and AUPR decreases from

0.792 to 0.623. The Brier score increases from 0.134 to 0.207, while ECE10 rises from 0.029 to 0.150. End-to-end BCE with $\lambda = 0.5$ also degrades: AUROC decreases from 0.876 to 0.761 and AUPR from 0.822 to 0.644; Brier increases from 0.123 to 0.194 and ECE10 from 0.024 to 0.118. Thus, calibration-aware training does not remove sensitivity to the particular shift. In this configuration, $\lambda = 0.5$ has better in-distribution metrics than predictive-only training, but its shifted metrics remain substantially worse.

The same broad effect appears for the small DREAM5 model. Predictive-only BCE has mean AUROC 0.834 and AUPR 0.736 in-distribution, falling to 0.730 and 0.570 under the shifted inputs. Its Brier score rises from 0.152 to 0.240 and ECE10 from 0.038 to 0.204. End-to-end training at $\lambda = 0.5$ begins with lower in-distribution ranking scores and also exhibits a large calibration deterioration under shift. Isotonic regression, in particular, should not be expected to remain calibrated after the test feature distribution changes when the mapping was fitted only on the original validation distribution.

The SERGIO results are less affected in ranking by this particular transformation (Figure 5.5). For the medium predictive-only model, mean AUROC changes from approximately 0.630 to 0.622, and AUPR from 0.142 to 0.132. ECE10 changes from approximately 0.011 to 0.006 in the saved summary, although the corresponding standard deviations are non-negligible. End-to-end configurations show relatively small ranking changes but do not uniformly improve calibration under the shift. The difference between the SERGIO and DREAM5 responses indicates that the same numerical transformation can have different effects depending on the feature distribution and learned predictor. It should not be interpreted as evidence that SERGIO is generally more robust than biological data: the two candidate sets differ in size, class prevalence, and data-generating process.

The complete results for both datasets, model capacities, methods, and evaluation splits are given in Appendix B.4, Table B.4. The controlled-shift experiment supports a limited but useful conclusion: calibration-aware training may improve the in-distribution calibration–ranking trade-off for some settings, but it does not guarantee robustness to changes in expression-feature scale and offset. Calibration is distribution-dependent, and a calibrator fitted on one distribution may become miscalibrated after a shift. A practical deployment would therefore require validation data representative of the target condition or a justified recalibration procedure.

5.5. Exploratory Cross-Network Transfer on DREAM5

5.5.1. Design and Interpretation

The final experiment trains a model on DREAM5 Network 1 and evaluates it both on a held-out subset from that source network and on candidate pairs from DREAM5 Network 3. The source split uses seed 42. The comparison includes predictive-only BCE, end-to-end calibration-aware BCE at $\lambda = 0.5$ and $\lambda = 0.1$, and temperature scaling fitted on source validation pre-

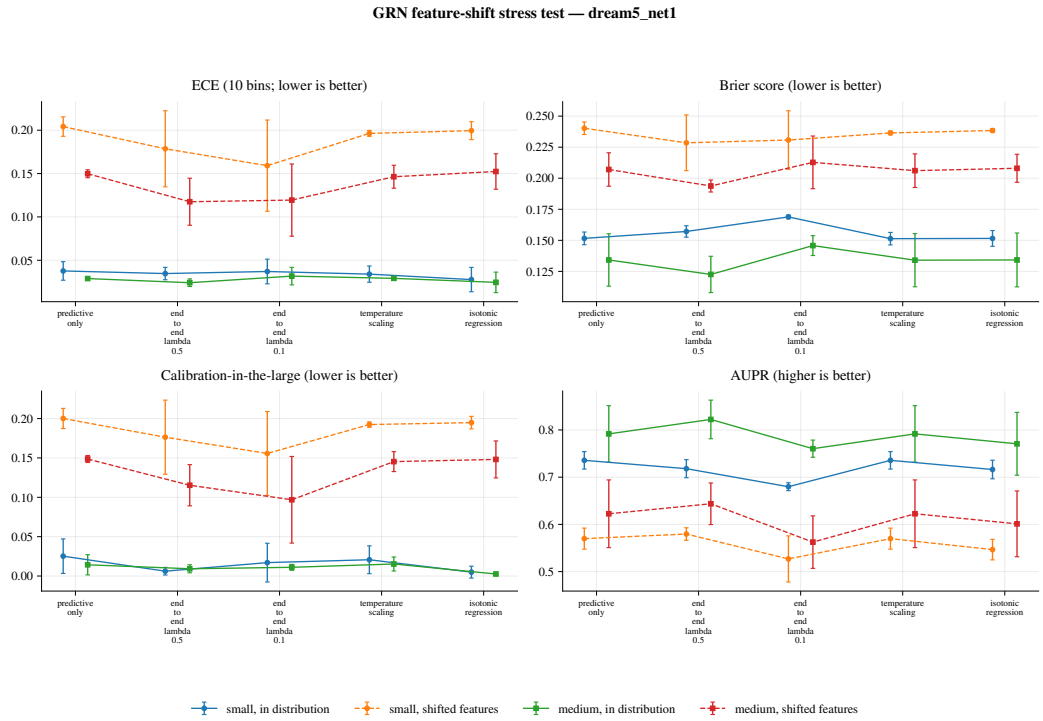


Figure 5.4.: Controlled feature-shift results for DREAM5 Network 1. Solid and dashed traces distinguish the original and shifted held-out inputs; error bars summarise variation across three seeds.

dictions. Unlike the controlled affine perturbation, cross-network transfer changes the data source and gold-standard network. It can reflect differences in expression distributions, network structure, organism, reference-label coverage, and positive prevalence. It is therefore reported as exploratory cross-network distribution shift, not as a controlled pure-covariate-shift experiment.

5.5.2. Results

The outputs show a substantial decline in ranking performance when the Network 1 model is evaluated on Network 3. The source-test BCE model obtains AUROC 0.834 and AUPR 0.746, whereas cross-network AUROC falls to 0.606 and AUPR to 0.251. The target candidate set has a lower positive prevalence (0.172) than the source test set (0.333), which contributes to the reduction in AUPR and also changes the interpretation of calibration metrics.

On the source test set, end-to-end training with $\lambda = 0.5$ reduces the absolute calibration-in-the-large gap to 0.006 and gives ECE10 0.040, compared with 0.048 for predictive-only BCE. On Network 3, however, its calibration-in-the-large gap rises to 0.168 and ECE10 to 0.183; the Brier score is 0.200. The $\lambda = 0.1$ model has cross-network AUROC 0.652 and AUPR 0.310, better ranking than the other tested transfer settings, but its calibration-in-the-large gap

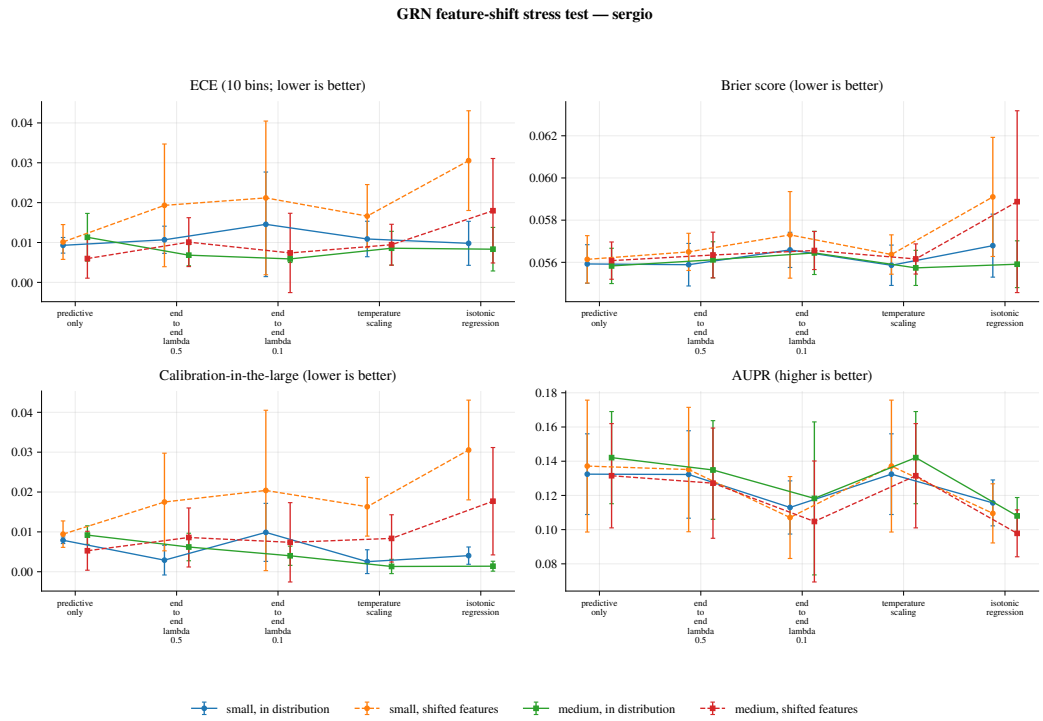


Figure 5.5.: Controlled feature-shift results for SERGIO. Metrics are evaluated on the same held-out pairs before and after the affine feature transformation.

remains high at 0.162 and ECE10 is 0.176. This illustrates that relative ranking performance and calibration can move differently under transfer.

Temperature scaling slightly improves the source-test Brier score and ECE10 in the saved results, but it does not restore calibration on Network 3: its calibration-in-the-large gap is 0.098 and ECE10 is 0.149, while AUROC and AUPR remain unchanged from predictive-only BCE. Since temperature scaling is fitted using source validation labels, this is expected to be a limited correction when the target prevalence and network differ. The target metrics should be interpreted descriptively because the source and target candidate sets were sampled separately and the target labels come from a different reference network.

All eight source-test and target-network evaluations are reproduced in Appendix B.5, Table B.5. This cross-network experiment provides a cautionary result: source-domain calibration does not automatically transfer to a different network. End-to-end calibration-aware training may change the ranking behaviour, but it did not eliminate the target-domain calibration gap in this single-seed exploratory analysis. More seeds, multiple transfer directions, and a carefully harmonised candidate-pair protocol would be needed before making a stronger claim.

Table 5.2.: DREAM5 cross-network transfer from Network 1 to Network 3. Values are from the saved single-seed output. Lower Brier/ECE/calibration-in-the-large are preferred; higher AUROC/AUPR are preferred.

Evaluation	Method	AUROC	AUPR	Brier	ECE10	Cal.-in-large
Source test	BCE	0.834	0.746	0.151	0.048	0.048
Source test	BCE + temperature	0.834	0.746	0.151	0.041	0.035
Source test	End-to-end, $\lambda = 0.5$	0.816	0.729	0.155	0.040	0.006
Source test	End-to-end, $\lambda = 0.1$	0.804	0.706	0.163	0.029	0.013
Network 3	BCE	0.606	0.251	0.183	0.152	0.083
Network 3	BCE + temperature	0.606	0.251	0.180	0.149	0.098
Network 3	End-to-end, $\lambda = 0.5$	0.619	0.264	0.200	0.183	0.168
Network 3	End-to-end, $\lambda = 0.1$	0.652	0.310	0.198	0.176	0.162

5.6. Discussion

First, calibration-aware training can be competitive with predictive-only training, but its effects depend on the dataset, capacity, base loss, and calibration weight. The medium BCE model trained with $\lambda = 0.5$ performed well on DREAM5 Network 1, improving mean AUPR and Brier score relative to the predictive-only medium baseline in the saved three-seed results. This pattern did not generalise uniformly to SERGIO or to the small model. The results therefore support treating λ as an empirical design choice, not assuming that stronger calibration weighting is always beneficial.

Second, post-hoc calibration can improve empirical calibration metrics with little or no change to ranking. This is especially clear for temperature scaling, whose monotonic transformation preserves the ordering of predictions. Isotonic regression can also reduce ECE, but its flexible mapping may be sensitive to the amount and distribution of validation data. In either case, improvements on a validation-like test set do not guarantee calibration after a distribution shift.

Third, the two shift experiments show that calibration is not a property of the model independent of the evaluation distribution. The affine stress test substantially degrades DREAM5 performance, while cross-network transfer produces a pronounced decline in ranking and calibration. The latter is particularly important for GRN inference, where changes in species, network topology, experimental design, and label prevalence can occur together.

6. Conclusion, Limitations, and Future Work

6.1. Conclusion

This thesis investigated whether calibration can be treated as a structural constraint during learning rather than as a correction applied only after a predictive model has been trained. The work combined a theoretical discussion of calibration-regularised empirical risk minimisation with controlled simulation-supervised experiments and an application to gene regulatory network (GRN) edge prediction. The overall conclusion is conditional: end-to-end calibration can improve the reliability of predicted probabilities, and in some settings can improve predictive quality at the same time, but its benefits depend on the data-generating process, the optimisation procedure, the calibration weight, and the evaluation distribution.

The theoretical chapters framed calibration-aware learning through a composite objective that combines a predictive loss with a differentiable calibration surrogate. This formulation provides a way to interpret calibration as an inductive bias that changes which solutions are favoured during training. The analysis connected this view to the calibration–refinement decomposition and to complexity-based and PAC-Bayesian generalisation arguments. These arguments motivate the approach, but they do not imply that every calibration penalty will reduce generalisation error: the conclusion depends on the assumptions and on whether the restricted hypothesis class actually improves the relevant trade-off.

The controlled experiments provide the clearest evidence because the true posterior is known. In the one-dimensional simulation with 30 random seeds, setting $\lambda = 0.1$ reduced posterior MSE from 0.0189 to 0.0101 at $m = 10$, from 0.0482 to 0.0127 at $m = 100$, and from 0.0378 to 0.0129 at $m = 1000$, relative to predictive-only training. ECE decreased from 0.069 to 0.046, from 0.122 to 0.055, and from 0.114 to 0.055, respectively. The reported paired tests found significant differences in 25 of 27 metric–capacity comparisons after correction. The Brier refinement term also decreased in the reported comparisons, indicating that the calibration gains were not necessarily obtained by sacrificing refinement. However, the experiments and ablations qualify a simple regularisation interpretation: the convex weighting changes the effective gradient scale under full-batch gradient descent, and a matched learning-rate control reproduced much of the observed pattern. At $\lambda = 0$, where the predictive loss no longer anchors the solution, performance deteriorated sharply; for $m = 1000$, posterior MSE was 0.0580, with across-seed variance 2.1×10^{-3}. Calibration should therefore complement, not replace, a proper predictive objective.

The controlled covariate-shift experiment further suggests that calibration-aware training can help when the input distribution changes while the conditional label mechanism is held fixed. At $m = 1000$, the predictive-only model's posterior MSE increased from 0.0390 in-distribution to 0.0418 under shift, whereas the $\lambda = 0.05$ model changed from 0.0120 to 0.0130. Across the experiment, 218 of 252 paired comparisons were significant after Bonferroni correction. These tests establish differences between regularisation settings within each evaluation distribution; they do not directly establish that the ID-to-shift degradation differs significantly between settings. The result is encouraging within the controlled setup, but it is not sufficient to claim general robustness under real biological distribution shifts.

The GRN experiments tested whether the approach remains useful when the true edge posterior is unknown. On DREAM5 Network 1, the medium-capacity BCE model trained end-to-end with $\lambda = 0.5$ improved AUROC from 0.857 to 0.882, AUPR from 0.787 to 0.833, and Brier score from 0.136 to 0.120, while ECE_{10} changed from 0.030 to 0.028. These results show that calibration-aware training can be competitive and, for this configuration, improved ranking as well as probability-sensitive metrics. The pattern was not consistent across all settings: for the small DREAM5 model, end-to-end training at $\lambda = 0.5$ reduced AUROC from 0.833 to 0.816 and AUPR from 0.734 to 0.718, while ECE_{10} changed from 0.037 to 0.038. On SERGIO, the medium BCE model's AUROC decreased from 0.629 to 0.611, and ECE_{10} increased from 0.008 to 0.011. Thus, the application results support tuning and evaluating the calibration–prediction trade-off rather than assuming that one setting is uniformly best.

Post-hoc calibration remained a useful comparison. For the small DREAM5 BCE model, isotonic regression reduced ECE_{10} from 0.037 to 0.026, while the Brier score remained 0.152 after rounding. Temperature scaling reduced ECE_{10} for the medium DREAM5 BCE model from 0.030 to 0.028 without changing its ranking metrics. These findings illustrate the complementary roles of the two approaches: post-hoc calibration can be an inexpensive correction, whereas end-to-end calibration can influence the learned predictor itself. Neither approach guarantees that calibration will persist when the evaluation distribution changes.

The shift experiments make this final point particularly clear. In the exploratory single-seed transfer from DREAM5 Network 1 to Network 3, the predictive-only BCE model's ECE_{10} increased from 0.048 on the source test set to 0.152 on Network 3. End-to-end training at $\lambda = 0.1$ yielded a target-network AUPR of 0.310, compared with 0.251 for predictive-only BCE, but its target ECE_{10} remained high at 0.176. This experiment is descriptive rather than conclusive, because the networks differ in prevalence and potentially in biological and technical properties, and only one seed was used.

Taken together, the thesis answers the central research question with a qualified yes. In the controlled setting, a suitable calibration-aware objective improved posterior accuracy and several calibration measures simultaneously. In the GRN application, the effects were more heterogeneous, and distribution shift remained a substantial challenge. The practical lesson is to evaluate probabilistic models using calibration diagnostics, proper scoring rules, and ranking metrics together, and to validate them on data representative of the intended deployment setting. A low ECE alone is not sufficient evidence of a useful or trustworthy model.

6.2. Limitations

Several limitations constrain the interpretation and generalisability of the findings.

Restricted controlled data-generating process. The main simulation uses a single one-dimensional problem with a smooth true posterior, a fixed training sample size of $n = 100$, and one model family. Even a model with $m = 10$ can represent the target closely, limiting what the experiment can reveal about capacity-dependent effects. The numerical results should not be assumed to transfer to high-dimensional problems, more irregular posteriors, or other model families.

Optimisation and objective confounding. The principal controlled experiments use full-batch gradient descent with a fixed learning rate and epoch budget. Under this optimiser, changing the convex weight λ also changes the scale of the composite gradient and therefore the effective step size. The matched-learning-rate ablation reproduced much of the observed effect, so the experiments do not isolate calibration regularisation as the sole mechanism. The soft-ECE gradient uses a straight-through approximation, with $B = 15$ bins and bandwidth $h = 0.08$; these choices were not systematically varied. Other optimisers, including adaptive methods, may behave differently.

Uneven statistical support across experiments. The main controlled comparison uses 30 seeds, but several supporting analyses—including training dynamics, base-loss comparisons, and the post-hoc comparison—use a single seed. The post-hoc comparison also uses a single split and different numbers of training points for the end-to-end and post-hoc pipelines. Consequently, not all observed differences have the same level of statistical support. In addition, most reported choices of λ use the test-set posterior MSE as an oracle criterion, except for the validation-based ablation. This makes the oracle results optimistic relative to a real application in which the true posterior is unavailable. Selecting λ from 20 validation points incurred an approximately 0.002–0.003 increase in posterior MSE in the reported experiment.

Calibration metrics are estimators. Binned ECE and MCE depend on the binning scheme and sample counts. In particular, MCE can be dominated by sparsely populated bins. A change in one metric therefore does not establish that every notion of calibration has improved. The experiments mitigate this issue by reporting multiple calibration diagnostics and proper scoring rules, but finite-sample estimates remain imperfect.

No known true posterior for the GRN tasks. Unlike the controlled experiment, the GRN datasets do not provide the true conditional probability that a candidate TF–target pair is a regulatory edge. The reported ECE, Brier score, and related quantities measure empirical behaviour on the constructed evaluation set; they cannot establish exact conditional calibration

or posterior MSE. The labels also represent the available gold-standard network, which may not contain every true regulatory interaction.

Candidate-pair sampling and dependence. The DREAM5 pair dataset uses sampled negative examples. Its positive prevalence therefore differs from the prevalence among all possible TF–target pairs, and calibration metrics are conditional on the sampled candidate set. Training and test pairs may share genes, so examples are not fully independent and the split does not directly measure generalisation to unseen genes. Results should not be compared directly with official DREAM5 challenge scores unless the candidate construction and evaluation protocol are aligned.

Artificial and confounded distribution shifts. The affine expression-feature transformation is a controlled stress test, not a validated representation of a particular biological intervention. Although labels are held fixed in this experiment, that design alone does not demonstrate that the formal covariate-shift assumption holds in a real biological setting. The Network 1-to-Network 3 transfer changes the data source and reference network and involves different positive prevalences; it therefore conflates changes in expression distributions, network structure, organism, label coverage, and prevalence. It cannot isolate covariate shift as the sole cause of performance degradation.

6.3. Future Work

The results suggest several concrete directions for further investigation.

Broaden the controlled simulation study. The theoretical and empirical claims should be tested across higher-dimensional inputs, less smooth and more heterogeneous true posteriors, multiple sample sizes, and model families with genuinely different approximation capacities. Repeating the experiments with additional random seeds and independently generated data-generating processes would help determine when calibration-aware training improves posterior estimation, and when it merely changes optimisation dynamics.

Separate regularisation from optimisation effects. Future experiments should compare the composite objective against predictive-only training under carefully matched optimisation conditions. This includes explicit learning-rate controls, adaptive optimisers, matched training budgets, and systematic variation of the calibration surrogate and its hyperparameters. The calibration-only failure at $\lambda = 0$ also motivates studying principled constraints or lower bounds on the contribution of the proper predictive loss. Such experiments would help identify whether improvements arise from calibration information, altered gradient scale, or their interaction.

Use validation-based model and weight selection. The oracle selection of λ based on test posterior MSE is useful for understanding the best attainable result in a known-posterior simulation, but is unavailable in practice. Future work should prioritise validation procedures that select λ using only observable quantities, assess their stability with small validation sets, and compare them with early stopping and post-hoc calibration. For biomedical applications, validation data should reflect the expected deployment population whenever possible.

Expand calibration evaluation. A more comprehensive evaluation should report multiple calibration estimators, proper scoring rules, and ranking measures together. It should examine sensitivity to binning, sample size, and class prevalence, and distinguish marginal calibration from conditional calibration. For GRN inference, calibration diagnostics should be accompanied by careful reporting of the candidate-pair sampling scheme and positive prevalence so that the meaning of the predicted probabilities is explicit.

Conduct systematic DREAM5 transfer experiments. The cross-network analysis should be expanded to multiple DREAM5 networks and transfer directions, with more random seeds and a harmonised TF–target candidate protocol. Where feasible, target-domain validation data should be used to assess recalibration. These experiments should separate changes in ranking from changes in probability reliability and should avoid attributing a combined biological and technical shift solely to covariate shift.

Strengthen the biological benchmark. Future GRN studies should compare the calibrated Transformer not only with its predictive-only counterpart but also with established inference methods such as GENIE3, GRNBoost2, ARACNE, and TIGRESS under a consistent protocol. Evaluations should consider gene-level or network-level holdouts to test generalisation to unseen genes or networks, and should examine whether calibrated edge probabilities support better prioritisation of candidate interactions for experimental follow-up. Where suitable biological validation is available, the analysis should also assess whether the model's probability estimates remain meaningful across conditions and datasets.

Explore alternative calibration objectives. The present study uses differentiable calibration surrogates within a composite objective. Further work could compare soft-binned objectives with kernel-based estimators and other differentiable calibration losses, investigate the effect of the predictive base loss, and evaluate whether calibration constraints can be designed to target specific deployment requirements. Any such extension should be assessed for both calibration and predictive utility, since optimising a calibration metric alone can produce an uninformative predictor.

In summary, the next step is not simply to increase the weight assigned to calibration. It is to identify the conditions under which calibration-aware learning offers a reproducible advantage, separate that advantage from optimisation artefacts, and test whether probability

reliability persists under the distribution changes encountered in biomedical applications.

A. Appendix: Additional Details Simulation-Supervised Experiments

A.1. Optional Task: Calibration under Controlled Covariate Shift

The optional task asks whether end-to-end calibration improves robustness when the input distribution changes, and how marginal and conditional calibration behave under that change. This experiment provides a controlled investigation using the same one-dimensional problem as the preceding sections, where the true posterior $\eta(x)$ is known. It complements, but does not replace, the cross-network DREAM5 analysis: here, the conditional label mechanism can be held fixed while the input distribution changes.

Experimental design. The training and in-distribution (ID) test inputs are sampled uniformly from $[-1, 1]$. For the shifted test distribution, the density is tilted according to $q(x) \propto \exp(\alpha x)$ with $\alpha = 1.5$, while the support remains $[-1, 1]$. The conditional distribution $P(Y = 1 \mid X = x) = \eta(x)$ is unchanged. This is therefore a controlled covariate-shift setting, rather than a setting in which the label mechanism also changes. Figure A.1 shows the two input distributions.

The experiment compares predictive-only training ($\lambda = 1$) with calibration-aware objectives at $\lambda \in \{0.5, 0.1, 0.05\}$. The models use seven Legendre degrees, $m \in \{5, 10, 100, 120, 200, 300, 1000\}$. Each configuration is evaluated over 30 random seeds on both the ID and shifted test distributions. The objective weights the predictive cross-entropy term by λ and the soft calibration penalty by $1 - \lambda$. Thus, smaller values of λ place greater emphasis on the calibration penalty. The tested values are intended to examine the effect of this trade-off, not to establish a universal optimal setting.

Evaluation metrics. Figure A.2 reports six metrics: top-label expected calibration error (ECE), a marginal probability calibration gap evaluated against the known posterior, conditional absolute error $\mathbb{E}_{Q_X}[|p(X) - \eta(X)|]$, posterior mean squared error $\mathbb{E}_{Q_X}[(p(X) - \eta(X))^2]$, Brier score, and cross-entropy. The posterior-based quantities can be evaluated directly against $\eta(x)$, without introducing Bernoulli label noise. Their values are nevertheless weighted by the test input distribution, so they may change under covariate shift even though $\eta(x)$ itself remains invariant. The marginal gap is a diagnostic based on the known posterior and should

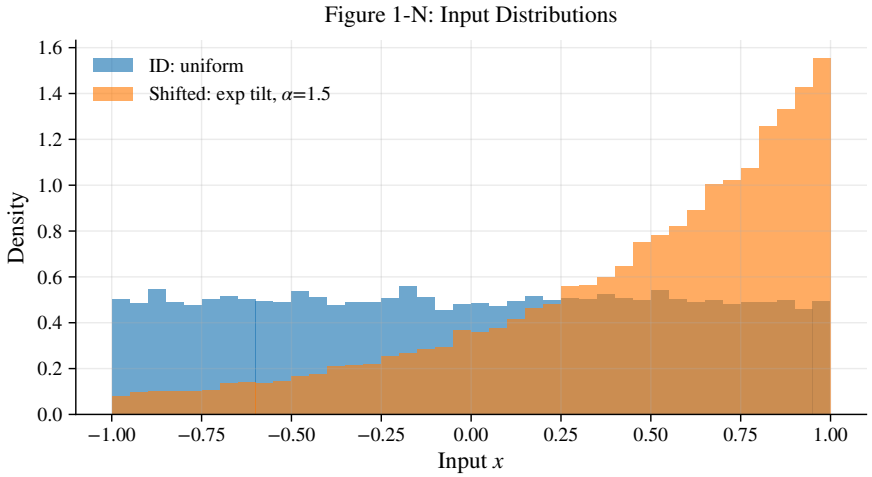


Figure A.1.: Input distributions in the controlled covariate-shift experiment. The ID distribution is uniform on $[-1, 1]$, whereas the shifted distribution applies an exponential tilt with $\alpha = 1.5$. The conditional label mechanism remains unchanged.

not be interpreted as a general guarantee of marginal calibration.

Results across capacities. Across the tested capacities, the predictive-only baseline generally exhibits larger calibration errors and posterior errors than the objectives with smaller λ. This pattern is visible on both the ID and shifted distributions. The difference is particularly clear for the more flexible models, where predictive-only training has difficulty maintaining the same level of calibration and posterior accuracy.

The results also show that the two smaller weights do not dominate each other on every metric and capacity. In particular, $\lambda = 0.05$ and $\lambda = 0.1$ give similar results in several settings. The appropriate conclusion is therefore not that one value is uniformly optimal, but that placing greater weight on the calibration penalty can improve the probability estimates in this controlled experiment.

Numerical comparison at the largest capacity. Table A.1 reports the mean and standard deviation for each metric at $m = 1000$. Reporting variability is important here: the standard deviations quantify how much the observed metrics vary across the 30 random seeds, whereas the means alone would hide this variation.

At $m = 1000$, the top-label ECE of the predictive-only baseline increases from 0.1115 on ID data to 0.1193 under shift. For $\lambda = 0.05$, it remains nearly unchanged, at 0.0505 and 0.0506, respectively. Posterior MSE shows a similar pattern: it increases from 0.0390 to 0.0418 for the baseline, compared with an increase from 0.0120 to 0.0130 for $\lambda = 0.05$. The marginal calibration gap and conditional absolute error are also lower for the calibration-aware objectives on both distributions.

Figure 1-O: Evaluation Metrics under Covariate Shift

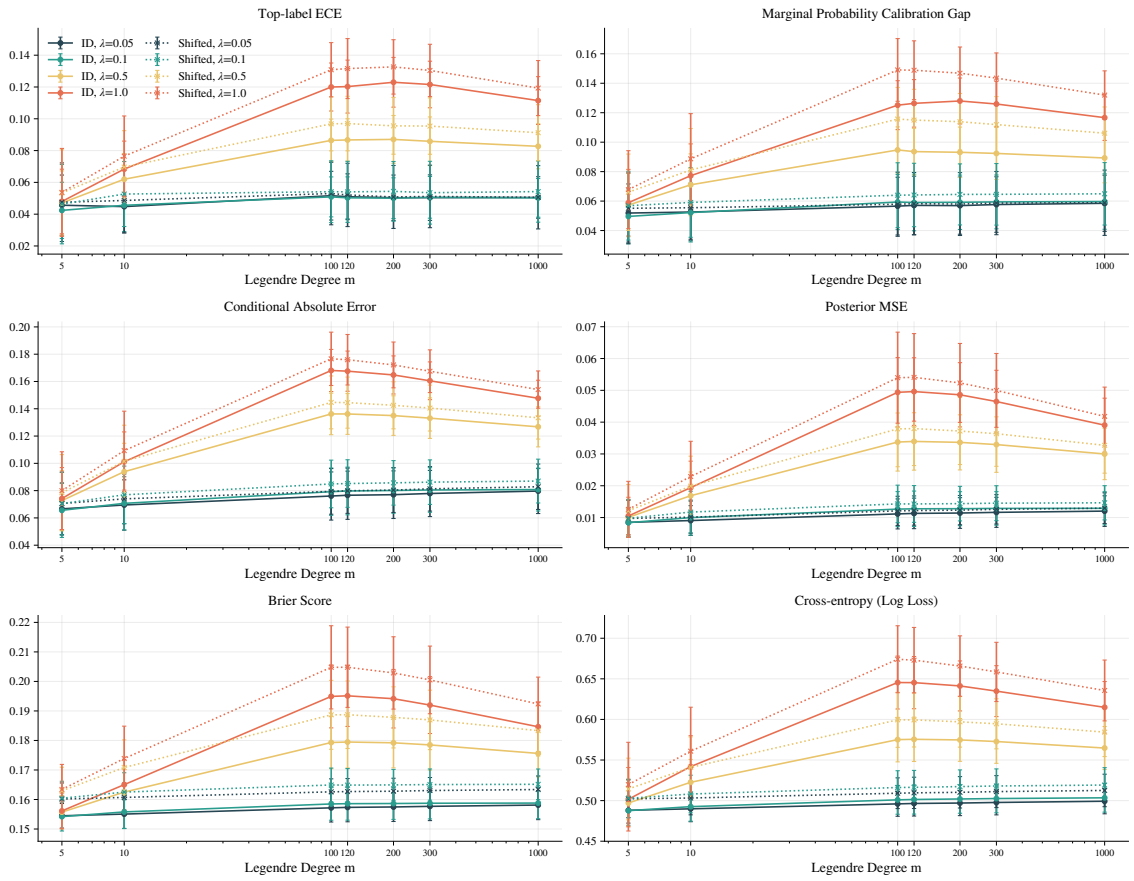


Figure A.2.: Evaluation metrics under the ID and shifted input distributions. Curves show means across 30 seeds and error bars show one standard deviation. Results are reported separately for each model capacity and objective weight.

Table A.1.: Evaluation metrics at Legendre degree $m = 1000$, reported as mean ± standard deviation across 30 seeds. Lower values indicate smaller errors for all reported metrics.

λ	Distribution	Top-label ECE	Marginal gap	Conditional error	MSE_η	Brier	Cross-entropy
1.00	ID	0.1115 ± 0.0150	0.1167 ± 0.0155	0.1477 ± 0.0132	0.0390 ± 0.0085	0.1846 ± 0.0084	0.6149 ± 0.0318
1.00	Shifted	0.1193 ± 0.0173	0.1320 ± 0.0165	0.1541 ± 0.0136	0.0418 ± 0.0092	0.1924 ± 0.0091	0.6357 ± 0.0375
0.50	ID	0.0827 ± 0.0142	0.0893 ± 0.0169	0.1268 ± 0.0147	0.0300 ± 0.0081	0.1756 ± 0.0080	0.5648 ± 0.0265
0.50	Shifted	0.0912 ± 0.0176	0.1061 ± 0.0178	0.1334 ± 0.0156	0.0327 ± 0.0088	0.1833 ± 0.0086	0.5844 ± 0.0301
0.10	ID	0.0503 ± 0.0121	0.0597 ± 0.0188	0.0808 ± 0.0157	0.0129 ± 0.0052	0.1588 ± 0.0053	0.5034 ± 0.0176
0.10	Shifted	0.0542 ± 0.0192	0.0649 ± 0.0212	0.0870 ± 0.0161	0.0146 ± 0.0054	0.1651 ± 0.0052	0.5192 ± 0.0214
0.05	ID	0.0505 ± 0.0131	0.0585 ± 0.0191	0.0797 ± 0.0165	0.0120 ± 0.0048	0.1581 ± 0.0049	0.4992 ± 0.0155
0.05	Shifted	0.0506 ± 0.0199	0.0589 ± 0.0222	0.0828 ± 0.0166	0.0130 ± 0.0048	0.1634 ± 0.0046	0.5125 ± 0.0200

These results are notable for two reasons. First, the calibration-aware objectives have lower errors on both distributions, so the comparison is not driven solely by a small change between ID and shifted performance. Second, the differences between the ID and shifted results are generally smaller for the calibration-aware objectives on the reported metrics. However, the latter observation is descriptive and does not, by itself, establish a statistically significant improvement in robustness.

Paired statistical comparisons. To assess whether the observed differences between regularization settings were statistically significant, paired Wilcoxon signed-rank tests were conducted across the 30 seeds. Each setting $\lambda \in \{0.05, 0.1, 0.5\}$ was compared with the baseline $\lambda = 1.0$, separately for each model capacity, evaluation distribution, and metric. Bonferroni correction is applied across the 252 comparisons recorded in the paired-test results.

After correction, 218 of the 252 comparisons are significant at the 5% level, the remaining comparisons are not significant after correction. The results therefore provide broad evidence that the choice of regularization strength affects the evaluated calibration and predictive metrics, although the significance of individual comparisons varies across capacities, metrics, and distributions. The complete test results, including raw and Bonferroni-adjusted p-values, are reported in Appendix A.2. These tests compare regularization settings within each distribution; they do not directly test whether the change from the in-distribution to the shifted setting differs significantly between regularization settings.

Interpretation and limitations. The controlled experiment provides evidence that calibration-aware training can improve calibration diagnostics and posterior accuracy under a change in the input distribution, while the conditional label mechanism remains fixed. At the tested smaller values of λ, the improvements are visible on both ID and shifted data, including at the largest model capacity. The findings are consistent with the hypothesis that incorporating a calibration penalty into training can mitigate the deterioration observed with predictive-only training in this synthetic setting.

The experiment does not establish that the same benefit will transfer to real biological networks. Cross-network transfer may involve changes in input representations, graph structure, and the conditional label mechanism, not only changes in the input distribution. The present results should therefore be read as a controlled investigation of the optional task, while the DREAM5 cross-network experiment remains a separate empirical question.

A.2. Paired Tests for the Covariate-Shift Experiment

This appendix reports paired Wilcoxon signed-rank tests comparing each regularization setting $\lambda \in \{0.05, 0.1, 0.5\}$ with the baseline $\lambda = 1.0$. Tests are conducted separately for each model capacity m, evaluation distribution (in-distribution (ID) or shifted), and metric. The

reported adjusted p-values use the Bonferroni correction as provided in the experiment's results file. Significance is assessed at $\alpha = 0.05$. The table contains 252 comparisons, of which 218 are significant after correction. These tests compare regularization settings within each distribution; they do not directly test the difference between the ID-to-shift change across settings.

Table A.2.: Paired Wilcoxon signed-rank tests for the covariate-shift experiment.

m	Distribution	Metric	Comparison	p_{raw}	p_{Bonf}	Significant
5	ID	Brier score	0.05 vs. 1.0	0.158	1.000	No
5	ID	Brier score	0.1 vs. 1.0	0.003	0.753	No
5	ID	Brier score	0.5 vs. 1.0	0.001	0.285	No
5	ID	Conditional error	0.05 vs. 1.0	0.016	1.000	No
5	ID	Conditional error	0.1 vs. 1.0	6.08e-04	0.153	No
5	ID	Conditional error	0.5 vs. 1.0	4.18e-04	0.105	No
5	ID	Cross-entropy	0.05 vs. 1.0	0.025	1.000	No
5	ID	Cross-entropy	0.1 vs. 1.0	8.72e-04	0.220	No
5	ID	Cross-entropy	0.5 vs. 1.0	3.45e-04	0.087	No
5	ID	Top-label ECE	0.05 vs. 1.0	0.529	1.000	No
5	ID	Top-label ECE	0.1 vs. 1.0	0.017	1.000	No
5	ID	Top-label ECE	0.5 vs. 1.0	0.019	1.000	No
5	ID	Marginal gap	0.05 vs. 1.0	0.021	1.000	No
5	ID	Marginal gap	0.1 vs. 1.0	3.05e-05	0.008	Yes
5	ID	Marginal gap	0.5 vs. 1.0	4.18e-04	0.105	No
5	ID	MSE_η	0.05 vs. 1.0	0.084	1.000	No
5	ID	MSE_η	0.1 vs. 1.0	0.002	0.399	No
5	ID	MSE_η	0.5 vs. 1.0	0.001	0.262	No
5	Shifted	Brier score	0.05 vs. 1.0	0.164	1.000	No
5	Shifted	Brier score	0.1 vs. 1.0	0.004	1.000	No
5	Shifted	Brier score	0.5 vs. 1.0	0.045	1.000	No
5	Shifted	Conditional error	0.05 vs. 1.0	0.129	1.000	No
5	Shifted	Conditional error	0.1 vs. 1.0	0.003	0.876	No
5	Shifted	Conditional error	0.5 vs. 1.0	0.023	1.000	No
5	Shifted	Cross-entropy	0.05 vs. 1.0	0.014	1.000	No
5	Shifted	Cross-entropy	0.1 vs. 1.0	0.002	0.509	No
5	Shifted	Cross-entropy	0.5 vs. 1.0	0.033	1.000	No
5	Shifted	Top-label ECE	0.05 vs. 1.0	0.124	1.000	No
5	Shifted	Top-label ECE	0.1 vs. 1.0	0.009	1.000	No
5	Shifted	Top-label ECE	0.5 vs. 1.0	0.516	1.000	No
5	Shifted	Marginal gap	0.05 vs. 1.0	0.003	0.645	No
5	Shifted	Marginal gap	0.1 vs. 1.0	1.70e-04	0.043	Yes
5	Shifted	Marginal gap	0.5 vs. 1.0	0.009	1.000	No
5	Shifted	MSE_η	0.05 vs. 1.0	0.253	1.000	No
5	Shifted	MSE_η	0.1 vs. 1.0	0.007	1.000	No
5	Shifted	MSE_η	0.5 vs. 1.0	0.050	1.000	No
10	ID	Brier score	0.05 vs. 1.0	3.73e-09	9.39e-07	Yes
10	ID	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
10	ID	Brier score	0.5 vs. 1.0	3.73e-09	9.39e-07	Yes
10	ID	Conditional error	0.05 vs. 1.0	3.73e-09	9.39e-07	Yes
10	ID	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
10	ID	Conditional error	0.5 vs. 1.0	3.73e-09	9.39e-07	Yes
10	ID	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
10	ID	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
10	ID	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
10	ID	Top-label ECE	0.05 vs. 1.0	1.99e-06	5.02e-04	Yes
10	ID	Top-label ECE	0.1 vs. 1.0	3.54e-08	8.92e-06	Yes
10	ID	Top-label ECE	0.5 vs. 1.0	3.54e-08	8.92e-06	Yes
10	ID	Marginal gap	0.05 vs. 1.0	9.98e-07	2.52e-04	Yes
10	ID	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes

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m	Distribution	Metric	Comparison	p_{raw}	p_{Bonf}	Significant
10	ID	Marginal gap	0.5 vs. 1.0	8.01e-08	2.02e-05	Yes
10	ID	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
10	ID	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
10	ID	MSE_η	0.5 vs. 1.0	3.73e-09	9.39e-07	Yes
10	Shifted	Brier score	0.05 vs. 1.0	2.61e-08	6.57e-06	Yes
10	Shifted	Brier score	0.1 vs. 1.0	3.73e-09	9.39e-07	Yes
10	Shifted	Brier score	0.5 vs. 1.0	5.59e-09	1.41e-06	Yes
10	Shifted	Conditional error	0.05 vs. 1.0	1.64e-07	4.13e-05	Yes
10	Shifted	Conditional error	0.1 vs. 1.0	3.73e-09	9.39e-07	Yes
10	Shifted	Conditional error	0.5 vs. 1.0	1.86e-08	4.69e-06	Yes
10	Shifted	Cross-entropy	0.05 vs. 1.0	9.31e-09	2.35e-06	Yes
10	Shifted	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
10	Shifted	Cross-entropy	0.5 vs. 1.0	3.73e-09	9.39e-07	Yes
10	Shifted	Top-label ECE	0.05 vs. 1.0	5.14e-06	0.001	Yes
10	Shifted	Top-label ECE	0.1 vs. 1.0	4.71e-07	1.19e-04	Yes
10	Shifted	Top-label ECE	0.5 vs. 1.0	2.76e-06	6.96e-04	Yes
10	Shifted	Marginal gap	0.05 vs. 1.0	3.79e-06	9.55e-04	Yes
10	Shifted	Marginal gap	0.1 vs. 1.0	3.15e-07	7.93e-05	Yes
10	Shifted	Marginal gap	0.5 vs. 1.0	1.64e-07	4.13e-05	Yes
10	Shifted	MSE_η	0.05 vs. 1.0	2.61e-08	6.57e-06	Yes
10	Shifted	MSE_η	0.1 vs. 1.0	3.73e-09	9.39e-07	Yes
10	Shifted	MSE_η	0.5 vs. 1.0	5.59e-09	1.41e-06	Yes
100	ID	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	ID	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
100	Shifted	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes

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Table A.2 – continued from previous page

m	Distribution	Metric	Comparison	p_{raw}	p_{Bonf}	Significant
120	ID	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	ID	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
120	Shifted	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	ID	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes

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m	Distribution	Metric	Comparison	p_{raw}	p_{Bonf}	Significant
200	Shifted	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
200	Shifted	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	ID	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
300	Shifted	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes

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m	Distribution	Metric	Comparison	p_{raw}	p_{Bonf}	Significant
1000	ID	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	ID	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Brier score	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Brier score	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Brier score	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Conditional error	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Conditional error	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Conditional error	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Cross-entropy	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Cross-entropy	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Cross-entropy	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Top-label ECE	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Top-label ECE	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Top-label ECE	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Marginal gap	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Marginal gap	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	Marginal gap	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	MSE_η	0.05 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	MSE_η	0.1 vs. 1.0	1.86e-09	4.69e-07	Yes
1000	Shifted	MSE_η	0.5 vs. 1.0	1.86e-09	4.69e-07	Yes

B. Appendix: Additional Details for GRN Calibration Experiments

B.1. Hyperparameters and Experimental Settings

Table B.1 summarizes the settings used in the main and controlled-shift experiments. The cross-network experiment is a separate exploratory run and uses a single seed.

Table B.1.: Hyperparameters and protocol for the GRN calibration experiments.

Setting	Value
Input representation	TF and target expression channels, each with at most 64 cells/conditions
Candidate-pair split	Stratified pair-level train/validation/test split
Main/controlled-shift seeds	42, 123, 2026
Cross-network seed	42
Architectures	Small and medium encoder-only Transformer
Base losses	Binary cross-entropy (BCE), Brier loss
End-to-end objective	$(1-\lambda)\widetilde{\mathcal{C}}_{\text{soft}} + \lambda\mathcal{L}_{\text{base}}$
λ values	1.0 (predictive-only), 0.5, 0.1
Post-hoc calibration	Temperature scaling and isotonic regression, fitted on validation predictions
Maximum epochs	30
Batch size	32
Learning rate	10^{-3}
Early stopping	Validation-based, patience 5
ECE evaluation	10 bins
SERGIO network	25 master regulators, 50 TFs, 100 target genes; 1 bin and 300 cells
SERGIO candidate pairs	5,000 TF–target pairs; 303 positives in the saved pair dataset
DREAM5 main candidate set	12,000 sampled candidate pairs for Network 1
DREAM5 cross-network	Network 1 source, Network 3 target; 12,000 candidate pairs in each saved dataset
Controlled feature shift	TF: $1.5x + 0.5$; target: $0.75x - 0.25$

The encoder layer uses batch-first sequence tensors and pre-layer normalisation (‘norm_first=True‘). The encoder feed-forward width is twice the model width. Mean pooling combines the encoded condition positions into a fixed-length representation before the prediction head.

Table B.2.: Exact Transformer architecture configurations from the experiment utility.

Component	Small	Medium
Input features per position	2	2
Input projection	Linear($2 \to 16$)	Linear($2 \to 32$)
Model width d_{model}	16	32
Attention heads	2	4
Transformer encoder layers	1	2
Encoder feed-forward width	32	64
Encoder dropout	0.1	0.1
Pooling	Mean over positions	Mean over positions
Prediction head	LayerNorm, Linear($16 \to 16$), GELU, Dropout(0.1), Linear($16 \to 1$)	LayerNorm, Linear($32 \to 32$), GELU, Dropout(0.1), Linear($32 \to 1$)
Output	One logit per TF–target pair	One logit per TF–target pair

B.2. Additional Notes on Data Preparation

For SERGIO, the expression matrix is stored with genes as rows and cells as columns. Non-finite values are replaced by zero, a $\log(1+x)$ transform is applied only when the non-negative expression values have a high upper percentile, and each gene is standardised across cells. A reproducibly selected subset of at most 64 cells is retained. The pair dataset enumerates every TF–target combination and assigns the label from the simulated TF-to-target adjacency matrix.

For DREAM5, the expression matrix is matched to its corresponding gold-standard edge list. The preparation utility maps gene identifiers to expression rows, constructs labelled candidate pairs, and samples negatives to limit the candidate set size. The saved main Network 1 pair set has 12,000 candidates. A separate Network 3 pair set was also prepared for the cross-network experiment. As the preparation notebook notes, a general TF-only candidate restriction is not applied unless an explicit TF list is supplied; this should be retained as a limitation when interpreting the results.

B.3. Complete Main Experiment Results

Table B.3 reproduces the complete seed-aggregated table of the main experiment. Each metric is reported as mean $\pm$ standard deviation over the three seeds (42, 123, and 2026). The table includes both model capacities, both base losses, all three objective weights, and both post-hoc calibration methods. For post-hoc methods, $\lambda = 1$ identifies the predictive-only model from which the validation-fitted calibrator was derived.

Table B.3.: Complete three-seed main experiment results: mean ± standard deviation.

Dataset	Model	Loss	Method	λ	AUROC	AUPR	Brier	ECE10
dream5_net1	medium	bce	end_to_end	0.1	0.8291 ± 0.0201	0.7474 ± 0.0376	0.1486 ± 0.0114	0.0275 ± 0.0067
dream5_net1	medium	bce	end_to_end	0.5	0.8819 ± 0.0237	0.8327 ± 0.0327	0.1200 ± 0.0119	0.0281 ± 0.0096
dream5_net1	medium	bce	isotonic_regression	1.0	0.8550 ± 0.0326	0.7645 ± 0.0588	0.1360 ± 0.0175	0.0269 ± 0.0053
dream5_net1	medium	bce	predictive_only	1.0	0.8570 ± 0.0321	0.7872 ± 0.0517	0.1362 ± 0.0175	0.0302 ± 0.0059
dream5_net1	medium	bce	temperature_scaling	1.0	0.8570 ± 0.0321	0.7872 ± 0.0517	0.1360 ± 0.0178	0.0284 ± 0.0055
dream5_net1	medium	brier	end_to_end	0.1	0.6990 ± 0.0447	0.5741 ± 0.0529	0.1959 ± 0.0128	0.0376 ± 0.0069
dream5_net1	medium	brier	end_to_end	0.5	0.8454 ± 0.0287	0.7764 ± 0.0462	0.1395 ± 0.0153	0.0249 ± 0.0054
dream5_net1	medium	brier	isotonic_regression	1.0	0.8707 ± 0.0172	0.7979 ± 0.0258	0.1254 ± 0.0093	0.0230 ± 0.0095
dream5_net1	medium	brier	predictive_only	1.0	0.8723 ± 0.0178	0.8171 ± 0.0264	0.1247 ± 0.0101	0.0216 ± 0.0063
dream5_net1	medium	brier	temperature_scaling	1.0	0.8723 ± 0.0178	0.8171 ± 0.0264	0.1247 ± 0.0101	0.0230 ± 0.0105
dream5_net1	small	bce	end_to_end	0.1	0.7943 ± 0.0088	0.6858 ± 0.0204	0.1662 ± 0.0035	0.0222 ± 0.0065
dream5_net1	small	bce	end_to_end	0.5	0.8164 ± 0.0109	0.7177 ± 0.0189	0.1573 ± 0.0045	0.0378 ± 0.0025
dream5_net1	small	bce	isotonic_regression	1.0	0.8298 ± 0.0133	0.7129 ± 0.0144	0.1518 ± 0.0060	0.0260 ± 0.0159
dream5_net1	small	bce	predictive_only	1.0	0.8330 ± 0.0149	0.7344 ± 0.0175	0.1518 ± 0.0052	0.0368 ± 0.0101
dream5_net1	small	bce	temperature_scaling	1.0	0.8330 ± 0.0149	0.7344 ± 0.0175	0.1515 ± 0.0051	0.0329 ± 0.0090
dream5_net1	small	brier	end_to_end	0.1	0.7404 ± 0.0425	0.6214 ± 0.0580	0.1826 ± 0.0105	0.0252 ± 0.0063
dream5_net1	small	brier	end_to_end	0.5	0.7994 ± 0.0161	0.6920 ± 0.0224	0.1639 ± 0.0064	0.0228 ± 0.0105
dream5_net1	small	brier	isotonic_regression	1.0	0.8248 ± 0.0204	0.7107 ± 0.0225	0.1526 ± 0.0095	0.0261 ± 0.0153
dream5_net1	small	brier	predictive_only	1.0	0.8262 ± 0.0214	0.7331 ± 0.0242	0.1525 ± 0.0093	0.0329 ± 0.0083
dream5_net1	small	brier	temperature_scaling	1.0	0.8262 ± 0.0214	0.7331 ± 0.0242	0.1522 ± 0.0092	0.0297 ± 0.0076
sergio	medium	bce	end_to_end	0.1	0.5902 ± 0.0370	0.1183 ± 0.0447	0.0564 ± 0.0010	0.0058 ± 0.0008
sergio	medium	bce	end_to_end	0.5	0.6111 ± 0.0193	0.1366 ± 0.0276	0.0562 ± 0.0009	0.0106 ± 0.0052
sergio	medium	bce	isotonic_regression	1.0	0.6135 ± 0.0163	0.1083 ± 0.0117	0.0561 ± 0.0013	0.0077 ± 0.0056
sergio	medium	bce	predictive_only	1.0	0.6287 ± 0.0061	0.1417 ± 0.0276	0.0558 ± 0.0009	0.0075 ± 0.0020
sergio	medium	bce	temperature_scaling	1.0	0.6287 ± 0.0061	0.1417 ± 0.0276	0.0558 ± 0.0009	0.0074 ± 0.0020
sergio	medium	brier	end_to_end	0.1	0.5647 ± 0.0522	0.0895 ± 0.0334	0.0568 ± 0.0007	0.0014 ± 0.0017
sergio	medium	brier	end_to_end	0.5	0.5892 ± 0.0376	0.1182 ± 0.0429	0.0564 ± 0.0010	0.0042 ± 0.0042
sergio	medium	brier	isotonic_regression	1.0	0.6042 ± 0.0030	0.1065 ± 0.0175	0.0562 ± 0.0024	0.0100 ± 0.0134
sergio	medium	brier	predictive_only	1.0	0.6226 ± 0.0036	0.1397 ± 0.0283	0.0559 ± 0.0009	0.0064 ± 0.0035
sergio	medium	brier	temperature_scaling	1.0	0.6226 ± 0.0036	0.1397 ± 0.0283	0.0559 ± 0.0009	0.0065 ± 0.0022
sergio	small	bce	end_to_end	0.1	0.6139 ± 0.0194	0.1130 ± 0.0155	0.0566 ± 0.0008	0.0146 ± 0.0131
sergio	small	bce	end_to_end	0.5	0.6223 ± 0.0062	0.1324 ± 0.0256	0.0559 ± 0.0010	0.0107 ± 0.0034
sergio	small	bce	isotonic_regression	1.0	0.6200 ± 0.0075	0.1152 ± 0.0126	0.0568 ± 0.0015	0.0098 ± 0.0055
sergio	small	bce	predictive_only	1.0	0.6201 ± 0.0051	0.1324 ± 0.0236	0.0559 ± 0.0009	0.0092 ± 0.0018
sergio	small	bce	temperature_scaling	1.0	0.6201 ± 0.0051	0.1324 ± 0.0236	0.0559 ± 0.0010	0.0108 ± 0.0045
sergio	small	brier	end_to_end	0.1	0.5649 ± 0.0438	0.0914 ± 0.0329	0.0568 ± 0.0007	0.0077 ± 0.0064
sergio	small	brier	end_to_end	0.5	0.5834 ± 0.0482	0.1008 ± 0.0370	0.0569 ± 0.0008	0.0164 ± 0.0078
sergio	small	brier	isotonic_regression	1.0	0.5799 ± 0.0566	0.0931 ± 0.0267	0.0568 ± 0.0015	0.0080 ± 0.0070
sergio	small	brier	predictive_only	1.0	0.5956 ± 0.0485	0.1008 ± 0.0316	0.0563 ± 0.0006	0.0045 ± 0.0041
sergio	small	brier	temperature_scaling	1.0	0.5956 ± 0.0485	0.1008 ± 0.0316	0.0563 ± 0.0006	0.0072 ± 0.0054

B.4. Complete Controlled Feature-Shift Results

Table B.4 shows all the results of the controlled-feature shift experiment. Results are reported as mean ± standard deviation across three seeds. The same held-out candidate pairs and labels are used for the in-distribution and affine-shifted evaluations.

B.5. Complete DREAM5 Cross-Network Results

Table B.5 displays the results from the cross-network experiment. This experiment uses one seed and is therefore exploratory; no across-seed standard deviation is available. The source is DREAM5 Network 1 (in silico) and the target is DREAM5 Network 3 (E. coli).

Table B.5.: Complete cross-network results, single seed.

Evaluation	Method	λ	Preval.	AUROC	AUPR	Brier	ECE10	Cal.-in-large	Score-cond. gap
source_test	BCE	1.0	0.3328	0.8335	0.7457	0.1515	0.0480	0.0480	0.0480
source_test	BCE+TS	1.0	0.3328	0.8335	0.7457	0.1509	0.0410	0.0352	0.0410
cross_network	BCE	1.0	0.1722	0.6060	0.2512	0.1828	0.1521	0.0834	0.1521
cross_network	BCE+TS	1.0	0.1722	0.6060	0.2512	0.1803	0.1488	0.0978	0.1488
source_test	end_to_end	0.5	0.3328	0.8159	0.7294	0.1553	0.0400	0.0062	0.0400
cross_network	end_to_end	0.5	0.1722	0.6189	0.2639	0.2003	0.1832	0.1682	0.1832
source_test	end_to_end	0.1	0.3328	0.8039	0.7055	0.1626	0.0291	0.0133	0.0291
cross_network	end_to_end	0.1	0.1722	0.6517	0.3096	0.1980	0.1762	0.1622	0.1762

Table B.4.: Complete controlled feature-shift results: mean ± standard deviation.

Dataset	Model	Method	Test split	AUROC	AUPR	Brier	ECE10	Cal.-in-large
dream5_net1	medium	end_to_end_lambda_0.1	in_distribution	0.8281 ± 0.0123	0.7603 ± 0.0182	0.1458 ± 0.0080	0.0317 ± 0.0101	0.0111 ± 0.0037
dream5_net1	medium	end_to_end_lambda_0.1	shifted_features	0.6989 ± 0.0479	0.5625 ± 0.0556	0.2128 ± 0.0212	0.1194 ± 0.0416	0.0969 ± 0.0550
dream5_net1	medium	end_to_end_lambda_0.5	in_distribution	0.8758 ± 0.0311	0.8222 ± 0.0409	0.1225 ± 0.0146	0.0241 ± 0.0044	0.0092 ± 0.0051
dream5_net1	medium	end_to_end_lambda_0.5	shifted_features	0.7607 ± 0.0324	0.6437 ± 0.0440	0.1937 ± 0.0048	0.1175 ± 0.0271	0.1153 ± 0.0262
dream5_net1	medium	isotonic_regression	in_distribution	0.8578 ± 0.0379	0.7707 ± 0.0665	0.1342 ± 0.0217	0.0245 ± 0.0118	0.0026 ± 0.0018
dream5_net1	medium	isotonic_regression	shifted_features	0.7529 ± 0.0301	0.6012 ± 0.0696	0.2080 ± 0.0113	0.1524 ± 0.0204	0.1481 ± 0.0235
dream5_net1	medium	predictive_only	in_distribution	0.8592 ± 0.0373	0.7917 ± 0.0597	0.1342 ± 0.0211	0.0289 ± 0.0010	0.0143 ± 0.0128
dream5_net1	medium	predictive_only	shifted_features	0.7570 ± 0.0308	0.6225 ± 0.0717	0.2070 ± 0.0135	0.1498 ± 0.0046	0.1486 ± 0.0044
dream5_net1	medium	temperature_scaling	in_distribution	0.8592 ± 0.0373	0.7917 ± 0.0597	0.1340 ± 0.0214	0.0291 ± 0.0018	0.0153 ± 0.0089
dream5_net1	medium	temperature_scaling	shifted_features	0.7570 ± 0.0308	0.6225 ± 0.0717	0.2061 ± 0.0135	0.1463 ± 0.0132	0.1453 ± 0.0127
dream5_net1	small	end_to_end_lambda_0.1	in_distribution	0.7899 ± 0.0059	0.6798 ± 0.0084	0.1689 ± 0.0015	0.0370 ± 0.0142	0.0170 ± 0.0246
dream5_net1	small	end_to_end_lambda_0.1	shifted_features	0.6892 ± 0.0230	0.5269 ± 0.0488	0.2307 ± 0.0236	0.1591 ± 0.0526	0.1557 ± 0.0532
dream5_net1	small	end_to_end_lambda_0.5	in_distribution	0.8168 ± 0.0112	0.7180 ± 0.0191	0.1571 ± 0.0046	0.0347 ± 0.0072	0.0062 ± 0.0048
dream5_net1	small	end_to_end_lambda_0.5	shifted_features	0.7278 ± 0.0120	0.5796 ± 0.0132	0.2285 ± 0.0224	0.1786 ± 0.0439	0.1763 ± 0.0471
dream5_net1	small	isotonic_regression	in_distribution	0.8318 ± 0.0155	0.7162 ± 0.0197	0.1515 ± 0.0063	0.0277 ± 0.0141	0.0050 ± 0.0076
dream5_net1	small	isotonic_regression	shifted_features	0.7227 ± 0.0292	0.5467 ± 0.0218	0.2384 ± 0.0016	0.1995 ± 0.0104	0.1949 ± 0.0080
dream5_net1	small	predictive_only	in_distribution	0.8336 ± 0.0150	0.7357 ± 0.0185	0.1516 ± 0.0051	0.0377 ± 0.0107	0.0252 ± 0.0219
dream5_net1	small	predictive_only	shifted_features	0.7299 ± 0.0268	0.5699 ± 0.0223	0.2402 ± 0.0051	0.2042 ± 0.0112	0.2001 ± 0.0128
dream5_net1	small	temperature_scaling	in_distribution	0.8336 ± 0.0150	0.7357 ± 0.0185	0.1513 ± 0.0050	0.0340 ± 0.0094	0.0207 ± 0.0177
dream5_net1	small	temperature_scaling	shifted_features	0.7299 ± 0.0268	0.5699 ± 0.0223	0.2365 ± 0.0015	0.1963 ± 0.0034	0.1924 ± 0.0034
sergio	medium	end_to_end_lambda_0.1	in_distribution	0.5902 ± 0.0369	0.1183 ± 0.0447	0.0564 ± 0.0010	0.0059 ± 0.0009	0.0040 ± 0.0024
sergio	medium	end_to_end_lambda_0.1	shifted_features	0.5878 ± 0.0528	0.1048 ± 0.0354	0.0566 ± 0.0009	0.0074 ± 0.0099	0.0074 ± 0.0099
sergio	medium	end_to_end_lambda_0.5	in_distribution	0.6208 ± 0.0223	0.1349 ± 0.0288	0.0561 ± 0.0009	0.0068 ± 0.0027	0.0062 ± 0.0035
sergio	medium	end_to_end_lambda_0.5	shifted_features	0.6111 ± 0.0104	0.1272 ± 0.0322	0.0563 ± 0.0011	0.0101 ± 0.0061	0.0086 ± 0.0074
sergio	medium	isotonic_regression	in_distribution	0.6158 ± 0.0215	0.1080 ± 0.0107	0.0559 ± 0.0011	0.0083 ± 0.0055	0.0014 ± 0.0012
sergio	medium	isotonic_regression	shifted_features	0.6277 ± 0.0046	0.0978 ± 0.0137	0.0589 ± 0.0043	0.0180 ± 0.0131	0.0177 ± 0.0135
sergio	medium	predictive_only	in_distribution	0.6296 ± 0.0091	0.1421 ± 0.0269	0.0558 ± 0.0008	0.0113 ± 0.0060	0.0092 ± 0.0023
sergio	medium	predictive_only	shifted_features	0.6221 ± 0.0095	0.1315 ± 0.0305	0.0561 ± 0.0009	0.0060 ± 0.0049	0.0053 ± 0.0049
sergio	medium	temperature_scaling	in_distribution	0.6296 ± 0.0091	0.1421 ± 0.0269	0.0557 ± 0.0008	0.0086 ± 0.0042	0.0013 ± 0.0018
sergio	medium	temperature_scaling	shifted_features	0.6221 ± 0.0095	0.1315 ± 0.0305	0.0562 ± 0.0007	0.0094 ± 0.0051	0.0084 ± 0.0059
sergio	small	end_to_end_lambda_0.1	in_distribution	0.6139 ± 0.0194	0.1130 ± 0.0155	0.0566 ± 0.0008	0.0146 ± 0.0131	0.0099 ± 0.0073
sergio	small	end_to_end_lambda_0.1	shifted_features	0.6071 ± 0.0314	0.1071 ± 0.0239	0.0573 ± 0.0021	0.0212 ± 0.0193	0.0204 ± 0.0201
sergio	small	end_to_end_lambda_0.5	in_distribution	0.6223 ± 0.0063	0.1322 ± 0.0256	0.0559 ± 0.0010	0.0107 ± 0.0034	0.0029 ± 0.0037
sergio	small	end_to_end_lambda_0.5	shifted_features	0.6185 ± 0.0261	0.1351 ± 0.0364	0.0565 ± 0.0009	0.0193 ± 0.0154	0.0175 ± 0.0122
sergio	small	isotonic_regression	in_distribution	0.6196 ± 0.0076	0.1156 ± 0.0134	0.0568 ± 0.0015	0.0098 ± 0.0055	0.0040 ± 0.0022
sergio	small	isotonic_regression	shifted_features	0.5889 ± 0.0035	0.1095 ± 0.0173	0.0591 ± 0.0028	0.0305 ± 0.0125	0.0305 ± 0.0125
sergio	small	predictive_only	in_distribution	0.6201 ± 0.0049	0.1324 ± 0.0236	0.0559 ± 0.0009	0.0093 ± 0.0020	0.0079 ± 0.0008
sergio	small	predictive_only	shifted_features	0.6137 ± 0.0189	0.1372 ± 0.0386	0.0561 ± 0.0011	0.0101 ± 0.0044	0.0094 ± 0.0033
sergio	small	temperature_scaling	in_distribution	0.6201 ± 0.0049	0.1324 ± 0.0236	0.0559 ± 0.0010	0.0109 ± 0.0044	0.0025 ± 0.0030
sergio	small	temperature_scaling	shifted_features	0.6137 ± 0.0189	0.1372 ± 0.0386	0.0564 ± 0.0009	0.0166 ± 0.0079	0.0163 ± 0.0074

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